GRADE 7 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 8) • Forward 2 (Grade 9)

61 question sets • 1525 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • 7.RP.A.1 • RATIOS & PROPORTIONAL RELATIONSHIPS

Compute unit rates associated with ratios of fractions

MAIN	Grade 7 main question	**7.RP.A.1**
BACKWARD 1	Find a unit rate from whole-number quantities	**6.RP.A.2**
BACKWARD 2	Divide a fraction by a fraction	**6.NS.A.1**
FORWARD 1	Compare unit rates shown in different representations	**8.EE.B.5**
FORWARD 2	Convert rates from one unit to another	**HSN-Q.A.1**

SET 1 • 7.RP.A.1

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour at a steady pace.

How many miles does she walk per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

2 1/4 miles per hour

GRADING NOTE 9/4 or 2.25 miles per hour is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q1/5

A recipe uses $\frac{2}{3}$ cup of sugar for every $\frac{1}{2}$ cup of flour.

How many cups of sugar are used per cup of flour?

Cups of sugar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

1 1/3 cups

GRADING NOTE 4/3 cups is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q2/5

Leo paints $\frac{5}{6}$ of a wall in $\frac{1}{2}$ hour.

How many walls does he paint per hour?

A. $\frac{3}{5}$ wall

B. $1\frac{2}{3}$ walls

C. $\frac{5}{12}$ wall

D. $2\frac{1}{2}$ walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1 2/3 walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q3/5

True or false?

A snail crawls $\frac{1}{8}$ meter in $\frac{1}{4}$ minute. Its speed is $\frac{1}{2}$ meter per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q4/5

True or false?

$\frac{2}{5}$ pound of cheese costs \$3. The price is \$1.20 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is \$3 ÷ 2/5 = \$7.50 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 1

Find a unit rate from whole-number quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A car travels 180 miles in 3 hours at a steady speed.

What is its speed in miles per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q1/5

6 notebooks cost $9.

What is the cost per notebook?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$1.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q2/5

A printer prints 120 pages in 8 minutes.

How many pages does it print per minute?

A. 112

B. 960

C. 15

D. 128

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

Jo reads 90 pages in 5 days at the same rate each day. That is 18 pages per day.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q4/5

True or false?

4 pounds of grapes cost $24. The price is $8 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is $6 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 2

Divide a fraction by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{3}{4} \div \frac{1}{2}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/2 = 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q1/5

Divide.

$\frac{5}{6} \div \frac{1}{3}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5/2 = 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q2/5

Divide.

$\frac{5}{8} \div \frac{1}{4}$

A. $1\frac{1}{4}$

B. $\frac{5}{32}$

C. $\frac{2}{5}$

D. $2\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{3} \div \frac{1}{3} = \frac{2}{9}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 1

Compare unit rates shown in different representations • Grade 8 • Expressions & Equations • 8.EE.B.5

Which relationship has the greatest unit rate?

A. y = 4.2x

B. A table with the pairs (2, 8) and (5, 20)

C. A line through (0, 0) and (3, 12.3)

D. y = 3.9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q1/5

26

QUESTION ID

S1-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4.2x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q1/5

Runner A: a graph of meters (y) against seconds (x) is a line through (0, 0) and (10, 45).

Runner B: d = 4.8t, where d is meters and t is seconds.

Which runner is faster, and by how many meters per second?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q2/5

27

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Runner B, by 0.3 meter per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q2/5

Plan A charges $0.15 per text message. The cost for Plan B is shown by a line through (0, 0) and (40, 5), where x is texts and y is dollars.

Which plan costs less per text?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q3/5

28

QUESTION ID

S1-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Plan B

GRADING NOTE Plan B costs $0.125 per text, which is less than $0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q3/5

True or false?

The graph of y = 2.5x is steeper than the graph of a proportional relationship that passes through (4, 9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q4/5

29

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q4/5

True or false?

A proportional relationship that passes through (6, 15) has a greater unit rate than y = 2.6x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q5/5

30

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 2

Convert rates from one unit to another • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1

A cheetah runs 66 feet per second.

How many miles per hour is this? (1 mile = 5,280 feet)

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q1/5

32

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q1/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 440 feet per minute. (1 mile = 5,280 feet)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q2/5

33

QUESTION ID

S1-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q2/5

A pump moves 3 gallons every $\frac{1}{4}$ minute.

How many gallons per hour is this?

Gallons per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q3/5

34

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

720

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q3/5

A runner covers 400 meters in 80 seconds at a steady pace.

What is the speed in kilometers per hour?

A. 18

B. 300

C. 0.3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q4/5

35

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

60 miles per hour is the same speed as 1 mile per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q5/5

36

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 60 miles per hour is 1 mile per minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q5/5

SET 2 • 7.RP.A.2.a • RATIOS & PROPORTIONAL RELATIONSHIPS

Decide whether two quantities in a table are in a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Complete a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Divide multi-digit decimals	**6.NS.B.3**
FORWARD 1	Decide whether a table shows a linear function	**8.F.A.3**
FORWARD 2	Recognize a quantity that changes at a constant rate per unit inte	**HSF-LE.A.1.b**

SET 2 • 7.RP.A.2.a

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship between x and y.

x	2	5	8
y	7	17.5	28

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q1/5

39

QUESTION ID

S2-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q1/5

Which table shows a proportional relationship?

A. x: 2, 4, 6 and y: 3, 5, 7

B. x: 1, 2, 3 and y: 5, 6, 7

C. x: 3, 6, 9 and y: 12, 24, 36

D. x: 1, 3, 5 and y: 4, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q2/5

40

QUESTION ID

S2-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. x: 3, 6, 9 and y: 12, 24, 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q2/5

The table shows the cost of renting a bike.

Is the cost proportional to the number of hours? Explain.

Hours	1	2	3
Cost ($)	4	9	14

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q3/5

41

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. The ratios 4/1, 9/2, and 14/3 are not equal.

GRADING NOTE Any explanation showing the cost-to-hours ratios are not all equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q3/5

Gallons of water and minutes are in a proportional relationship.

Find the two missing values in the table.

Minutes	2	4	6	?
Gallons	5	10	?	25

Missing values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q4/5

42

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

15 gallons for 6 minutes; 10 minutes for 25 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q4/5

True or false?

The table shows a proportional relationship between x and y.

x	1	2	4
y	3	6	10

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q5/5

43

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 1

Complete a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

What is the missing value?

Cups of flour	2	4	6
Cups of milk	3	?	9

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q1/5

45

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q1/5

The table shows equivalent ratios.

What is the missing value?

Boys	5	10	15
Girls	4	8	?

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q2/5

46

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q2/5

Which ratio is equivalent to 3 : 5?

A. 9 : 15

B. 6 : 8

C. 3 : 15

D. 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q3/5

47

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9 : 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q3/5

True or false?

2 : 7 and 6 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q4/5

48

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q4/5

True or false?

4 : 6 and 6 : 8 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q5/5

49

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 2

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

17.5 ÷ 5

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q1/5

51

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q1/5

Divide.

8.4 ÷ 2.4

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q2/5

52

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q2/5

Divide.

9.6 ÷ 3

A. 3.2

B. 32

C. 0.32

D. 3.02

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q3/5

53

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$28 \div 3.5 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q4/5

54

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$4.5 \div 1.5 = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q5/5

55

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 1

Decide whether a table shows a linear function • Grade 8 • Functions • 8.F.A.3

True or false?

The table shows a linear function.

x	0	1	2	3
y	4	9	14	19

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q1/5

57

QUESTION ID

S2-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q1/5

Which table shows a linear function that is NOT proportional?

A. x: 0, 1, 2 and y: 1, 2, 4

B. x: 0, 1, 2 and y: 0, 4, 8

C. x: 0, 1, 2 and y: 3, 5, 7

D. x: 0, 1, 2 and y: 0, 1, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q2/5

58

QUESTION ID

S2-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. x: 0, 1, 2 and y: 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q2/5

The table shows a linear function.

What is its rate of change?

x	1	2	3	4
y	2	5	8	11

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q3/5

59

QUESTION ID

S2-F1-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q3/5

Is the function in the table linear? Explain.

x	1	2	3	4
y	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q4/5

60

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

No. y increases by 3, then 5, then 7, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

A table with x-values 0, 1, 2 and y-values 7, 9, 11 shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q5/5

61

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is linear, but y is not 0 when x is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 2

Recognize a quantity that changes at a constant rate per unit interval • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

True or false?

In the table, y grows by equal differences over equal intervals of x.

x	0	2	4	6
y	5	11	17	23

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q1/5

63

QUESTION ID

S2-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q1/5

A tank holds 80 gallons and drains 6 gallons every minute.

What is the change in the amount of water per minute?

Change per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q2/5

64

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−6 gallons per minute

GRADING NOTE "It decreases by 6 gallons each minute" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q2/5

Which quantity changes at a constant rate per unit interval?

A. A price increases 5% each year.

B. A population doubles every year.

C. A bacteria count triples each hour.

D. A plant grows 2 cm every week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q3/5

65

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A plant grows 2 cm every week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q3/5

The table shows a linear function.

By how much does y change for each increase of 1 in x?

x	0	3	6	9
y	10	16	22	28

Change in y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q4/5

66

QUESTION ID

S2-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q4/5

True or false?

An account that earns 4% of its current balance each year grows by the same number of dollars each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q5/5

67

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q5/5

SET 3 • 7.RP.A.2.a • RATIOS & PROPORTIONAL RELATIONSHIPS

Decide whether a graph shows a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Plot the pairs from a ratio table	**6.RP.A.3.a**
BACKWARD 2	Find the coordinates of points in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Recognize linear functions whose graphs do not pass through the…	**8.F.A.3**
FORWARD 2	Interpret the intercepts of a graph in context	**HSF-IF.B.4**

SET 3 • 7.RP.A.2.a

MAIN

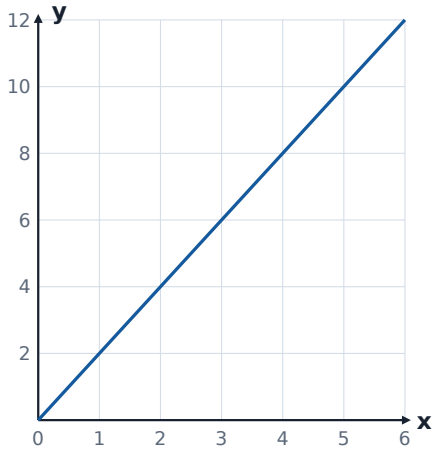
Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship between x and y.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q1/5

70

QUESTION ID

S3-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

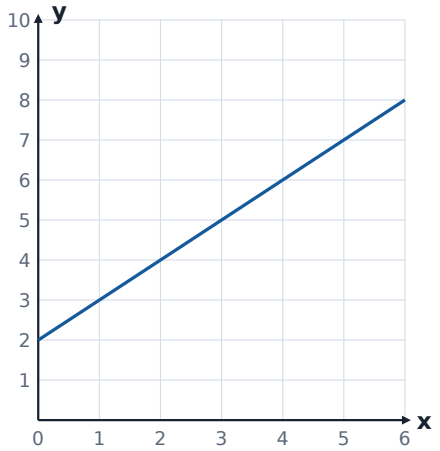
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ID S3-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q1/5

Is the relationship in the graph proportional? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q2/5

71

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. The line does not pass through the origin (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q2/5

Which graph shows a proportional relationship?

A. A straight line that crosses the y-axis at 3

B. Points that do not lie on one line

C. A curve through the origin

D. A straight line through the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q3/5

72

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A straight line through the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

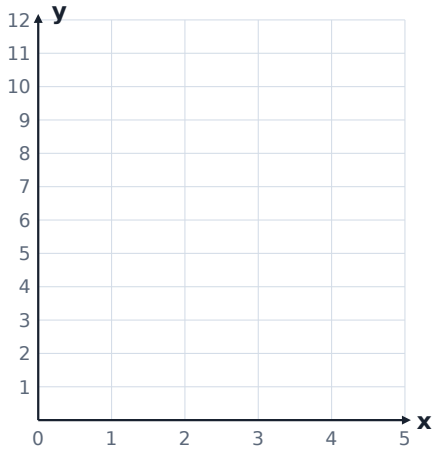
ID S3-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q3/5

Plot the points (1, 2.5), (2, 5), and (4, 10).

Do the points show a proportional relationship? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q4/5

73

QUESTION ID

S3-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (1, 2.5), (2, 5), and (4, 10)

Written: Yes. The points lie on a straight line through the origin; y ÷ x = 2.5 for every point.

GRADING NOTE Grade both: the three points plotted correctly, and "yes" with a reason (same ratio, or a line through the origin).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

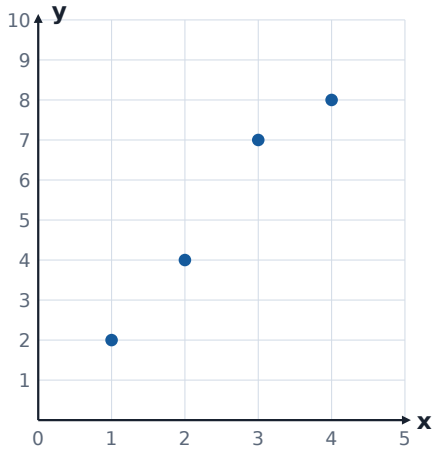
Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q4/5

True or false?

The points in the graph show a proportional relationship.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q5/5

74

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q5/5

SET 3 • 7.RP.A.2.a

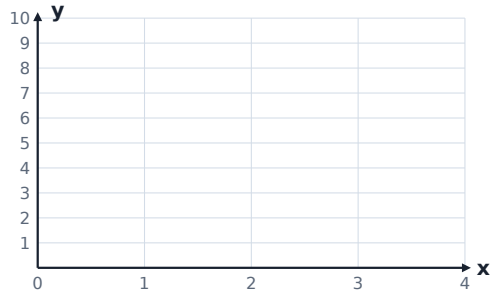
BACKWARD 1

Plot the pairs from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

Plot each pair (x, y) on the grid.

x	1	2	3
y	3	6	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q1/5

76

QUESTION ID

S3-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 3), (2, 6), and (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q1/5

A ratio table has the pairs (2, 5), (4, 10), and (6, ?).

What is the missing y-value?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q2/5

77

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q2/5

Which pair belongs in the same ratio table as (3, 4)?

A. (4, 5)

B. (9, 10)

C. (6, 7)

D. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q3/5

78

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q3/5

True or false?

The pairs (2, 6), (4, 12), and (5, 15) can all come from the same ratio table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q4/5

79

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q4/5

True or false?

To plot the pair (3, 7), you start at the origin, move 7 units right, and then move 3 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q5/5

80

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

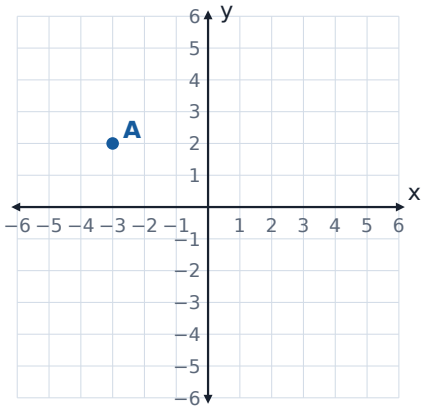
Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

BACKWARD 2

Find the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q1/5

82

QUESTION ID

S3-B2-Q1

TYPE

Short answer

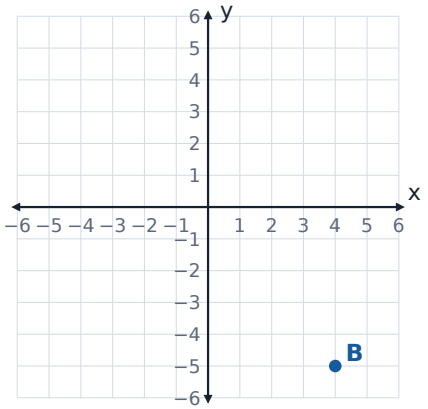
CORRECT ANSWER

(−3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q2/5

83

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q2/5

Which point is at the origin?

A. (1, 1)

B. (0, 1)

C. (1, 0)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q3/5

84

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q3/5

True or false?

The point (3, 0) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q4/5

85

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q4/5

True or false?

The point (0, 5) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q5/5

86

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q5/5

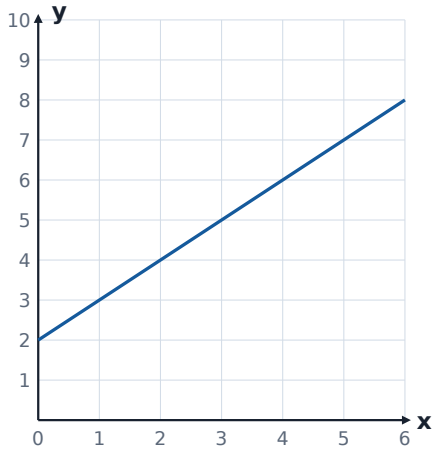
SET 3 • 7.RP.A.2.a

FORWARD 1

Recognize linear functions whose graphs do not pass through the origin • Grade 8 • Functions • 8.F.A.3

The graph shows a line.

Write its equation in the form $y = mx + b$.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q1/5

88

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q1/5

Which equation is linear but NOT proportional?

A. $y = x^2$

B. $y = \frac{3}{x}$

C. $y = 3x$

D. $y = 3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q2/5

89

QUESTION ID

S3-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. y = 3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q2/5

True or false?

The graph of y = 4x is a line through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q3/5

90

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q3/5

A line passes through (0, 3) and (2, 7).

Is the relationship proportional? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q4/5

91

QUESTION ID

S3-F1-Q4

TYPE

Short answer

CORRECT ANSWER

No. The line is linear, but it crosses the y-axis at 3, not at the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q4/5

True or false?

Every linear function is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q5/5

92

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

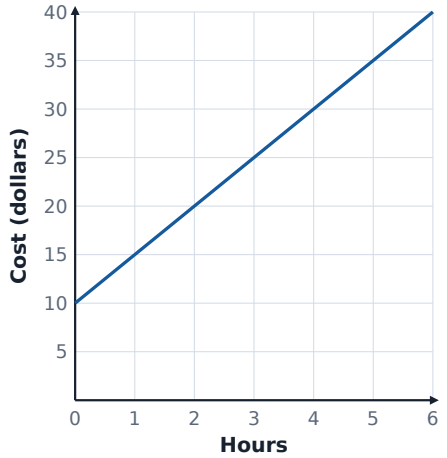
FORWARD 2

Interpret the intercepts of a graph in context • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows the cost of renting a kayak.

What does the point where the line meets the vertical axis mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q1/5

94

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

There is a $10 cost at 0 hours (a starting fee).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q1/5

The height of a candle after t hours is h = 24 − 3t centimeters.

After how many hours does the candle burn out (h = 0)?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q2/5

95

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q2/5

The water in a tank after m minutes is w = 50 − 5m gallons.

What does the w-intercept, 50, mean?

A. The tank starts with 50 gallons.

B. The tank drains 50 gallons per minute.

C. The tank is empty after 50 minutes.

D. The tank drains 5 gallons in 50 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q3/5

96

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The tank starts with 50 gallons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q3/5

True or false?

The cost of a rental is C = 15 + 2h dollars for h hours. The C-intercept, 15, is the cost when h = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q4/5

97

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q4/5

True or false?

For h = 24 − 3t, the h-intercept is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q5/5

98

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The h-intercept is 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b • RATIOS & PROPORTIONAL RELATIONSHIPS

Identify the constant of proportionality in tables, graphs, equations, and descriptions

MAIN	Grade 7 main question	**7.RP.A.2.b**
BACKWARD 1	Describe a unit rate	**6.RP.A.2**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Interpret the unit rate as the slope of a graph	**8.EE.B.5**
FORWARD 2	Write a linear function from two points or a description	**HSF-LE.A.2**

SET 4 • 7.RP.A.2.b

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality (y per x)?

x	2	4	6
y	5	10	15

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q1/5

101

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

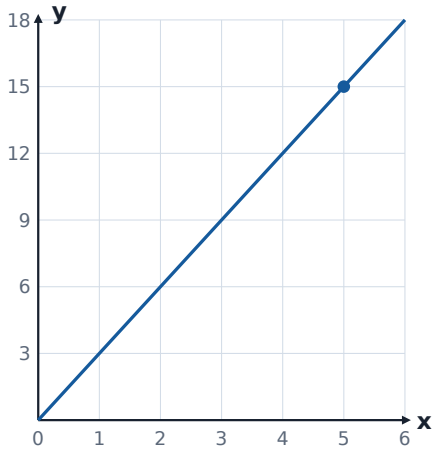
ID S4-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q1/5

The graph shows a proportional relationship.

What is the constant of proportionality?

k = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q2/5

102

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q2/5

What is the constant of proportionality in $y = \frac{2}{3}x$?

A. 2

B. $\frac{3}{2}$

C. 3

D. $\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q3/5

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q3/5

True or false?

6 pencils cost $2.10. The constant of proportionality, in dollars per pencil, is 0.35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q4/5

104

QUESTION ID

S4-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q4/5

Kim types 216 words in 4 minutes at a steady rate.

What is the constant of proportionality in words per minute?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q5/5

105

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 1

Describe a unit rate • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A pack of 8 juice boxes costs $6.

What is the cost per juice box?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q1/5

107

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q1/5

A recipe uses 4 cups of flour for 2 loaves.

Which is the unit rate?

A. 6 cups of flour per loaf

B. 8 cups of flour per loaf

C. $\frac{1}{2}$ cup of flour per loaf

D. 2 cups of flour per loaf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q2/5

108

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 2 cups of flour per loaf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q2/5

Lena swims 600 meters in 12 minutes at a steady pace.

How many meters does she swim per minute?

Meters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q3/5

109

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q3/5

True or false?

"$3 per pound" is a unit rate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q4/5

110

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 apples cost $4. The unit rate is $1.25 per apple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q5/5

111

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The unit rate is $0.80 per apple.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

In the table, y is always 4 times x.

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q1/5

113

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q1/5

Each ticket costs $7.

Which equation gives the cost c of t tickets?

A. $t = 7c$

B. $c = \frac{t}{7}$

C. $c = 7t$

D. $c = t + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q2/5

114

QUESTION ID

S4-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. c = 7t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q2/5

Write an equation for y in terms of x.

x	2	3	5
y	10	15	25

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q3/5

115

QUESTION ID

S4-B2-Q3

TYPE

Short answer

CORRECT ANSWER

y = 5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q3/5

True or false?

For y = 6x, when x = 3, y = 18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q4/5

116

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q4/5

True or false?

In d = 50t, d is the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q5/5

117

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE t is the independent variable; d depends on t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b

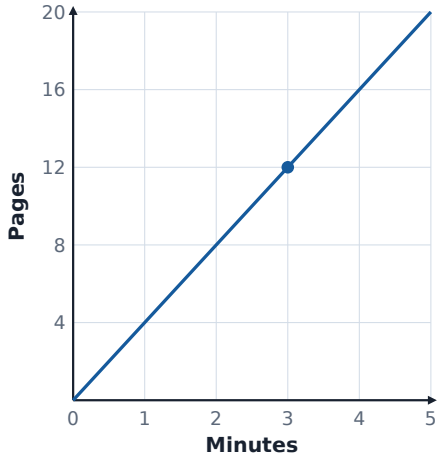
FORWARD 1

Interpret the unit rate as the slope of a graph • Grade 8 • Expressions & Equations • 8.EE.B.5

The graph shows a proportional relationship between minutes and pages read.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q1/5

119

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 0) and (8, 6).

What is its slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q2/5

120

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

GRADING NOTE 0.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q2/5

The graph of a proportional relationship has a slope of 1.5 liters per minute.

Which point is on the graph?

A. (4, 6)

B. (6, 4)

C. (1.5, 1)

D. (4, 5.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q3/5

121

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q3/5

True or false?

The slope of the graph of y = 0.8x is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q4/5

122

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q4/5

True or false?

A line through the origin and (2, 9) has a slope of $\frac{2}{9}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q5/5

123

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The slope is 9/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

FORWARD 2

Write a linear function from two points or a description • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A linear function passes through (1, 5) and (3, 11).

Write the function.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q1/5

125

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q1/5

A gym charges a $20 joining fee plus $15 each month.

Write a function for the total cost C after m months.

C(m) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q2/5

126

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15m + 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q2/5

Which linear function passes through (0, −4) and (2, 2)?

A. f(x) = 3x + 2

B. f(x) = −4x + 3

C. f(x) = 3x − 4

D. f(x) = 2x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q3/5

127

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. f(x) = 3x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

The linear function through (2, 7) and (4, 11) is f(x) = 2x + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q4/5

128

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

The linear function through (0, 1) and (5, 11) is f(x) = 5x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q5/5

129

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The slope is 2, so f(x) = 2x + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q5/5

Represent a proportional relationship with an equation

MAIN	Grade 7 main question	**7.RP.A.2.c**
BACKWARD 1	Write an equation for a dependent variable	**6.EE.C.9**
BACKWARD 2	Solve unit rate problems	**6.RP.A.3.b**
FORWARD 1	Construct a linear function y = mx + b from a situation	**8.F.B.4**
FORWARD 2	Write an equation in two variables for a situation	**HSA-CED.A.2**

SET 5 • 7.RP.A.2.c

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Peaches cost $2.40 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q1/5

132

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

c = 2.4p

GRADING NOTE c = 2.40p is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q1/5

A car travels 29 miles per gallon of gas.

Which equation gives the miles m for g gallons?

A. $m = 29g$

B. $m = g + 29$

C. $g = 29m$

D. $m = \frac{g}{29}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q2/5

133

QUESTION ID

S5-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. m = 29g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	4	8	12
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q3/5

134

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = 0.75x

GRADING NOTE y = (3/4)x is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q3/5

True or false?

Notebooks cost $1.75 each. The equation c = 1.75n gives the cost c of n notebooks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q4/5

135

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q4/5

True or false?

A printer prints 12 pages every 3 minutes. The equation p = 12m gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q5/5

136

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is p = 4m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 1

Write an equation for a dependent variable • Grade 6 • Expressions & Equations • 6.EE.C.9

A worker earns $18 per hour.

Write an equation for the earnings e after h hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q1/5

138

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

e = 18h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q1/5

Which equation matches the table?

x	1	2	3
y	9	18	27

A. y = x + 8

B. y = 3x

C. x = 9y

D. y = 9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q2/5

139

QUESTION ID

S5-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. y = 9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q2/5

Water fills a tank at 6 liters per minute.

Write an equation for the liters L after t minutes.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q3/5

140

QUESTION ID

S5-B1-Q3

TYPE

Short answer

CORRECT ANSWER

L = 6t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q3/5

True or false?

In e = 18h, the value of e depends on the value of h.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q4/5

141

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q4/5

True or false?

The point (2, 7) is on the graph of y = 5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q5/5

142

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 2

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

5 bags of rice cost $15.

At the same rate, how much do 8 bags cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q1/5

144

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q1/5

A machine makes 140 parts in 7 hours.

At the same rate, how many parts does it make in 10 hours?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q2/5

145

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q2/5

A bus travels at 45 miles per hour.

How long does it take to travel 135 miles?

A. 3 hours

B. 2 hours

C. 4 hours

D. 6 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q3/5

146

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q3/5

True or false?

At $4 per pound, 6 pounds cost $24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q4/5

147

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q4/5

True or false?

A printer prints 12 pages per minute. It takes 6 minutes to print 60 pages.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q5/5

148

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It takes 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 1

Construct a linear function y = mx + b from a situation • Grade 8 • Functions • 8.F.B.4

A plumber charges $60 for a visit plus $45 per hour.

Write an equation for the cost y after x hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q1/5

150

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = 45x + 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q1/5

The table shows a linear function.

Write its equation in the form y = mx + b.

x	0	1	2
y	7	10	13

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q2/5

151

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q2/5

A line has a slope of 2 and passes through (0, −5).

Which is its equation?

A. y = −5x + 2

B. y = −2x − 5

C. y = 2x − 5

D. y = 2x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q3/5

152

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q3/5

True or false?

A pool has 300 gallons and fills at 20 gallons per minute. The equation y = 20x + 300 gives the gallons y after x minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q4/5

153

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q4/5

True or false?

A candle 15 cm tall burns 2 cm per hour. Its height after x hours is y = 15x − 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q5/5

154

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The height is y = −2x + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 2

Write an equation in two variables for a situation • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2

Pens cost $2 each and markers cost $3 each. Kai spends exactly $30.

Write an equation using p for pens and m for markers.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q1/5

156

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2p + 3m = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q1/5

Adult tickets cost $12 and child tickets cost $8. A group pays $96.

Write an equation using a for adult tickets and c for child tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q2/5

157

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12a + 8c = 96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q2/5

A farm has 40 animals. Every animal is a chicken or a goat.

Which equation relates the number of chickens c and goats g?

A. cg = 40

B. c + g = 40

C. 2c + 4g = 40

D. c = 40g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q3/5

158

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. c + g = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q3/5

True or false?

The equation 5x + 2y = 50 can describe buying x items at $5 each and y items at $2 each for a total of $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q4/5

159

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q4/5

True or false?

p = 6 and m = 4 is a solution of 2p + 3m = 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q5/5

160

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q5/5

Explain what a point on the graph of a proportional relationship means

MAIN	Grade 7 main question	**7.RP.A.2.d**
BACKWARD 1	Explain a pair in a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Read values from the graph of an equation	**6.EE.C.9**
FORWARD 1	Interpret the rate of change and initial value of a linear function	**8.F.B.4**
FORWARD 2	Calculate the average rate of change over an interval	**HSF-IF.B.6**

SET 6 • 7.RP.A.2.d

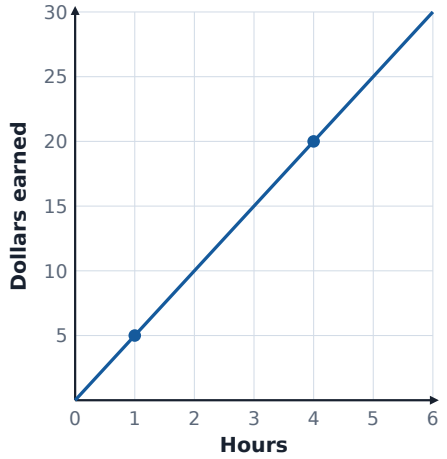
MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the money Raj earns.

What does the point (4, 20) mean in this situation?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q1/5

163

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

Raj earns $20 for working 4 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

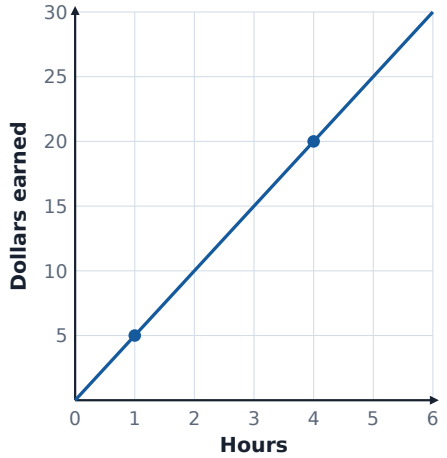
ID S6-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q1/5

The graph shows the money Raj earns.

What does the point (1, 5) tell you?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q2/5

164

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

Raj earns $5 per hour. This is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q2/5

The graph shows the money Raj earns.

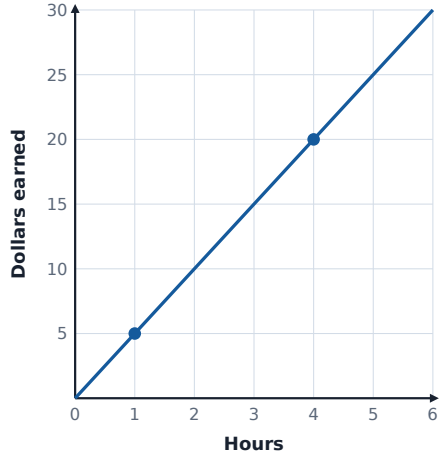
What does the point (0, 0) mean?

A. Raj earns $0 when he works 0 hours.

B. Raj earns $5 per hour.

C. Raj works 0 hours to earn $5.

D. Raj starts with $5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q3/5

165

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Raj earns $0 when he works 0 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q3/5

True or false?

A proportional graph shows cups of rice (x) and cups of water (y). The point (1, 3.5) means 3.5 cups of water are used for each cup of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q4/5

166

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q4/5

True or false?

A car's distance graph passes through (2, 110), where x is hours and y is miles. This means the car travels 110 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q5/5

167

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 110 miles in 2 hours, or 55 miles per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q5/5

SET 6 • 7.RP.A.2.d

BACKWARD 1

Explain a pair in a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows the cost of apples.

What does the pair (2, 6) mean?

Pounds	1	2	3
Cost ($)	3	6	9

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q1/5

169

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2 pounds of apples cost $6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q1/5

The table shows the time a swimmer takes.

What does the pair (4, 10) mean?

Laps	2	4	6
Minutes	5	10	15

A. Each lap takes 4 minutes.

B. 4 laps take 10 minutes.

C. 10 laps take 4 minutes.

D. Each lap takes 10 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q2/5

170

QUESTION ID

S6-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 4 laps take 10 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q2/5

True or false?

A table shows hours (x) and miles walked (y). The pair (3, 12) means 12 miles are walked in 3 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q3/5

171

QUESTION ID

S6-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q3/5

One ticket costs $8.

What pair (tickets, cost) shows the cost of 5 tickets?

Pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q4/5

172

QUESTION ID

S6-B1-Q4

TYPE

Short answer

CORRECT ANSWER

(5, 40)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q4/5

True or false?

A table shows boxes (x) and pencils (y). The pair (2, 24) means each box holds 2 pencils.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q5/5

173

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 2 boxes hold 24 pencils.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 7.RP.A.2.d

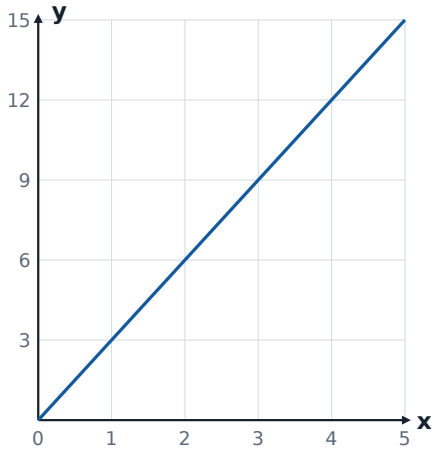
BACKWARD 2

Read values from the graph of an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

The graph shows y = 3x.

What is y when x = 4?

y = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q1/5

175

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

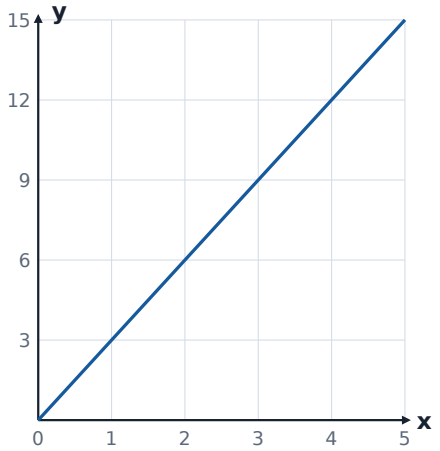
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q1/5

The graph shows y = 3x.

What is x when y = 6?



x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q2/5

176

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q2/5

Which point is on the graph of y = 2x?

A. (6, 3)

B. (4, 6)

C. (3, 6)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q3/5

177

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q3/5

True or false?

The point (5, 15) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q4/5

178

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q4/5

True or false?

The point (1, 4) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q5/5

179

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 1

Interpret the rate of change and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A taxi ride costs y = 2.5x + 4 dollars for x miles.

What does the 4 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q1/5

181

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A $4 starting fee (the cost at 0 miles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q1/5

A taxi ride costs y = 2.5x + 4 dollars for x miles.

What does the 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q2/5

182

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The cost increases by $2.50 for each mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q2/5

A tank holds $y = 200 - 8x$ gallons after x minutes.

What is the rate of change?

A. −8 gallons per minute

B. 200 gallons per minute

C. 8 gallons

D. −200 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q3/5

183

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8 gallons per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q3/5

True or false?

A plant's height after x weeks is y = 0.5x + 3 centimeters. The plant was 3 cm tall at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q4/5

184

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q4/5

True or false?

In y = 12x + 30, the initial value is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q5/5

185

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 2

Calculate the average rate of change over an interval • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$

What is the average rate of change of f from x = 1 to x = 3?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q1/5

187

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q1/5

The table shows the height of a balloon.

What is the average rate of change from 2 seconds to 4 seconds?

Time (s)	0	2	4
Height (m)	0	40	60

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q2/5

188

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 meters per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q2/5

A car is 30 miles from home at 1 p.m. and 150 miles from home at 3 p.m.

What is its average rate of change in distance?

A. 75 miles per hour

B. 60 miles per hour

C. 50 miles per hour

D. 120 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q3/5

189

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 60 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q3/5

True or false?

For f(x) = 2x + 1, the average rate of change over any interval is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q4/5

190

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ equals the average rate of change from $x = 2$ to $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q5/5

191

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 2 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q5/5

SET 7 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Solve multistep ratio problems

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Use ratio language to describe two quantities	**6.RP.A.1**
BACKWARD 2	Use a table of equivalent ratios to solve a problem	**6.RP.A.3.a**
FORWARD 1	Solve linear equations with rational coefficients	**8.EE.C.7.b**
FORWARD 2	Create one-variable equations to solve ratio problems	**HSA-CED.A.1**

SET 7 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A recipe uses flour and sugar in a ratio of 3 : 2. Lee uses $7\frac{1}{2}$ cups of flour.

How many cups of sugar does she need?

Cups of sugar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q1/5

194

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q1/5

A club has boys and girls in a ratio of 4 : 5. There are 36 members.

How many girls are in the club?

Girls: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q2/5

195

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q2/5

Ali mixes blue and yellow paint in a ratio of 2 : 3. He makes 15 liters of paint.

How many liters are blue?

A. 10 liters

B. 9 liters

C. 7.5 liters

D. 6 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q3/5

196

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q3/5

True or false?

On a map, 2 cm represents 15 km. A distance of 7 cm on the map represents 52.5 km.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q4/5

197

QUESTION ID

S7-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q4/5

True or false?

Juice concentrate and water are mixed in a ratio of 1 : 4. To make 6 liters of juice, you need 6 liters of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q5/5

198

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You need 4.8 liters of water.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 1

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A class has 12 girls and 15 boys.

Write the ratio of boys to girls.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q1/5

200

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15 : 12

GRADING NOTE 5 : 4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q1/5

For every 3 red tiles, there are 5 blue tiles.

What is the ratio of blue tiles to all tiles?

A. 5 : 8

B. 5 : 3

C. 8 : 5

D. 3 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q2/5

201

QUESTION ID

S7-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 5 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q2/5

A team won 9 games and lost 6 games. There were no ties.

Write the ratio of wins to games played.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q3/5

202

QUESTION ID

S7-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9 : 15

GRADING NOTE 3 : 5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q3/5

True or false?

There are 2 cats for every 7 dogs. The ratio of dogs to cats is 7 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q4/5

203

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q4/5

True or false?

A bowl has 4 apples and 9 oranges and no other fruit. The ratio of apples to all the fruit is 4 : 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q5/5

204

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ratio is 4 : 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 2

Use a table of equivalent ratios to solve a problem • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Lemonade uses 2 cups of mix for every 5 cups of water.

How many cups of water are needed for 8 cups of mix?

Cups of water: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q1/5

206

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q1/5

A store sells 3 notebooks for $4.

At the same rate, how much do 12 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q2/5

207

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q2/5

Which pair belongs in a table of ratios equivalent to 3 : 4?

A. 9 : 16

B. 12 : 15

C. 6 : 7

D. 15 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q3/5

208

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q3/5

True or false?

A table of ratios equivalent to 5 : 2 can include 20 : 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q4/5

209

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q4/5

True or false?

A table of ratios equivalent to 2 : 3 can include 8 : 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q5/5

210

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 1

Solve linear equations with rational coefficients • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for b.

$5b = 3(b + 6)$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q1/5

212

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q1/5

Solve for x.

$\frac{2}{3}x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q2/5

213

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$0.4(x - 5) = 2$

A. $x = 0.5$

B. $x = 3$

C. $x = 10$

D. $x = 5.8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q3/5

214

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q3/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x - 2 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q4/5

215

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q4/5

True or false?

$x = 4$ is the solution of $2(x + 3) = 5x - 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q5/5

216

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 2

Create one-variable equations to solve ratio problems • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1

The ratio of red marbles to green marbles is 3 : 5. There are 14 more green marbles than red marbles.

Write and solve an equation to find the number of red marbles.

Red marbles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q1/5

218

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

21

GRADING NOTE Example equation: 5x − 3x = 14, so x = 7 and there are 3(7) = 21 red marbles. Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q1/5

A 60-meter rope is cut into two pieces in a ratio of 2 : 3.

Write and solve an equation to find the length of the longer piece.

Longer piece: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q2/5

219

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

36 meters

GRADING NOTE Example equation: 2x + 3x = 60, so x = 12. Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q2/5

The ages of two cousins are in a ratio of 4 : 7. The sum of their ages is 66.

Which equation can find x, where the ages are 4x and 7x?

A. $4x \cdot 7x = 66$

B. $4x + 7x = 66$

C. $4 + 7x = 66$

D. $7x - 4x = 66$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q3/5

220

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4x + 7x = 66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q3/5

True or false?

The ratio of a to b is 2 : 5, and a + b = 35. Then a = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q4/5

221

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q4/5

True or false?

The ratio of a to b is 1 : 3, and b − a = 12. Then a = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q5/5

222

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE a = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q5/5

Find a percent increase or a percent decrease

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of a quantity	**6.RP.A.3.c**
BACKWARD 2	Find the whole from a part and a percent	**6.RP.A.3.c**
FORWARD 1	Compare quantities written in scientific notation — nearest relate…	**8.EE.A.3**
FORWARD 2	Model a quantity that changes by a constant percent rate	**HSF-LE.A.1.c**

SET 8 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A town's population grew from 2,500 to 3,000.

What is the percent increase?

Percent increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q1/5

225

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q1/5

The price of a jacket dropped from $80 to $60.

What is the percent decrease?

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q2/5

226

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q2/5

A plant grew from 40 cm to 46 cm tall.

What is the percent increase?

A. 15%

B. 13%

C. 87%

D. 6%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q3/5

227

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q3/5

True or false?

A score that falls from 50 to 40 is a 20% decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q4/5

228

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q4/5

True or false?

A price that rises from $60 to $75 is a 15% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q5/5

229

QUESTION ID

S8-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a 25% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 30% of 70?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q1/5

231

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q1/5

What is 15% of $80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q2/5

232

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q2/5

What is 4% of 250?

A. 40

B. 100

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q3/5

233

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q3/5

True or false?

75% of 40 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q4/5

234

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q4/5

True or false?

20% of 45 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q5/5

235

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20% of 45 is 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 2

Find the whole from a part and a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

12 is 40% of what number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

237

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

9 students in a class walk to school. That is 25% of the class.

How many students are in the class?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

238

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

18 is 60% of what number?

A. 48

B. 12

C. 10.8

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

239

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

15 is 50% of 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

240

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

8 is 20% of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

241

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8 is 50% of 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 7.RP.A.3

FORWARD 1

Compare quantities written in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

City A has about 3×10^6 people. Town B has about 6×10^4 people.

How many times as many people live in City A as in Town B?

Times as many: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q1/5 • NEAREST RELATED

243

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write 4,500,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q2/5 • NEAREST RELATED

244

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which number is the greatest?

A. 9×10^6

B. 8×10^5

C. 2×10^7

D. 1×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q3/5 • NEAREST RELATED

245

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

8×10^8 is 100 times as large as 8×10^6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q4/5 • NEAREST RELATED

246

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

3×10^{-2} is greater than 3×10^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q5/5 • NEAREST RELATED

247

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 8 • 7.RP.A.3

FORWARD 2

Model a quantity that changes by a constant percent rate • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A town of 10,000 people grows by 5% each year.

What is the population after 2 years?

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q1/5

249

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11,025

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q1/5

Which situation shows growth by a constant percent rate?

A. A plant grows 2 cm each week.

B. An account earns 3% of its current balance each year.

C. A worker earns $15 more each year.

D. A tank fills 4 gallons each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q2/5

250

QUESTION ID

S8-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. An account earns 3% of its current balance each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q2/5

A car worth $20,000 loses 10% of its value each year.

What is it worth after 2 years?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q3/5

251

QUESTION ID

S8-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$16,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q3/5

True or false?

A quantity that is multiplied by 1.08 each year grows by 8% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q4/5

252

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q4/5

True or false?

A quantity that decreases by 10% each year loses the same number of units each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q5/5

253

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q5/5

Solve problems with markups, discounts, sales tax, and tips

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Multiply decimals	**6.NS.B.3**
BACKWARD 2	Find a percent of an amount of money	**6.RP.A.3.c**
FORWARD 1	Solve equations to find an original price	**8.EE.C.7.b**
FORWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**

SET 9 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A meal costs $45. Kai leaves a 20% tip.

What is the total cost of the meal with the tip?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q1/5

256

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q1/5

A bike costs $120. It is on sale for 25% off. Then 8% sales tax is added to the sale price.

What is the final cost?

Final cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q2/5

257

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$97.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q2/5

A store buys a lamp for $30 and marks the price up by 60%.

What is the selling price?

A. $36

B. $18

C. $90

D. $48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q3/5

258

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q3/5

True or false?

A pair of shoes costs $64. At 15% off, the sale price is $54.40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q4/5

259

QUESTION ID

S9-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q4/5

True or false?

A shirt costs $35. With 6% sales tax, the total is $37.20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q5/5

260

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total is $37.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 1

Multiply decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

0.15 × 80

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q1/5

262

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q1/5

Multiply.

1.08 × 25

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q2/5

263

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q2/5

Multiply.

0.06×45

A. 270

B. 2.7

C. 0.27

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q3/5

264

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$0.2 \times 36.5 = 7.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q4/5

265

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$1.5 \times 4.2 = 6.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q5/5

266

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 2

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 10% of $56?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q1/5

268

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$5.60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q1/5

What is 25% of $48?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q2/5

269

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q2/5

What is 8% of $50?

A. $0.40

B. $4

C. $8

D. $40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q3/5

270

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q3/5

True or false?

20% of $35 is $7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q4/5

271

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q4/5

True or false?

5% of $60 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q5/5

272

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 1

Solve equations to find an original price • Grade 8 • Expressions & Equations • 8.EE.C.7.b

After a 20% discount, a game costs $36. The equation 0.8p = 36 models this.

What was the original price p?

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q1/5

274

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q1/5

A meal plus a 15% tip cost $46. The equation $1.15m = 46$ models this.

What was the cost m of the meal?

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q2/5

275

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q2/5

Solve for x.

1.06x = 53

A. x = 5

B. x = 51.94

C. x = 50

D. x = 56.18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q3/5

276

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q3/5

True or false?

If 0.75p = 30, then p = 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q4/5

277

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q4/5

True or false?

If 1.25p = 50, then p = 62.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q5/5

278

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE p = 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 2

Interpret the parts of an expression in context • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A price p is discounted 10%. Then 7% sales tax is added. The final cost is 1.07(0.9p).

What does 0.9p represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q1/5

280

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The price after the 10% discount (90% of p).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q1/5

The final cost of an item is 1.07(0.9p).

What does the factor 1.07 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q2/5

281

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Adding 7% sales tax (paying 107% of the discounted price).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q2/5

A store's selling price is p + 0.4p, where p is what the store paid.

What does 0.4p represent?

A. The selling price

B. The amount of a 40% discount

C. The price the store paid

D. The amount of the 40% markup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q3/5

282

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The amount of the 40% markup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q3/5

True or false?

The expression 0.85p gives the price after a 15% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q4/5

283

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q4/5

True or false?

The expression 1.2p gives the price after a 20% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q5/5

284

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1.2p is a 20% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q5/5

SET 10 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Solve problems with simple interest, commissions, and fees

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Evaluate a formula	**6.EE.A.2.c**
BACKWARD 2	Write a percent as a rate per 100 and as a decimal	**6.RP.A.3.c**
FORWARD 1	Model a growing balance with a linear function	**8.F.B.4**
FORWARD 2	Compare simple interest with compound interest	**HSF-LE.A.1.a**

SET 10 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ella deposits $800 in an account that earns 3% simple interest per year.

How much interest does she earn in 4 years?

Interest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q1/5

287

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q1/5

A salesperson earns a 6% commission. She sells $2,450 of products.

What is her commission?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q2/5

288

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$147

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q2/5

A ticket website charges a 2.5% fee on a $600 order.

What is the fee?

A. $150

B. $15

C. $25

D. $1.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q3/5

289

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q3/5

True or false?

$1,500 borrowed at 5% simple interest per year for 2 years has $150 in interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q4/5

290

QUESTION ID

S10-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q4/5

True or false?

A 4% fee on $250 is $1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q5/5

291

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The fee is $10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 1

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use I = prt.

Find I when p = 200, r = 0.05, and t = 3.

I = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q1/5

293

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q1/5

Use A = lw.

Find A when l = 12 and w = 7.5.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q2/5

294

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q2/5

Use d = rt.

Find d when r = 55 and t = 4.

A. 13.75

B. 220

C. 51

D. 59

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q3/5

295

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 220

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q3/5

True or false?

For P = 2l + 2w, when l = 6 and w = 4, P = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q4/5

296

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q4/5

True or false?

For C = 5n + 3, when n = 4, C = 32.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q5/5

297

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE C = 23.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 2

Write a percent as a rate per 100 and as a decimal • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 6% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q1/5

299

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q1/5

Write 2.5% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q2/5

300

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.025

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q2/5

Which decimal is equal to 45%?

A. 4.5

B. 0.45

C. 45

D. 0.045

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q3/5

301

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q3/5

True or false?

$3\% = \frac{3}{100} = 0.03$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q4/5

302

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q4/5

True or false?

8% = 0.8

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q5/5

303

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 1

Model a growing balance with a linear function • Grade 8 • Functions • 8.F.B.4

An account starts with $500 and earns $20 in simple interest each year.

Write a function for the balance B after t years.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q1/5

305

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

20t + 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q1/5

A salesperson earns $300 a week plus a 5% commission on sales s.

Write a function for the weekly pay P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q2/5

306

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.05s + 300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q2/5

A loan balance after t years is B = 1,200 + 36t.

What is the rate of change?

A. $36 per year

B. $1,236 per year

C. $1,200 per year

D. 3% per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q3/5

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $36 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

For B = 20t + 500, the balance after 3 years is $560.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q4/5

308

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

For P = 0.05s + 300, the initial value is 0.05.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q5/5

309

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 2

Compare simple interest with compound interest • Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

$1,000 earns 5% simple interest per year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q1/5

311

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$1,100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q1/5

$1,000 earns 5% interest compounded once each year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q2/5

312

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$1,102.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q2/5

Which kind of interest makes a balance grow by equal differences over equal time intervals?

A. Both kinds

B. Compound interest

C. Simple interest

D. Neither kind

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q3/5

313

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Simple interest

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q3/5

True or false?

With compound interest, a balance grows by equal factors over equal time intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q4/5

314

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q4/5

True or false?

After 1 year, $500 at 4% compound interest (compounded yearly) earns more than $500 at 4% simple interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q5/5

315

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE After 1 year, both earn $20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

Grade 9 (High School) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q5/5

SET 11 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find the percent error of an estimate or a measurement

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find the absolute value of a number	**6.NS.C.7.c**
BACKWARD 2	Write a fraction as a percent	**6.RP.A.3.c**
FORWARD 1	Compute with numbers in scientific notation — nearest related (n…	**8.EE.A.4**
FORWARD 2	Choose a level of accuracy that fits a measurement	**HSN-Q.A.3**

SET 11 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ty estimates that 50 people are at a meeting. The actual number is 40.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q1/5

318

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q1/5

A scale shows 4.8 kg for a bag whose actual mass is 5 kg.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q2/5

319

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

4%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q2/5

A student measures a line that is actually 20 cm long as 19 cm.

What is the percent error?

A. 95%

B. 5%

C. 5.3%

D. 1%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q3/5

320

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q3/5

True or false?

A guess of 90 for an actual value of 100 has a percent error of 10%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q4/5

321

QUESTION ID

S11-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q4/5

True or false?

To find percent error, you divide the size of the error by the estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q5/5

322

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You divide by the actual value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 1

Find the absolute value of a number • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-12|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q1/5

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q1/5

Find the value.

$|-4.5|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q2/5

325

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q2/5

Which number has the greatest absolute value?

A. −3

B. 8

C. −9

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q3/5

326

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q3/5

True or false?

$|-15| = 15$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q4/5

327

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q4/5

True or false?

$|-2| < |1|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q5/5

328

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 2

Write a fraction as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write $\frac{3}{20}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q1/5

330

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q1/5

Write $\frac{7}{25}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q2/5

331

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

28%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q2/5

Which percent is equal to $\frac{9}{12}$?

A. 75%

B. 80%

C. 9%

D. 12%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q3/5

332

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{8} = 12.5\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q4/5

333

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{5} = 25\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q5/5

334

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 7.RP.A.3

FORWARD 1

Compute with numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Multiply. Write the answer in scientific notation.

$(3 \times 10^4) \times (2 \times 10^3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q1/5 • NEAREST RELATED

336

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q1/5 • NEAREST RELATED

Divide. Write the answer in scientific notation.

$(8 \times 10^{9}) \div (4 \times 10^{3})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q2/5 • NEAREST RELATED

337

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2×10^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q2/5 • NEAREST RELATED

Add.

$(2.5 \times 10^5) + (4 \times 10^4)$

A. 6.5×10^5

B. 2.9×10^5

C. 2.9×10^4

D. 6.5×10^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q3/5 • NEAREST RELATED

338

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.9×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$(5 \times 10^3) \times (3 \times 10^2) = 1.5 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q4/5 • NEAREST RELATED

339

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$(6 \times 10^8) \div (2 \times 10^2) = 3 \times 10^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q5/5 • NEAREST RELATED

340

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 3×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 11 • 7.RP.A.3

FORWARD 2

Choose a level of accuracy that fits a measurement • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3

A ruler is marked in millimeters. A length is read as 12.4 cm.

To what place value is this measurement given?

Place: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q1/5

342

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Tenths of a centimeter (the nearest millimeter)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q1/5

A scale measures mass to the nearest 0.1 kg.

Which reading is reported to an accuracy that fits this scale?

A. 3 kg

B. 3.4 kg

C. 3.4172 kg

D. 3.41 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q2/5

343

QUESTION ID

S11-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 3.4 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q2/5

A track is measured as 400 m. The measurement could be off by as much as 2 m.

What is the greatest possible percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q3/5

344

QUESTION ID

S11-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q3/5

True or false?

A length measured with a meter stick marked only in centimeters should not be reported as 1.2347 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q4/5

345

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q4/5

True or false?

Writing more decimal places always makes a measurement more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q5/5

346

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a • THE NUMBER SYSTEM

Describe situations in which opposite quantities combine to make 0

MAIN	Grade 7 main question	**7.NS.A.1.a**
BACKWARD 1	Use positive and negative numbers to represent quantities	**6.NS.C.5**
BACKWARD 2	Recognize opposite numbers	**6.NS.C.6.a**
FORWARD 1	Use opposite exponents with the properties of integer exponents…	**8.EE.A.1**
FORWARD 2	Explain the steps used to solve an equation	**HSA-REI.A.1**

SET 12 • 7.NS.A.1.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.a

Ari has $15 in his wallet. He spends all $15.

Write an addition expression for this situation and find its value.

Expression and value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q1/5

349

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

15 + (−15) = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q1/5

Which situation combines to make 0?

A. Gaining 5 points, then gaining 5 more points

B. A temperature drop of 6°, then another drop of 6°

C. Climbing up 8 floors, then down 8 floors

D. Walking 3 miles east, then 2 miles west

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q2/5

350

QUESTION ID

S12-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Climbing up 8 floors, then down 8 floors

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q2/5

A diver is 12 feet below the surface of the water (−12 feet).

How many feet must she rise to be at the surface (0 feet)?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q3/5

351

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q3/5

True or false?

A gain of 9 yards followed by a loss of 9 yards is a total change of 0 yards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q4/5

352

QUESTION ID

S12-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q4/5

True or false?

A deposit of $40 followed by a withdrawal of $30 leaves an account balance unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q5/5

353

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 1

Use positive and negative numbers to represent quantities • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 20 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q1/5

355

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q1/5

Write an integer for a gain of 7 yards.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q2/5

356

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE +7 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q2/5

Which integer represents a withdrawal of $35?

A. 35

B. −35

C. −3.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q3/5

357

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A temperature of 4 degrees below zero can be written as $-4°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q4/5

358

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q4/5

True or false?

The integer −10 represents a deposit of $10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q5/5

359

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 2

Recognize opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 8?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q1/5

361

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q1/5

What is the opposite of −13?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q2/5

362

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

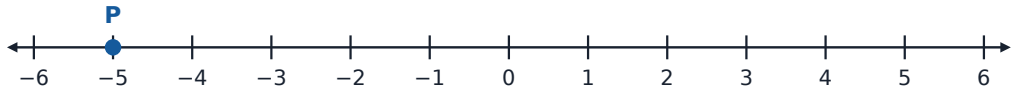
13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q2/5

Point P is at −5.

Which number is the same distance from 0 on the other side of 0?



A. 5 **B.** −10 **C.** 10 **D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q3/5

363

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 6 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q4/5

364

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q4/5

True or false?

0 and 1 are opposites.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q5/5

365

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a

FORWARD 1

Use opposite exponents with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Find the value.

$5^{3} \times 5^{-3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q1/5 • NEAREST RELATED

367

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write as a single power. Then find its value.

$2^{4} \times 2^{-6}$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q2/5 • NEAREST RELATED

368

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$2^{-2} = 1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which expression is equal to 1?

A. $7^2 \times 7^2$

B. $7^{-2} \times 7^{-2}$

C. $7^{-2} + 7^2$

D. $7^{-2} \times 7^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q3/5 • NEAREST RELATED

369

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $7^{-2} \times 7^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$10^{-2} = \frac{1}{100}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q4/5 • NEAREST RELATED

370

QUESTION ID

S12-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$3^{-2} = -9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q5/5 • NEAREST RELATED

371

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^{-2} = 1/9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 12 • 7.NS.A.1.a

FORWARD 2

Explain the steps used to solve an equation • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

To solve $x - 9 = 4$, Lia adds 9 to both sides.

What property of equality justifies this step?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q1/5

373

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The addition property of equality

GRADING NOTE Any answer saying that adding the same number to both sides keeps the equation true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q1/5

Solve x + 15 = 6.

What number do you add to both sides, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q2/5

374

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Add −15 to both sides; x = −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q2/5

Which step removes the 5 when solving 3x + 5 = 20?

A. Multiply both sides by −5.

B. Divide both sides by 5.

C. Add −5 to both sides.

D. Add 5 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q3/5

375

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Add −5 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q3/5

True or false?

Adding −4 to both sides of x + 4 = 10 gives x = 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q4/5

376

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q4/5

True or false?

Adding 8 to both sides of $x - 8 = 3$ gives $x = -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q5/5

377

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $x = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b • THE NUMBER SYSTEM

Add rational numbers on a number line

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Find and position integers on a number line	**6.NS.C.6.c**
BACKWARD 2	Add decimals	**5.NBT.B.7**
FORWARD 1	Locate irrational numbers on a number line	**8.NS.A.2**
FORWARD 2	Decide whether a sum is rational or irrational	**HSN-RN.B.3**

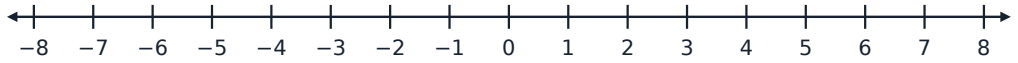
SET 13 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

Use the number line to add.

$-3 + 7$



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q1/5

380

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q1/5

Add.

$-6 + (-4)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q2/5

381

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

−10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q2/5

Start at 2 on a number line and move 5 units to the left.

Which sum does this show?

A. −5 + (−2)

B. −2 + 5

C. 2 + 5

D. 2 + (−5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q3/5

382

QUESTION ID

S13-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 + (−5)

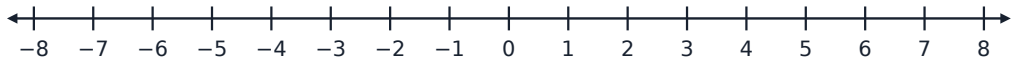
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q3/5

Show −2 + 6 on the number line. Start at −2 and draw an arrow 6 units long.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q4/5

383

QUESTION ID

S13-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: An arrow that starts at −2 and ends at 4 (6 units to the right)

Written: 4

GRADING NOTE Grade both: the arrow from −2 to 4, and the sum 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q4/5

True or false?

$-4.5 + 2 = -2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q5/5

384

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

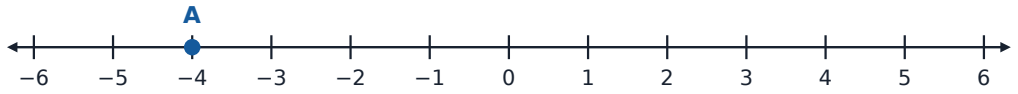
Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 1

Find and position integers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number is at point A?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q1/5

386

QUESTION ID

S13-B1-Q1

TYPE

Short answer

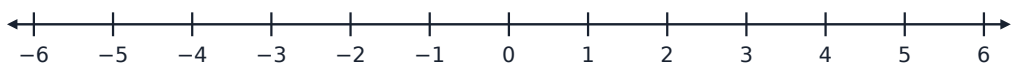
CORRECT ANSWER

−4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q1/5

Plot −3 and 5 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q2/5

387

QUESTION ID

S13-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points at −3 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q2/5

Which number is farthest to the left on a number line?

A. −2

B. 0

C. 3

D. −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q3/5

388

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q3/5

True or false?

−1 is to the right of −5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q4/5

389

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q4/5

True or false?

−6 is between −3 and 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q5/5

390

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 2

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.75 + 2.5

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q1/5

392

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q1/5

Add.

0.8 + 4.65

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q2/5

393

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q2/5

Add.

12.4 + 3.06

A. 46.4

B. 15.10

C. 12.71

D. 15.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q3/5

394

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$2.7 + 1.3 = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q4/5

395

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q4/5

True or false?

5.6 + 0.04 = 6

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q5/5

396

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 1

Locate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is √20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q1/5

398

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

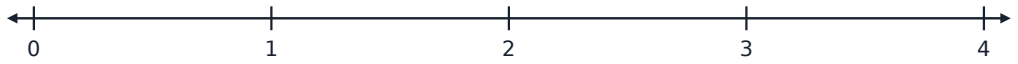
4 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q1/5

Plot an approximate location of √2 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q2/5

399

QUESTION ID

S13-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A point between 1 and 2, a little less than halfway (about 1.4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q2/5

Which is the best approximation of √50 to the nearest tenth?

A. 25

B. 7.1

C. 7.5

D. 7.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q3/5

400

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q3/5

True or false?

√10 is between 3 and 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q4/5

401

QUESTION ID

S13-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q4/5

True or false?

π is less than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q5/5

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 2

Decide whether a sum is rational or irrational • Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is 3 + √2 rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q1/5

404

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q1/5

Is $\frac{1}{2} + \frac{3}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q2/5

405

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; 5/4

GRADING NOTE 1 1/4 or 1.25 is also correct for the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q2/5

Which sum is irrational?

A. $\sqrt{9} + 4$

B. $0.5 + \frac{1}{2}$

C. $-2 + 2$

D. $\sqrt{3} + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q3/5

406

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{3} + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q4/5

407

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q4/5

True or false?

The sum of a rational number and an irrational number can be rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q5/5

408

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b • THE NUMBER SYSTEM

Use additive inverses: a number plus its opposite is 0

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Opposites are the same distance from 0	**6.NS.C.6.a**
BACKWARD 2	Interpret absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Tell whether a linear equation has one, no, or infinitely many solut…	**8.EE.C.7.a**
FORWARD 2	Solve linear equations in one variable	**HSA-REI.B.3**

SET 14 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

What number added to −9.5 gives 0?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q1/5

411

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

9.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q1/5

Fill in the blank.

$\frac{3}{4}$ + ___ = 0

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q2/5

412

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

−3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q2/5

Which sum is equal to 0?

A. $-2.6 + (-2.6)$

B. $-2.6 + 2.6$

C. $2.6 + 0$

D. $-2.6 + 6.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q3/5

413

QUESTION ID

S14-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2.6 + 2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q3/5

True or false?

For any rational number p, −p + p = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q4/5

414

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q4/5

True or false?

The additive inverse of −7 is −7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q5/5

415

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The additive inverse of −7 is 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 1

Opposites are the same distance from 0 • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q1/5

417

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q1/5

Which pair of numbers are opposites?

A. −11 and 11

B. −11 and −11

C. 11 and $\frac{1}{11}$

D. 1 and −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q2/5

418

QUESTION ID

S14-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. −11 and 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q2/5

How far is −9 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q3/5

419

QUESTION ID

S14-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q3/5

True or false?

$-\frac{2}{3}$ and $\frac{2}{3}$ are the same distance from 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q4/5

420

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q4/5

True or false?

The opposite of 0 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q5/5

421

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 2

Interpret absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-18|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q1/5

423

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q1/5

A diver is at −30 feet.

How far is she from the surface (0 feet)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q2/5

424

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q2/5

Which number has an absolute value of 6?

A. $\frac{1}{6}$

B. -6

C. $-\frac{1}{6}$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q3/5

425

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q3/5

True or false?

$|-3.5| = |3.5|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q4/5

426

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q4/5

True or false?

$|-20| = -20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q5/5

427

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 1

Tell whether a linear equation has one, no, or infinitely many solutions • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$3x + 5 = 3x - 2$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q1/5

429

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$2(x + 4) = 2x + 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q2/5

430

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q2/5

Which equation has exactly one solution?

A. $4x + 1 = 2x + 7$

B. $2(2x) = 4x$

C. $4x + 1 = 4x + 7$

D. $4x + 1 = 4x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q3/5

431

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $4x + 1 = 2x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q3/5

True or false?

$x + 6 = x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q4/5

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q4/5

True or false?

$5x - 5x = 0$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q5/5

433

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is true for every x, so it has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 2

Solve linear equations in one variable • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x.

$4x - 7 = -2x + 11$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q1/5

435

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q1/5

Solve for x.

$-3(x - 2) = 15$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q2/5

436

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q2/5

Solve for x.

$\frac{x}{5} + 3 = -1$

A. x = −20

B. x = −4

C. x = 10

D. x = 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q3/5

437

QUESTION ID

S14-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = −20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $6 + 2x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q4/5

438

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

$x = 4$ is the solution of $7 - x = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q5/5

439

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b • THE NUMBER SYSTEM

Interpret sums of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Use signed numbers in a real-world situation	**6.NS.C.5**
BACKWARD 2	Compare signed numbers in a real-world situation	**6.NS.C.7.b**
FORWARD 1	Model a changing quantity with a linear function	**8.F.B.4**
FORWARD 2	Evaluate a function and interpret the value in context	**HSF-IF.A.2**

SET 15 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

The temperature was −5°F at dawn. It rose 12°F by noon.

What was the temperature at noon?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q1/5

442

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

7°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q1/5

A submarine is at −240 feet. It rises 85 feet.

What is its new position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q2/5

443

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

−155 feet

GRADING NOTE "155 feet below sea level" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q2/5

Jo's account balance is −$18. She deposits $25.

Which sum gives her new balance?

A. −18 + (−25) = −43

B. 18 + 25 = 43

C. −18 + 25 = 7

D. 18 + (−25) = −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q3/5

444

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −18 + 25 = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q3/5

True or false?

A football team loses 6 yards, then gains 4 yards. The total change is −2 yards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q4/5

445

QUESTION ID

S15-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q4/5

True or false?

A hiker 350 feet below sea level climbs 400 feet. She ends 50 feet below sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q5/5

446

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE She ends 50 feet above sea level.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 1

Use signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.5

A stock's price changed by −$3.25.

Was this a gain or a loss, and of how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q1/5

448

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A loss of $3.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q1/5

Which number describes 15 meters above sea level?

A. $\frac{1}{15}$

B. 15

C. 0

D. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q2/5

449

QUESTION ID

S15-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q2/5

Write a number for a temperature of 8 degrees below zero.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q3/5

450

QUESTION ID

S15-B1-Q3

TYPE

Short answer

CORRECT ANSWER

−8°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q3/5

True or false?

In a bank account, a balance of −$50 can mean $50 is owed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q4/5

451

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q4/5

True or false?

A temperature of −2° is warmer than 0°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q5/5

452

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 2

Compare signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.7.b

Which temperature is colder, −8°C or −3°C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q1/5

454

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8°C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q1/5

Which statement is true?

A. $-7 < -12$

B. $-12 > -7$

C. $-12 < -7$

D. $-7 = -12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q2/5

455

QUESTION ID

S15-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $-12 < -7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q2/5

Write an inequality to compare elevations of −15 feet and −9 feet.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q3/5

456

QUESTION ID

S15-B2-Q3

TYPE

Short answer

CORRECT ANSWER

−15 < −9

GRADING NOTE −9 > −15 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q3/5

True or false?

A balance of −$20 is less than a balance of −$5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q4/5

457

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q4/5

True or false?

−4°F is colder than −10°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q5/5

458

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 1

Model a changing quantity with a linear function • Grade 8 • Functions • 8.F.B.4

A diver starts at −10 feet and descends 4 feet per minute.

Write a function for her position d after t minutes.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q1/5

460

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−4t − 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q1/5

The temperature is −6°F and rises 3°F per hour.

Write a function for the temperature T after h hours.

T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q2/5

461

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3h − 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q2/5

An account balance starts at −$40 and goes up $15 per week.

What is the balance after 4 weeks?

A. $20

B. −$100

C. −$20

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q3/5

462

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q3/5

True or false?

For T = 3h − 6, the temperature reaches 0°F after 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q4/5

463

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q4/5

True or false?

For d = −4t − 10, the starting position is −4 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q5/5

464

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The starting position is −10 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 2

Evaluate a function and interpret the value in context • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2

$h(t) = -16t^2 + 64t$ gives the height, in feet, of a ball after t seconds.

Find h(1).

h(1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q1/5

466

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

48 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q1/5

T(h) = −2 + 1.5h gives the temperature in °C after h hours.

Find T(6). What does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q2/5

467

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7; the temperature is 7°C after 6 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q2/5

$f(x) = 3x - 8$

What is $f(-2)$?

A. 14

B. -14

C. -2

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q3/5

468

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q3/5

True or false?

If $g(x) = -x + 5$, then $g(-3) = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q4/5

469

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2 - 4$, then $f(-3) = -13$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q5/5

470

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(-3) = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c • THE NUMBER SYSTEM

Subtract rational numbers by adding the additive inverse

MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Subtract decimals	**5.NBT.B.7**
BACKWARD 2	Subtract fractions with unlike denominators	**5.NF.A.1**
FORWARD 1	Find a rate of change from two points with negative coordinates	**8.F.B.4**
FORWARD 2	Subtract polynomials	**HSA-APR.A.1**

SET 16 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite as an addition expression. Then find the value.

$4 - 9$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q1/5

473

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$4 + (-9) = -5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q1/5

Subtract.

$-7 - (-12)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q2/5

474

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q2/5

Which expression is equal to $-3 - 8$?

A. $3 + (-8)$

B. $3 + 8$

C. $-3 + 8$

D. $-3 + (-8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q3/5

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3 + (−8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q3/5

True or false?

$-2.5 - 4 = -6.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q4/5

476

QUESTION ID

S16-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q4/5

True or false?

$\frac{1}{2} - (-\frac{1}{4}) = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q5/5

477

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 1

Subtract decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Subtract.

8.2 − 3.75

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q1/5

479

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q1/5

Subtract.

10 − 6.38

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q2/5

480

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q2/5

Subtract.

$5.4 - 2.9$

A. 8.3

B. 2.5

C. 3.5

D. 2.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q3/5

481

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$7.06 - 2.5 = 4.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q4/5

482

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$9.3 - 0.7 = 9.6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q5/5

483

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 2

Subtract fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$\frac{5}{6} - \frac{1}{4}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q1/5

485

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q1/5

Subtract.

$3\frac{1}{2} - 1\frac{2}{3}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q2/5

486

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 5/6

GRADING NOTE 11/6 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q2/5

Subtract.

$\frac{3}{4} - \frac{2}{5}$

A. $\frac{7}{20}$

B. $\frac{1}{20}$

C. $\frac{1}{2}$

D. $\frac{5}{9}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q3/5

487

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q4/5

488

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q4/5

True or false?

$\frac{4}{5} - \frac{1}{3} = \frac{3}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q5/5

489

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 1

Find a rate of change from two points with negative coordinates • Grade 8 • Functions • 8.F.B.4

Find the rate of change of the line through (−2, 5) and (3, −5).

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q1/5

491

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through $(-4, -1)$ and $(2, 8)$.

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q2/5

492

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/2

GRADING NOTE 1.5 or 1 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q2/5

What is the slope of the line through (1, −3) and (−1, 5)?

A. 1

B. 4

C. −1

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q3/5

493

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q3/5

True or false?

The slope of the line through (−3, −2) and (0, 4) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q4/5

494

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The slope of the line through (2, −1) and (5, −7) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q5/5

495

GRADING NOTE The slope is −2.

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 2

Subtract polynomials • Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Subtract.

$(5x^2 - 3x + 2) - (2x^2 - 7x - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q1/5

497

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^2 + 4x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q1/5

Subtract.

$(4x - 9) - (-x + 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q2/5

498

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5x − 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q2/5

Subtract.

$(x^2 + 2x) - (3x^2 - 5x)$

A. $-2x^2 + 7x$

B. $-2x^2 - 3x$

C. $-2x^2 + 3x$

D. $4x^2 + 7x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q3/5

499

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-2x^2 + 7x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q3/5

True or false?

$(6x + 1) - (2x - 5) = 4x + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q4/5

500

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q4/5

True or false?

$(3x^2 - x) - (x^2 + x) = 2x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q5/5

501

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is $2x^2 - 2x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c • THE NUMBER SYSTEM

Find the distance between two rational numbers as the absolute value of their difference

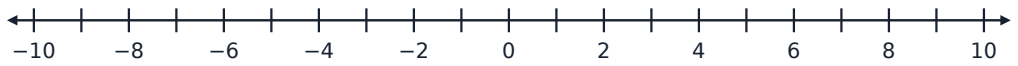
MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
BACKWARD 2	Position fractions and decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Use the Pythagorean theorem to find the distance between two po…	**8.G.B.8**
FORWARD 2	Use coordinates to find the perimeter of a polygon	**HSG-GPE.B.7**

SET 17 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Find the distance between −6 and 9 on the number line.



Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q1/5

504

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q1/5

Find the distance between −4.5 and −1.2 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q2/5

505

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

3.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q2/5

Which expression gives the distance between −3 and 8 on a number line?

A. −3 − 8

B. |−3 − 8|

C. −3 + 8

D. |−3 + 8|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q3/5

506

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. |−3 − 8|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q3/5

True or false?

$|2 - (-5)| = 7$, so 2 and -5 are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q4/5

507

QUESTION ID

S17-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q4/5

True or false?

The distance between −10 and −2 on a number line is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q5/5

508

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The distance is 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 1

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

Find the distance between (−3, 4) and (5, 4).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q1/5

510

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q1/5

Find the distance between (2, −6) and (2, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q2/5

511

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q2/5

Which two points are 5 units apart?

A. (1, −2) and (1, 3)

B. (1, −2) and (6, 3)

C. (−2, 1) and (2, 1)

D. (1, −2) and (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q3/5

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, −2) and (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q3/5

True or false?

(−4, −1) and (−4, 6) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q4/5

513

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q4/5

True or false?

(−5, 2) and (3, 2) are 2 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q5/5

514

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 8 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

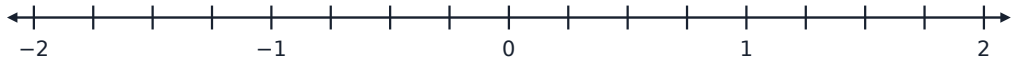
Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 2

Position fractions and decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot -1.5 and $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q1/5

516

QUESTION ID

S17-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at −1.5 and 3/4 (0.75)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a number between −2 and −1.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q2/5

517

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1.5

GRADING NOTE Any number greater than −2 and less than −1 is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q2/5

Which number is between −0.5 and 0?

A. −0.75

B. −1

C. 0.25

D. −0.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q3/5

518

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −0.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q3/5

True or false?

$-\frac{1}{2}$ is to the left of 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q4/5

519

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q4/5

True or false?

−2.3 is to the right of −2 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q5/5

520

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 1

Use the Pythagorean theorem to find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q1/5

522

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q1/5

Find the distance between (−2, −3) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q2/5

523

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q2/5

What is the distance between (0, 0) and (5, 12)?

A. 13 units

B. 7 units

C. 60 units

D. 17 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q3/5

524

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q3/5

True or false?

The distance between (−1, 1) and (2, 5) is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q4/5

525

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q5/5

526

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The distance is √18, about 4.24 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 2

Use coordinates to find the perimeter of a polygon • Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A triangle has vertices (0, 0), (6, 0), and (6, 8).

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q1/5

528

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q1/5

A rectangle has vertices (−3, −2), (5, −2), (5, 4), and (−3, 4).

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q2/5

529

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

28 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q2/5

A triangle has vertices (−2, 1), (2, 1), and (2, 4).

What is its perimeter?

A. 10 units

B. 6 units

C. 12 units

D. 7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q3/5

530

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q3/5

True or false?

A square with vertices (1, 1), (4, 1), (4, 4), and (1, 4) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q4/5

531

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q4/5

True or false?

A triangle with vertices (0, 0), (8, 0), and (0, 6) has a perimeter of 14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q5/5

532

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter is 24 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

Grade 9 (High School) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d • THE NUMBER SYSTEM

Use properties of operations to add and subtract rational numbers

MAIN	Grade 7 main question	**7.NS.A.1.d**
BACKWARD 1	Apply properties to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Add multi-digit decimals	**6.NS.B.3**
FORWARD 1	Solve equations by collecting like terms	**8.EE.C.7.b**
FORWARD 2	Use the structure of an expression to rewrite it	**HSA-SSE.A.2**

SET 18 • 7.NS.A.1.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.d

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum by grouping numbers to make the work easier. Show your work.

$-8 + 13.5 + 8$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q1/5

535

QUESTION ID

S18-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

13.5

GRADING NOTE REQUIRED WORK: Any grouping that uses the commutative and associative properties correctly. Example: $(-8 + 8) + 13.5 = 0 + 13.5$. Other valid groupings are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q1/5

Find the sum.

$\frac{3}{4} + (-2) + \frac{1}{4}$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q2/5

536

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q2/5

Which property is shown?

$-5 + 2.7 = 2.7 + (-5)$

A. Associative property of addition

B. Additive inverse property

C. Commutative property of addition

D. Additive identity property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q3/5

537

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Commutative property of addition

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q3/5

True or false?

$-6 + (4 + 6) = (-6 + 6) + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q4/5

538

QUESTION ID

S18-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q4/5

True or false?

$7 - 3 = 3 - 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q5/5

539

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtraction is not commutative: $4 \neq -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 1

Apply properties to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Use the distributive property to rewrite 3(x + 5).

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q1/5

541

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q1/5

Combine like terms.

4y + 2 + 6y

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q2/5

542

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10y + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q2/5

Which expression is equivalent to 2(4 + n)?

A. 4 + 2n

B. 6 + n

C. 8 + 2n

D. 8 + n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q3/5

543

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8 + 2n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q3/5

True or false?

$5 + (a + 7) = (5 + 7) + a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q4/5

544

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q4/5

True or false?

$6(x + 2) = 6x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q5/5

545

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 2

Add multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

14.62 + 7.9

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q1/5

547

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

22.52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q1/5

Add.

0.75 + 3.6 + 1.25

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q2/5

548

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q2/5

Add.

9.08 + 2.5

A. 34.08

B. 11.58

C. 9.33

D. 11.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q3/5

549

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 11.58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$2.25 + 3.75 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q4/5

550

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

$4.5 + 0.55 = 5.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q5/5

551

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 1

Solve equations by collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$-2.5x + 4 + 1.5x = -3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q1/5

553

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{3}x - 5 + \frac{2}{3}x = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q2/5

554

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q2/5

Solve for x.

$3x - 7 + x = 2x + 5$

A. $x = 12$

B. $x = 2$

C. $x = -1$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q3/5

555

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q3/5

True or false?

$x = -4$ is the solution of $5x + 3 - 2x = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q4/5

556

QUESTION ID

S18-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q4/5

True or false?

$x = 2$ is the solution of $-x + 6 + 3x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q5/5

557

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 2

Use the structure of an expression to rewrite it • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Rewrite $x^2 - 9$ as a product of two factors.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q1/5

559

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 3)(x - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q1/5

Factor out the greatest common factor.

$4x^2 + 12x$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q2/5

560

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4x(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q2/5

Which expression is equivalent to $x^2 - 25$?

A. $(x - 5)(x + 5)$

B. $(x + 5)^2$

C. $(x - 5)^2$

D. $(x - 25)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q3/5

561

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x - 5)(x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x^2 + 6x + 9 = (x + 3)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q4/5

562

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q4/5

True or false?

$x^2 - 16 = (x - 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q5/5

563

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 16 = (x - 4)(x + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a • THE NUMBER SYSTEM

Multiply rational numbers using the rules for signs

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Find powers of negative numbers	**8.EE.A.1**
FORWARD 2	Multiply polynomials	**HSA-APR.A.1**

SET 19 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

-6×7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q1/5

566

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

−42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q1/5

Multiply.

$-2.5 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q2/5

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q2/5

Multiply.

$-\frac{2}{3} \times \frac{9}{10}$

A. $-\frac{3}{5}$

B. $\frac{3}{5}$

C. $-\frac{27}{20}$

D. $-\frac{11}{13}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q3/5

568

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q3/5

True or false?

$(-1)(-1) = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q4/5

569

QUESTION ID

S19-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q4/5

True or false?

The product of three negative numbers is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q5/5

570

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{3}{5} \times \frac{10}{9}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q1/5

572

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q1/5

Multiply.

$\frac{4}{7} \times 14$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q2/5

573

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q2/5

Multiply.

$\frac{5}{6} \times \frac{3}{4}$

A. $\frac{5}{8}$

B. $\frac{3}{8}$

C. $\frac{15}{10}$

D. $\frac{8}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q3/5

574

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q4/5

575

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q4/5

True or false?

$\frac{3}{4} \times 8 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q5/5

576

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.4 × 3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q1/5

578

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q1/5

Multiply.

0.6×0.7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q2/5

579

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q2/5

Multiply.

1.25 × 8

A. 100

B. 1

C. 9.25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q3/5

580

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q3/5

True or false?

$0.3 \times 0.3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q4/5

581

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q5/5

582

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 1

Find powers of negative numbers • Grade 8 • Expressions & Equations • 8.EE.A.1

Find the value.

$(-2)^5$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q1/5

584

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q1/5

Find the value.

$(-3)^2 \times (-3)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q2/5

585

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q2/5

Which expression has a positive value?

A. -4^4

B. $(-4)^5$

C. $(-4)^3$

D. $(-4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q3/5

586

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $(-4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q3/5

True or false?

$(-5)^2 = 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q4/5

587

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q4/5

True or false?

$-2^4 = 16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q5/5

588

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2^4 = -(2^4) = -16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 2

Multiply polynomials • Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$-3x(2x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q1/5

590

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6x^2 + 15x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q1/5

Multiply.

$(x - 4)(x + 6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q2/5

591

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x - 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q2/5

Multiply.

$(-2x)(-7x)$

A. $14x$

B. $14x^2$

C. $-14x^2$

D. $-9x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q3/5

592

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $14x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q3/5

True or false?

$(x - 3)(x - 3) = x^2 - 6x + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q4/5

593

QUESTION ID

S19-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q4/5

True or false?

$-4(x - 2) = -4x - 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q5/5

594

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-4(x - 2) = -4x + 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

Grade 9 (High School) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a • THE NUMBER SYSTEM

Interpret products of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions and mixed numbers in word problems	**5.NF.B.6**
BACKWARD 2	Explain the meaning of signed numbers and 0	**6.NS.C.5**
FORWARD 1	Solve real-world equations with negative rates	**8.EE.C.7.b**
FORWARD 2	Create a one-variable equation for a situation with a negative rate	**HSA-CED.A.1**

SET 20 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

A stock's price drops $1.75 each day for 6 days.

What is the total change in price?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q1/5

597

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

−$10.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q1/5

A diver descends 2.5 feet each second.

What is the change in her position after 8 seconds?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q2/5

598

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

−20 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q2/5

A tank loses 3 gallons each minute.

Which product gives the change in the amount of water after 12 minutes?

A. $12 \times 3 = 36$

B. $12 \times (-3) = -36$

C. $12 + (-3) = 9$

D. $-12 \times (-3) = 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q3/5

599

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $12 \times (-3) = -36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q3/5

True or false?

The temperature falls 4°F each hour for 5 hours. The total change is −20°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q4/5

600

QUESTION ID

S20-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q4/5

True or false?

An account balance changes by −$15 each week. The change over 3 weeks is $45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q5/5

601

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The change is −$45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions and mixed numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats for each batch.

How many cups of oats are needed for 6 batches?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q1/5

603

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 1/2

GRADING NOTE 9/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q1/5

A path is $2\frac{1}{2}$ miles long. Sam walks $\frac{2}{5}$ of it.

How far does he walk?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q2/5

604

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q2/5

A board is $\frac{7}{8}$ yard long. Ty cuts a piece that is $\frac{1}{2}$ as long as the board.

How long is the piece?

A. $1\frac{3}{8}$ yards

B. $\frac{7}{4}$ yards

C. $\frac{8}{14}$ yard

D. $\frac{7}{16}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q3/5

605

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7/16 yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$\frac{2}{3}$ of a 12-foot rope is 8 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q4/5

606

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q4/5

True or false?

4 bags each hold $\frac{2}{3}$ pound of nuts. Together they hold 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q5/5

607

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They hold 8/3 = 2 2/3 pounds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 2

Explain the meaning of signed numbers and 0 • Grade 6 • The Number System • 6.NS.C.5

In a game, a score of 0 means no points won or lost.

What does a score of −12 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q1/5

609

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12 points were lost (12 points below 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q1/5

Sea level is 0 meters.

Which elevation is 30 meters below sea level?

A. −30 m

B. 30 m

C. 0 m

D. −3 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q2/5

610

QUESTION ID

S20-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. −30 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q2/5

What does a change of −$8 in a bank account mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q3/5

611

QUESTION ID

S20-B2-Q3

TYPE

Short answer

CORRECT ANSWER

$8 was taken out of the account.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q3/5

True or false?

A temperature change of −5° means it got colder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q4/5

612

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q4/5

True or false?

On a horizontal number line, positive numbers are to the left of 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q5/5

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 1

Solve real-world equations with negative rates • Grade 8 • Expressions & Equations • 8.EE.C.7.b

A balloon at 300 m descends 12.5 m each minute. Solve $300 - 12.5t = 175$ to find when it is at 175 m.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q1/5

615

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q1/5

A diver at −8 feet descends 1.5 feet each second. Solve $-8 - 1.5t = -26$ to find when she is at −26 feet.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q2/5

616

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q2/5

Solve for x.

$-4x + 10 = -14$

A. $x = -6$

B. $x = -1$

C. $x = 6$

D. $x = 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q3/5

617

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q3/5

True or false?

t = 5 is the solution of 20 − 3t = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q4/5

618

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q4/5

True or false?

$x = -3$ is the solution of $-2x - 7 = 13$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q5/5

619

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 2

Create a one-variable equation for a situation with a negative rate • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1

A phone battery is at 90% and drops 6% each hour.

Write and solve an equation to find when it reaches 30%.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q1/5

621

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

90 − 6h = 30; h = 10 hours

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q1/5

A submarine at −50 m dives 20 m each minute.

Write and solve an equation to find when it reaches −170 m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q2/5

622

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−50 − 20t = −170;
t = 6 minutes

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q2/5

A store's profit was −$400 and goes up $125 each month.

Which equation finds the month m when the profit is $600?

A. −400 + 125m = 600

B. 125 − 400m = 600

C. −400 − 125m = 600

D. 400 + 125m = 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q3/5

623

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

−400 + 125m = 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q3/5

True or false?

A 72°F room cools 4°F each hour. The equation $72 - 4t = 40$ finds when the room is 40°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q4/5

624

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q4/5

True or false?

The solution of $72 - 4t = 40$ is $t = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q5/5

625

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b • THE NUMBER SYSTEM

Divide integers and explain the sign of a quotient

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide multi-digit whole numbers	**6.NS.B.2**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
FORWARD 2	Decide whether a product is rational or irrational	**HSN-RN.B.3**

SET 21 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

Divide.

$-48 \div 6$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q1/5

628

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q1/5

Divide.

$-63 \div (-9)$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q2/5

629

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q2/5

Which fraction is equal to $-\frac{3}{4}$?

A. $\frac{-3}{4}$

B. $\frac{4}{-3}$

C. $\frac{-3}{-4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q3/5

630

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (−3)/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q3/5

True or false?

$$\frac{-10}{2} = \frac{10}{-2} = -5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q4/5

631

QUESTION ID

S21-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q4/5

True or false?

$-7 \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q5/5

632

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q5/5

SET 21 • 7.NS.A.2.b

BACKWARD 1

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

Divide.

1,872 ÷ 24

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q1/5

634

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q1/5

Divide.

3,915 ÷ 45

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q2/5

635

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

87

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q2/5

Divide.

2,436 ÷ 12

A. 203

B. 2,030

C. 23

D. 230

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q3/5

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 203

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q3/5

True or false?

5,040 ÷ 35 = 144

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q4/5

637

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q4/5

True or false?

1,950 ÷ 15 = 13

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q5/5

638

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 7.NS.A.2.b

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{4}{5} \div \frac{2}{15}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q1/5

640

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q1/5

Divide.

$3 \div \frac{3}{8}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q2/5

641

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q2/5

Divide.

$\frac{7}{10} \div \frac{1}{5}$

A. $\frac{2}{7}$

B. $\frac{7}{50}$

C. $1\frac{2}{5}$

D. $3\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q3/5

642

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q3/5

True or false?

$\frac{3}{8} \div \frac{3}{4} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q4/5

643

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$\frac{5}{6} \div \frac{5}{12} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q5/5

644

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is $\frac{-5}{8}$ rational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q1/5

646

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. It is a quotient of the integers −5 and 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q1/5

Which number is irrational?

A. $\sqrt{7}$

B. 0.25

C. $\frac{-22}{7}$

D. $-\sqrt{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q2/5

647

QUESTION ID

S21-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. √7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q2/5

Write -2.4 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q3/5

QUESTION ID

S21-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$-12/5$

GRADING NOTE Any equivalent quotient, such as $-24/10$, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q4/5

649

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q4/5

True or false?

√25 is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q5/5

650

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 7.NS.A.2.b

FORWARD 2

Decide whether a product is rational or irrational • Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is $-3 \times \sqrt{5}$ rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q1/5

652

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q1/5

Is $\frac{-2}{3} \times \frac{9}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q2/5

653

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; −3/2

GRADING NOTE −1 1/2 or −1.5 is also correct for the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q2/5

Which product is rational?

A. $-1 \times \pi$

B. $5 \times \sqrt{2}$

C. $2 \times \sqrt{3}$

D. $\sqrt{2} \times \sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q3/5

654

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{2} \times \sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q3/5

True or false?

The product of a nonzero rational number and an irrational number is irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q4/5

655

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q4/5

True or false?

The quotient of two rational numbers (with a nonzero divisor) can be irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q5/5

656

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b • THE NUMBER SYSTEM

Interpret quotients of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide fractions in word problems	**6.NS.A.1**
BACKWARD 2	Divide unit fractions and whole numbers in word problems	**5.NF.B.7.c**
FORWARD 1	Find a rate of change from a description	**8.F.B.4**
FORWARD 2	Calculate and interpret a negative average rate of change	**HSF-IF.B.6**

SET 22 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

A stock lost $18 in value over 6 days at a steady rate.

What was the change in value per day?

Change per day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q1/5

659

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

−$3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q1/5

A diver's position changed by −45 feet in 9 seconds at a steady rate.

What was the change in position per second?

Change per second: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q2/5

660

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q2/5

The temperature changed by −14°F over 4 hours at a steady rate.

What was the change per hour?

A. −10°F

B. 3.5°F

C. −3.5°F

D. −56°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q3/5

661

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −3.5°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q3/5

True or false?

A balance changes by −$120 over 8 weeks at a steady rate. The change is −$15 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q4/5

662

QUESTION ID

S22-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q4/5

True or false?

A hiker descends 600 feet in 4 hours at a steady rate. Her change in elevation is 150 feet per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q5/5

663

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The change is −150 feet per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 22 • MAIN • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 1

Divide fractions in word problems • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{4}$-cup servings are in 6 cups of rice?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q1/5

665

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q1/5

A $\frac{2}{3}$-mile trail is split into 4 equal parts.

How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q2/5

666

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/6 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q2/5

A strip $\frac{5}{6}$ meter long is cut into pieces $\frac{1}{12}$ meter long.

How many pieces are there?

A. $\frac{5}{72}$

B. 12

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q3/5

667

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q3/5

True or false?

There are 6 half-pound portions in 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q4/5

668

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q4/5

True or false?

3 people share $\frac{3}{4}$ pound of nuts equally. Each gets $\frac{1}{2}$ pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q5/5

669

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each gets 1/4 pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

BACKWARD 2

Divide unit fractions and whole numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$-pound burgers can be made from 5 pounds of meat?

Burgers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q1/5

671

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q1/5

3 friends share $\frac{1}{2}$ pound of grapes equally.

How much does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q2/5

672

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/6 pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q2/5

2 people share $\frac{1}{3}$ of a pizza equally.

How much of the whole pizza does each person get?

A. $\frac{2}{3}$ of the pizza

B. $\frac{1}{6}$ of the pizza

C. 6 pizzas

D. $\frac{3}{2}$ of the pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q3/5

673

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6 of the pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q3/5

True or false?

$4 \div \frac{1}{5} = 20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q4/5

674

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{8} \div 2 = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q5/5

675

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 1

Find a rate of change from a description • Grade 8 • Functions • 8.F.B.4

A tank held 240 gallons at 2:00 and 180 gallons at 2:15. It drained at a constant rate.

What is the rate of change in gallons per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q1/5

677

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−4 gallons per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q1/5

A plane descends from 30,000 feet to 21,000 feet in 15 minutes at a constant rate.

What is the rate of change in feet per minute?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q2/5

678

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−600 feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q2/5

A line passes through (0, 50) and (10, 20).

What is its rate of change?

A. $-\frac{1}{3}$

B. 3

C. -30

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q3/5

679

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q3/5

True or false?

A balance that goes from $80 to $20 in 12 weeks at a constant rate changes by −$5 per week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q4/5

680

QUESTION ID

S22-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q4/5

True or false?

A temperature that goes from 10° to −6° in 4 hours at a constant rate changes by −1° per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q5/5

681

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It changes by −4° per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 7.NS.A.2.b

FORWARD 2

Calculate and interpret a negative average rate of change • Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6

The table shows temperatures.

What is the average rate of change from 0 hours to 6 hours?

Time (h)	0	3	6
Temperature (°F)	12	3	−6

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q1/5

683

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−3°F per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q1/5

$f(x) = -x^2 + 10$

What is the average rate of change from $x = 1$ to $x = 3$?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q2/5

684

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q2/5

A company's profit was $5,000 in year 2 and $2,000 in year 5.

What is the average rate of change?

A. −$600 per year

B. $1,000 per year

C. −$3,000 per year

D. −$1,000 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q3/5

685

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −$1,000 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q3/5

True or false?

A negative average rate of change means the quantity decreased over the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q4/5

686

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q4/5

True or false?

For f(x) = 5 − 2x, the average rate of change from x = 0 to x = 4 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q5/5

687

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

Grade 9 (High School) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c • THE NUMBER SYSTEM

Use properties of operations to multiply and divide rational numbers

MAIN	Grade 7 main question	**7.NS.A.2.c**
BACKWARD 1	Apply the distributive property	**6.EE.A.3**
BACKWARD 2	Use the commutative and associative properties of multiplication	**3.OA.B.5**
FORWARD 1	Solve equations by multiplying by a reciprocal	**8.EE.C.7.b**
FORWARD 2	Rewrite expressions with rational exponents	**HSN-RN.A.2**

SET 23 • 7.NS.A.2.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.c

Use properties to multiply. Show how you grouped or ordered the factors.

$-4 \times 7.3 \times 25$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q1/5

690

QUESTION ID

S23-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

−730

GRADING NOTE REQUIRED WORK: Any correct use of the commutative and associative properties. Example: $(-4 \times 25) \times 7.3 = -100 \times 7.3$. Other valid groupings are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q1/5

Multiply.

$\frac{3}{5} \times (-12) \times \frac{5}{3}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q2/5

691

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

−12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q2/5

Which property is shown?

$-2 \times (5 \times 3.1) = (-2 \times 5) \times 3.1$

A. Multiplicative identity property

B. Associative property of multiplication

C. Distributive property

D. Commutative property of multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q3/5

692

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Associative property of multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q3/5

True or false?

$-6 \times (10 + \frac{1}{2}) = -60 + (-3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q4/5

693

QUESTION ID

S23-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q4/5

True or false?

$-8 \times \frac{1}{8} = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q5/5

694

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 23 • MAIN • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 1

Apply the distributive property • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite using the distributive property.

7(2y + 3)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q1/5

696

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14y + 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q1/5

Use the distributive property to find 8 × 47 as 8 × (40 + 7).

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q2/5

697

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

376

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q2/5

Which expression is equivalent to 5(x + 9)?

A. 5x + 14

B. x + 45

C. 5x + 45

D. 5x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q3/5

698

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5x + 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$4(3 + 2.5) = 12 + 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q4/5

699

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$3(a + 4) = 3a + 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q5/5

700

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

BACKWARD 2

Use the commutative and associative properties of multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

Fill in the blank.

$6 \times 4 = 4 \times ___$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

Find 2 × 7 × 5 by first multiplying 2 × 5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

703

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

Which expression is equal to (3 × 2) × 4?

A. 3 × 2 + 4

B. 3 + (2 × 4)

C. 3 × (2 × 4)

D. (3 + 2) × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

704

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3 × (2 × 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

True or false?

$8 \times 3 = 3 \times 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

705

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

True or false?

$(2 \times 3) \times 5 = 2 \times 3 + 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

706

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 1

Solve equations by multiplying by a reciprocal • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$-\frac{3}{4}x = 18$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q1/5

708

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q1/5

Solve for x.

$-0.6x = -4.2$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q2/5

709

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q2/5

Solve for x.

$\frac{2}{5}(x - 10) = -6$

A. x = −5

B. x = 25

C. x = −25

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q3/5

710

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q3/5

True or false?

$x = -10$ is the solution of $-\frac{1}{2}x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q4/5

711

QUESTION ID

S23-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $-1.5x = 4.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q5/5

712

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 23 • FORWARD 1 • Q5/5

SET 23 • 7.NS.A.2.c

FORWARD 2

Rewrite expressions with rational exponents • Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2

Find the value.

27^(1/3)

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q1/5

714

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q1/5

Write √x using a rational exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q2/5

715

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x^(1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q2/5

Which number is equal to 16^(1/2)?

A. 4

B. 32

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q3/5

716

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q3/5

True or false?

$(x^3)^{(1/3)} = x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q4/5

717

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q4/5

True or false?

25^(1/2) = 12.5

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q5/5

718

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 25^(1/2) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

Grade 9 (High School) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 23 • FORWARD 2 • Q5/5

SET 24 • 7.NS.A.2.d • THE NUMBER SYSTEM

Use long division to write a rational number as a terminating or repeating decimal

MAIN	Grade 7 main question	**7.NS.A.2.d**
BACKWARD 1	Write fractions with denominators of 10 or 100 as decimals	**4.NF.C.6**
BACKWARD 2	Divide multi-digit whole numbers	**6.NS.B.2**
FORWARD 1	Write a repeating decimal as a fraction	**8.NS.A.1**
FORWARD 2	Report a decimal result to a sensible level of accuracy	**HSN-Q.A.3**

SET 24 • 7.NS.A.2.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q1/5

721

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

0.375

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q1/5

Use long division to write $\frac{5}{11}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q2/5

722

QUESTION ID

S24-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

0.4545... (the digits 45 repeat)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

A bar over 45 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q2/5

Which decimal is equal to $-\frac{7}{12}$?

A. −0.58333... (the 3 repeats)

B. −0.712

C. −1.714...

D. −0.5833 (the decimal ends)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q3/5

723

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −0.58333... (the 3 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q3/5

True or false?

$\frac{2}{9} = 0.222...$, a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q4/5

724

QUESTION ID

S24-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q4/5

True or false?

$\frac{7}{20}$ is a repeating decimal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q5/5

725

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7/20 = 0.35, a terminating decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 24 • MAIN • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 1

Write fractions with denominators of 10 or 100 as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{7}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q1/5

727

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q1/5

Write $\frac{43}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q2/5

728

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{6}{100}$?

A. 0.06

B. 0.006

C. 0.6

D. 6.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q3/5

729

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{10} = 0.9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q4/5

730

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q4/5

True or false?

$\frac{25}{100} = 2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q5/5

731

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.C.6 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

BACKWARD 2

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

7,425 ÷ 25

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q1/5

733

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

297

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q1/5

Divide.

4,368 ÷ 52

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q2/5

734

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q2/5

Divide.

9,072 ÷ 36

A. 2,520

B. 252

C. 25.2

D. 242

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q3/5

735

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 252

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

6,912 ÷ 48 = 144

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q4/5

736

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

2,925 ÷ 75 = 49

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q5/5

737

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 1

Write a repeating decimal as a fraction • Grade 8 • The Number System • 8.NS.A.1

Write 0.777... (the 7 repeats) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q1/5

739

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q1/5

Write 0.1818... (the digits 18 repeat) as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q2/5

740

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/11

GRADING NOTE 18/99 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q2/5

Which fraction is equal to 0.333... (the 3 repeats)?

A. $\frac{1}{3}$

B. $\frac{3}{100}$

C. $\frac{3}{10}$

D. $\frac{33}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q3/5

741

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q3/5

True or false?

0.999... (the 9 repeats) is equal to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q4/5

742

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q4/5

True or false?

0.4545... (the digits 45 repeat) is equal to $\frac{45}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q5/5

743

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is equal to 45/99 = 5/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 7.NS.A.2.d

FORWARD 2

Report a decimal result to a sensible level of accuracy • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3

A 10-foot board is cut into 3 equal pieces. The tape measure reads to the nearest hundredth of a foot.

Report the length of each piece to that accuracy.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q1/5

745

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.33 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q1/5

$\frac{2}{7}$ = 0.285714... Round it to the nearest thousandth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q2/5

746

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.286

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q2/5

A scale reads to the nearest 0.1 kg. 5 kg of flour is split into 3 equal bags.

Which report of each bag's mass fits the scale?

A. 1.666666 kg

B. 1.7 kg

C. 2 kg

D. 1.67 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q3/5

747

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.7 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q3/5

True or false?

Reporting $\frac{1}{3}$ hour as 0.33 hour is accurate to the nearest hundredth of an hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q4/5

748

QUESTION ID

S24-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q4/5

True or false?

A length reported as 4.0 cm is less precise than a length reported as 4 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q5/5

749

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 24 • FORWARD 2 • Q5/5

SET 25 • 7.NS.A.3 • THE NUMBER SYSTEM

Solve real-world problems with the four operations on rational numbers

MAIN	Grade 7 main question	**7.NS.A.3**
BACKWARD 1	Multiply and divide decimals in word problems	**6.NS.B.3**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve real-world problems with scientific notation	**8.EE.A.4**
FORWARD 2	Use units to solve multistep problems	**HSN-Q.A.1**

SET 25 • 7.NS.A.3

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.3

A freezer is at 4°C. It cools 1.5°C every hour.

What is the temperature after 6 hours?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q1/5

752

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

−5°C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q1/5

Maya owes $42.60. She pays it off in 6 equal payments.

How much is each payment?

Payment: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q2/5

753

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

$7.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q2/5

A diver is at −18 m. She rises $\frac{1}{3}$ of the distance to the surface.

What is her new position?

A. −6 m

B. 12 m

C. −12 m

D. −24 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q3/5

754

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q3/5

True or false?

A hiker is at −86 m. She climbs 120 m. She ends 34 m above sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q4/5

755

QUESTION ID

S25-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q4/5

True or false?

Three stock prices change by −$2.50, +$4, and −$3.25. The total change is +$1.75.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q5/5

756

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total change is −$1.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 25 • MAIN • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 1

Multiply and divide decimals in word problems • Grade 6 • The Number System • 6.NS.B.3

Gas costs $3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q1/5

758

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$46.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q1/5

A 7.5-meter rope is cut into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q2/5

759

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q2/5

A 4.8-liter jug fills cups that hold 0.3 liter each.

How many cups can it fill?

A. 14.4

B. 16

C. 160

D. 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q3/5

760

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q3/5

True or false?

A 2.5-pound bag of nuts costs $3.20 per pound. The bag costs $8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q4/5

761

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q4/5

True or false?

$9.60 shared equally by 4 people is $2.60 each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q5/5

762

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each gets $2.40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 7.NS.A.3

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 4.5 = 12

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q1/5

764

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q1/5

Solve for y.

6y = 39

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q2/5

765

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q2/5

Solve for n.

$\frac{2}{3}n = 10$

A. $n = 30$

B. $n = \frac{20}{3}$

C. $n = 15$

D. $n = 9\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q3/5

766

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q3/5

True or false?

m = 8 is the solution of m − 2.75 = 5.25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q4/5

767

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q4/5

True or false?

k = 3 is the solution of 4k = 1.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q5/5

768

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is k = 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 1

Solve real-world problems with scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A hard drive holds 2×10^{12} bytes. A file is 4×10^{8} bytes.

How many files of this size fit on the drive?

Files: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q1/5

770

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5,000

GRADING NOTE 5×10^{3} is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q1/5

Earth is about 1.5×10^8 km from the Sun. Mars is about 2.3×10^8 km from the Sun.

About how much farther from the Sun is Mars? Write the answer in scientific notation.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q2/5

771

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q2/5

A cell is 2×10^{-5} m wide.

How wide are 3,000 of these cells placed side by side?

A. 6×10^{-8} m

B. 5×10^{-2} m

C. 6×10^{-2} m

D. 6×10^{2} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q3/5

772

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6×10^{-2} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q3/5

True or false?

$(6 \times 10^4) \times 5 = 3 \times 10^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q4/5

773

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q4/5

True or false?

$(9 \times 10^6) \div (3 \times 10^2) = 3 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q5/5

774

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 3×10^4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 7.NS.A.3

FORWARD 2

Use units to solve multistep problems • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1

A car uses 0.04 gallon of gas per mile. Gas costs $3.50 per gallon.

How much does the gas cost for a 250-mile trip?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q1/5

776

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q1/5

A faucet leaks 0.2 liter per hour.

How many liters leak in 1 week?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q2/5

777

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

33.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q2/5

A worker earns $18.50 per hour. She works 7.5 hours per day for 4 days.

How much does she earn?

A. $555

B. $74

C. $540

D. $138.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q3/5

778

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $555

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q3/5

True or false?

A speed of 12 meters per second is 43.2 kilometers per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q4/5

779

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q4/5

True or false?

Walking 3 miles per hour for 45 minutes covers 2.5 miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q5/5

780

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It covers 2.25 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Add and subtract linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Combine like terms	**6.EE.A.3**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Solve equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 2	Interpret the terms, factors, and coefficients of an expression	**HSA-SSE.A.1.a**

SET 26 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$(3x + 5) + (2x - 8)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q1/5

783

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

5x − 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q1/5

Simplify.

$(4.5y - 2) - (1.5y + 6)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q2/5

784

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

3y − 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q2/5

Simplify.

$\frac{1}{2}a + 3 - \frac{3}{4}a + 1$

A. $-\frac{1}{4}a + 4$

B. $-\frac{1}{4}a + 2$

C. $\frac{1}{4}a + 4$

D. $\frac{5}{4}a + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q3/5

785

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −(1/4)a + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q3/5

True or false?

$(6m - 1) - (2m - 4) = 4m + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q4/5

786

QUESTION ID

S26-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q4/5

True or false?

$-3k + 7 + 5k - 2 = 8k + 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q5/5

787

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The expression simplifies to 2k + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 26 • MAIN • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 1

Combine like terms • Grade 6 • Expressions & Equations • 6.EE.A.3

Combine like terms.

5n + 3n + 4

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q1/5

789

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q1/5

Combine like terms.

2y + 7 + y

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q2/5

790

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3y + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q2/5

Which expression is equivalent to x + x + x + 6?

A. 9x

B. 3x + 2

C. 3x + 6

D. $x^3 + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q3/5

791

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q3/5

True or false?

4b + 2b = 6b

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q4/5

792

QUESTION ID

S26-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q4/5

True or false?

$3a + 2 = 5a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q5/5

793

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 7.EE.A.1

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Are 2(x + 3) and 2x + 6 equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q1/5

795

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Using the distributive property, 2(x + 3) = 2x + 6 for every value of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q1/5

Which expression is equivalent to y + y + y?

A. 3y

B. y + 3

C. y^3

D. 3 + y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q2/5

796

QUESTION ID

S26-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q2/5

Write an expression with no parentheses that is equivalent to 4(n + 1).

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q3/5

797

QUESTION ID

S26-B2-Q3

TYPE

Short answer

CORRECT ANSWER

4n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q3/5

True or false?

5m − 2m and 3m are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q4/5

798

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q4/5

True or false?

a + 4 and 4a are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q5/5

799

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 1

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$2(3x - 1) - (x + 4) = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q1/5

801

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{2}(4x + 6) = 3x - 5$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q2/5

802

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q2/5

Solve for x.

$0.5(x + 8) - 2x = 1$

A. $x = 6$

B. $x = 3$

C. $x = 2$

D. $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q3/5

803

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q3/5

True or false?

$x = 5$ is the solution of $3(x - 2) = 2x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q4/5

804

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q4/5

True or false?

$x = 1$ is the solution of $4x - (x + 2) = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q5/5

805

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 1 • Q5/5

SET 26 • 7.EE.A.1

FORWARD 2

Interpret the terms, factors, and coefficients of an expression • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

What is the coefficient of x in $5x^2 - 3x + 8$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q1/5

807

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q1/5

How many terms are in $4a^3 - 2a + 7$?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q2/5

808

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q2/5

Which is a factor of $7(x + 2)^2$?

A. x^2

B. $(x + 2)^2$

C. 14

D. 7x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q3/5

809

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x + 2)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q3/5

True or false?

In −6y + 9, the constant term is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q4/5

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q4/5

True or false?

In $3x^2 + x$, the coefficient of x is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q5/5

811

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The coefficient of x is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 7.EE.A.1 • EXPRESSIONS & EQUATIONS

Factor and expand linear expressions with rational coefficients

MAIN	Grade 7 main question	**7.EE.A.1**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Identify the parts of an expression	**6.EE.A.2.b**
FORWARD 1	Expand to decide how many solutions an equation has	**8.EE.C.7.a**
FORWARD 2	Factor quadratic expressions	**HSA-SSE.A.2**

SET 27 • 7.EE.A.1

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$-4(2x - 3)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q1/5

814

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-8x + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q1/5

Factor out the greatest common factor.

6y + 15

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q2/5

815

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

3(2y + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q2/5

Which expression is equivalent to $\frac{1}{2}(8a - 6)$?

A. $8a - 3$

B. $4a - 3$

C. $4a - 6$

D. $16a - 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q3/5

816

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4a − 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q3/5

True or false?

$12m - 18 = 6(2m - 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q4/5

817

QUESTION ID

S27-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q4/5

True or false?

$-5(n + 2) = -5n + 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q5/5

818

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-5(n + 2) = -5n - 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 27 • MAIN • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q1/5

820

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q1/5

Use the greatest common factor to write 30 + 42 as a product.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q2/5

821

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6(5 + 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q2/5

Which expression is equal to 18 + 27?

A. 9(2 + 3)

B. 9(2 + 27)

C. 9(18 + 3)

D. 3(6 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q3/5

822

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9(2 + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q3/5

True or false?

The greatest common factor of 16 and 40 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q4/5

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q4/5

True or false?

$20 + 35 = 5(4 + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q5/5

824

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $20 + 35 = 5(4 + 7)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 7.EE.A.1

BACKWARD 2

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms are in 3x + 7y − 2?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q1/5

826

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q1/5

What is the coefficient of w in 9 + 5w?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q2/5

827

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q2/5

Which part of 4(n + 6) is a sum?

A. 4n

B. 4

C. n + 6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q3/5

828

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

In 2(a + b), the factor 2 multiplies the sum a + b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q4/5

829

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

In 8k − 3, the term 8k is a constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q5/5

830

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 7.EE.A.1

FORWARD 1

Expand to decide how many solutions an equation has • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x - 2) = 4x - 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q1/5

832

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$-3(x + 1) = -3x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q2/5

833

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q2/5

Which equation has no solution?

A. 2(x + 5) = x + 10

B. 2(x + 5) = 3x

C. 2(x + 5) = 2x + 10

D. 2(x + 5) = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q3/5

834

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2(x + 5) = 2x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q3/5

True or false?

$6(x - 1) = 2(3x - 3)$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q4/5

835

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q4/5

True or false?

5(x + 1) = 5x + 1 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q5/5

836

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 27 • FORWARD 1 • Q5/5

SET 27 • 7.EE.A.1

FORWARD 2

Factor quadratic expressions • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 7x + 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q1/5

838

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q1/5

Factor completely.

$3x^2 - 12$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q2/5

839

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3(x + 2)(x − 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q2/5

Factor.

$x^2 - 2x - 15$

A. $(x - 5)(x - 3)$

B. $(x - 15)(x + 1)$

C. $(x + 5)(x - 3)$

D. $(x - 5)(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q3/5

840

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x − 5)(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

$2x^2 + 10x = 2x(x + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q4/5

841

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q5/5

842

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 5x + 6 = (x + 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 7.EE.A.2 • EXPRESSIONS & EQUATIONS

Rewrite an expression to show how the quantities in a problem are related

MAIN	Grade 7 main question	**7.EE.A.2**
BACKWARD 1	Write expressions from words	**6.EE.A.2.a**
BACKWARD 2	Write and interpret numerical expressions	**5.OA.A.2**
FORWARD 1	Write equivalent expressions with the properties of integer expon…	**8.EE.A.1**
FORWARD 2	Choose an equivalent form of an expression to reveal a property	**HSA-SSE.B.3**

SET 28 • 7.EE.A.2

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.A.2

A shirt costs p dollars. The price is raised by 30%.

Write the new price as a single term.

New price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q1/5

845

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

1.3p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle's length is 4 cm more than its width w.

Write its perimeter in two equivalent ways.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q2/5

846

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

2w + 2(w + 4) and 4w + 8

GRADING NOTE Any two correct equivalent expressions earn credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q2/5

A price p is lowered by 15%.

Which expression shows the new price?

A. 1.15p

B. 0.85p

C. 0.15p

D. p − 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q3/5

847

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.85p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q3/5

True or false?

n + 0.08n = 1.08n shows that increasing n by 8% is the same as multiplying n by 1.08.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q4/5

848

QUESTION ID

S28-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q4/5

True or false?

The price after a 20% discount on a price of d dollars is d − 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q5/5

849

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is 0.8d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 28 • MAIN • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 1

Write expressions from words • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "5 less than y."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q1/5

851

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q1/5

Write an expression for "the product of 8 and n."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q2/5

852

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q2/5

Which expression means "the sum of x and 3, times 4"?

A. 4(x + 3)

B. x + 3 × 4

C. 4x + 3

D. 4 + x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q3/5

853

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q3/5

True or false?

"Twice a number m" can be written as 2m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q4/5

854

QUESTION ID

S28-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q4/5

True or false?

"7 fewer than k" can be written as 7 − k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q5/5

855

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is k − 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 7.EE.A.2

BACKWARD 2

Write and interpret numerical expressions • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 6 and 9, then multiply by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q1/5

857

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 × (6 + 9)

GRADING NOTE (6 + 9) × 2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q1/5

How does 5 × (24 + 13) compare with 24 + 13?

A. It is 5 more.

B. It is 5 less.

C. It is 5 times as large.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q2/5

858

QUESTION ID

S28-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. It is 5 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q2/5

Write an expression for "subtract 4 from 20, then divide by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q3/5

859

QUESTION ID

S28-B2-Q3

TYPE

Short answer

CORRECT ANSWER

(20 − 4) ÷ 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q3/5

True or false?

(7 + 8) × 3 is three times as large as 7 + 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q4/5

860

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q4/5

True or false?

$10 \times (5 - 2)$ is 10 more than $5 - 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q5/5

861

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 1

Write equivalent expressions with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$4^{5} \times 4^{-2}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

863

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4^{3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q1/5

Write as a single power.

$(2^3)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2^{12}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q2/5

Which number is equal to $5^6 \div 5^8$?

A. 25

B. 5^{14}

C. $\frac{1}{25}$

D. -25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q3/5

True or false?

$7^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

866

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q4/5

True or false?

$3^2 \times 3^3 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

867

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 \times 3^3 = 3^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 28 • FORWARD 1 • Q5/5

SET 28 • 7.EE.A.2

FORWARD 2

Choose an equivalent form of an expression to reveal a property • Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3

Factor $f(x) = x^2 - 6x + 8$ to find the values of x where $f(x) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q1/5

869

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q1/5

A price p is raised 10%, and then the new price is raised 10% again: p(1.1)(1.1).

Write this as a single factor times p, and give the total percent increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q2/5

870

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.21p; a 21% increase

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q2/5

A value after t years is 50(1.03)^t.

What is the yearly percent increase?

A. 50%

B. 3%

C. 0.3%

D. 103%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q3/5

871

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q3/5

True or false?

Writing $x^2 - 9$ as $(x - 3)(x + 3)$ shows that the expression equals 0 when $x = 3$ or $x = -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q4/5

872

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q4/5

True or false?

A 10% increase followed by a 10% decrease leaves a price unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q5/5

873

QUESTION ID

S28-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The final price is 0.99 times the original price.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

Grade 9 (High School) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Solve multistep problems with positive and negative rational numbers in any form

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Interpret a fraction as division	**5.NF.B.3**
BACKWARD 2	Find a percent of a quantity	**6.RP.A.3.c**
FORWARD 1	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 2	Represent constraints with equations or inequalities	**HSA-CED.A.3**

SET 29 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Jon earns $16.50 per hour. He gets a raise of $\frac{1}{5}$ of his wage. Then he works 30 hours.

How much does he earn for the 30 hours?

Earnings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q1/5

876

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

$594

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A board is $8\frac{1}{4}$ feet long. Ana cuts off 2.5 feet. She cuts the rest into 3 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q2/5

877

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

1 11/12 feet

GRADING NOTE 23/12 feet, or about 1.92 feet, is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q2/5

A 60-gallon tank is $\frac{3}{4}$ full. Then 12.5 gallons are used.

How many gallons are left?

A. 57.5

B. 47.5

C. 32.5

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q3/5

878

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q3/5

True or false?

A $40 sweater is 25% off. Mia pays with a $50 bill. Her change is $20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q4/5

879

QUESTION ID

S29-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q4/5

True or false?

Leo runs $2\frac{1}{2}$ miles on Monday and 3.4 miles on Tuesday. He runs 6.4 miles in all.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q5/5

880

QUESTION ID
S29-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE He runs 5.9 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 29 • MAIN • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 1

Interpret a fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $\frac{3}{4}$ as a division expression. Then find its decimal value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q1/5

882

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3 ÷ 4 = 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q1/5

5 friends share 2 pizzas equally.

How much pizza does each friend get?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q2/5

883

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2/5 of a pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q2/5

Which number is equal to 7 ÷ 8?

A. 7.8

B. $\frac{7}{8}$

C. $\frac{8}{7}$

D. 0.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q3/5

884

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q3/5

True or false?

$\frac{9}{2} = 9 \div 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q4/5

885

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q4/5

True or false?

$3 \div 5 = \frac{5}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q5/5

886

QUESTION ID

S29-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 7.EE.B.3

BACKWARD 2

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 35% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q1/5

888

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q1/5

What is 12.5% of 64?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q2/5

889

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 150% of 20?

A. 130

B. 30

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q3/5

890

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q3/5

True or false?

60% of 25 is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q4/5

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q4/5

True or false?

5% of 300 is 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q5/5

892

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5% of 300 is 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 1

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

Adult tickets cost $9 and child tickets cost $5. Someone buys 20 tickets for $140.

How many adult tickets were bought?

Adult tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q1/5

894

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q1/5

Plan A costs $20 plus $0.10 per minute. Plan B costs $0.35 per minute.

For how many minutes do the plans cost the same?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q2/5

895

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q2/5

Two numbers have a sum of 30 and a difference of 8.

What is the larger number?

A. 15

B. 11

C. 19

D. 22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q3/5

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q3/5

True or false?

$x = 3$ and $y = 5$ is a solution of both $x + y = 8$ and $2x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q4/5

897

QUESTION ID

S29-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q4/5

True or false?

$x = 4$ and $y = 2$ is a solution of both $x + y = 6$ and $x - y = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q5/5

898

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 29 • FORWARD 1 • Q5/5

SET 29 • 7.EE.B.3

FORWARD 2

Represent constraints with equations or inequalities • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Write an inequality for m muffins and l loaves.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q1/5

900

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$0.75m + 2.5l \leq 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q1/5

A baker has 24 cups of flour. Each muffin uses $\frac{3}{4}$ cup and each loaf uses 2.5 cups.

Can she make 12 muffins and 7 loaves? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q2/5

901

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. 12 muffins and 7 loaves need 9 + 17.5 = 26.5 cups, which is more than 24 cups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q2/5

A trip budget is at most $150. Gas costs $3.50 per gallon (g) and meals cost $12 each (f).

Which inequality shows the budget?

A. $3.5g + 12f \leq 150$

B. $12g + 3.5f \leq 150$

C. $3.5 + 12 \leq 150gf$

D. $3.5g + 12f \geq 150$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q3/5

902

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3.5g + 12f ≤ 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q3/5

True or false?

m = 8 and l = 4 makes 0.75m + 2.5l ≤ 24 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q4/5

903

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q4/5

True or false?

m = 10 and l = 9 makes 0.75m + 2.5l ≤ 24 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q5/5

904

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.75(10) + 2.5(9) = 30, which is more than 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 7.EE.B.3 • EXPRESSIONS & EQUATIONS

Use estimation to judge whether an answer is reasonable

MAIN	Grade 7 main question	**7.EE.B.3**
BACKWARD 1	Round decimals	**5.NBT.A.4**
BACKWARD 2	Use rounding to check whether an answer is reasonable	**4.OA.A.3**
FORWARD 1	Estimate very large or very small quantities with powers of 10	**8.EE.A.3**
FORWARD 2	Define appropriate quantities and units for a model	**HSN-Q.A.2**

SET 30 • 7.EE.B.3

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate 19.8 × 4.1 by rounding each number to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q1/5

907

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

80

GRADING NOTE 20 × 4 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q1/5

Kim says 48.7 + 31.2 = 799.

Use estimation to explain whether her answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q2/5

908

QUESTION ID

S30-M-Q2

TYPE

Short answer

CORRECT ANSWER

Not reasonable. 49 + 31 is about 80, so the sum should be about 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q2/5

A meal costs $38.75. Ana adds a 20% tip.

Which is the best estimate of the total?

A. About $40

B. About $58

C. About $78

D. About $47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q3/5

909

QUESTION ID

S30-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. About $47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q3/5

True or false?

$6\frac{7}{8} \times 3\frac{1}{10}$ is about 7×3, so a product of about 21 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q4/5

910

QUESTION ID

S30-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q4/5

True or false?

A $59.99 jacket is 30% off. A sale price of about $18 is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q5/5

911

QUESTION ID

S30-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sale price should be about $42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 30 • MAIN • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 1

Round decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 7.86 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q1/5

913

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q1/5

Round 12.348 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q2/5

914

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q2/5

Round 4.165 to the nearest hundredth.

A. 4.17

B. 4.1

C. 4.2

D. 4.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q3/5

915

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q3/5

True or false?

3.97 rounded to the nearest tenth is 4.0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q4/5

916

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q4/5

True or false?

15.52 rounded to the nearest whole number is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q5/5

917

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 7.EE.B.3

BACKWARD 2

Use rounding to check whether an answer is reasonable • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

Estimate 398 + 211 by rounding each number to the nearest hundred.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q1/5

919

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q1/5

Lia says 52 × 4 = 2,008.

Round 52 to the nearest ten to decide whether her answer is reasonable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q2/5

920

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Not reasonable.
50 × 4 = 200, so the answer should be about 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q2/5

Round each number to the nearest hundred.

What is the best estimate of 689 − 312?

A. 300

B. 400

C. 500

D. 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q3/5

921

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q3/5

True or false?

Rounding 49 to the nearest ten gives an estimate of 300 for 49 × 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q4/5

922

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q4/5

True or false?

Rounding each number to the nearest hundred gives an estimate of 1,200 for 812 − 397.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q5/5

923

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The estimate is 800 − 400 = 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 1

Estimate very large or very small quantities with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Estimate 38,900,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q1/5

925

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q1/5

Estimate 0.00062 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q2/5

926

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6×10^{-4}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q2/5

About how many times as large as 3×10^5 is 9×10^8?

A. 300

B. 3

C. 30

D. 3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q3/5

927

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q3/5

True or false?

3×10^8 is a reasonable estimate for 296,000,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q4/5

928

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} is about 6,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q5/5

929

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $6 \times 10^{-3} = 0.006$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 7.EE.B.3

FORWARD 2

Define appropriate quantities and units for a model • Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2

A heart beats about 70 times per minute.

About how many times does it beat in 1 day? Round to the nearest ten thousand.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q1/5

931

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

100,000

GRADING NOTE 70 × 1,440 = 100,800.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q1/5

Which unit best describes the distance between two cities: millimeters, meters, or kilometers?

Unit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q2/5

932

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q2/5

Which quantity is most useful for estimating the cost to paint the walls of a room?

A. The volume of the room

B. The height of the door

C. The number of windows

D. The area of the walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q3/5

933

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The area of the walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

Gallons of water used per day is a reasonable quantity for modeling a family's water use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q4/5

934

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q4/5

True or false?

The color of a car is a useful quantity for modeling how much fuel it uses on a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q5/5

935

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

Grade 9 (High School) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 30 • FORWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form px + q = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve one-step equations	**6.EE.B.7**
BACKWARD 2	Write an expression with a variable for a situation	**6.EE.B.6**
FORWARD 1	Solve equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 31 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges a $25 sign-up fee plus $18 per month. Ty has paid $151 in all.

Write and solve an equation to find the number of months m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q1/5

938

QUESTION ID
S31-M-Q1

TYPE
Short answer

CORRECT ANSWER
18m + 25 = 151; m = 7

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q1/5

Solve for x.

$4x + 9 = -11$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q2/5

939

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q2/5

Ava has $60. She buys movie tickets for $7.50 each and has $15 left.

Which equation finds the number of tickets t?

A. $60t - 7.5 = 15$

B. $7.5 - 60t = 15$

C. $7.5t + 60 = 15$

D. $60 - 7.5t = 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q3/5

940

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60 − 7.5t = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q3/5

True or false?

$x = 6$ is the solution of $\frac{1}{2}x + 3 = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q4/5

941

QUESTION ID

S31-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q4/5

True or false?

A plumber charges $45 plus $30 per hour. A bill of $165 means the plumber worked 5 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q5/5

942

QUESTION ID

S31-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The plumber worked 4 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 31 • MAIN • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 1

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 13 = 41

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q1/5

944

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q1/5

Solve for y.

9y = 72

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q2/5

945

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q2/5

Solve for n.

$n - 6.4 = 3.6$

A. $n = 10$

B. $n = 2.8$

C. $n = -2.8$

D. $n = 23.04$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q3/5

946

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. n = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q3/5

True or false?

k = 7 is the solution of 5k = 35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q4/5

947

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q4/5

True or false?

p = 9 is the solution of p + 4 = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q5/5

948

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

BACKWARD 2

Write an expression with a variable for a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

A ticket costs $12.

Write an expression for the cost of t tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q1/5

950

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q1/5

Ana has 5 more stickers than Ben. Ben has b stickers.

Write an expression for the number of stickers Ana has.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q2/5

951

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

b + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q2/5

A pizza has s slices. 4 friends share it equally.

Which expression gives the number of slices each friend gets?

A. $s - 4$

B. $4s$

C. $s + 4$

D. $\frac{s}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q3/5

952

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. s/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q3/5

True or false?

A pen costs p dollars. 3 pens cost 3p dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q4/5

953

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q4/5

True or false?

Mo is m years old. His age in 6 years is 6m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q5/5

954

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE His age in 6 years is m + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 1

Solve equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges $40 plus $5 per day. Store B charges $25 plus $8 per day.

After how many days do the stores charge the same amount?

Days: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q1/5

956

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q1/5

Solve for x.

$7x - 4 = 3x + 20$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q2/5

957

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q2/5

Solve for x.

$-2x + 9 = 4x - 15$

A. $x = -4$

B. $x = 1$

C. $x = 4$

D. $x = 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q3/5

958

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q3/5

True or false?

$x = -1$ is the solution of $5x + 3 = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q4/5

959

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q4/5

True or false?

$x = 2$ is the solution of $6x - 1 = 4x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q5/5

960

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 31 • FORWARD 1 • Q5/5

SET 31 • 7.EE.B.4.a

FORWARD 2

Rearrange a formula to solve for a chosen quantity • Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2l + 2w for l.

l = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q1/5

962

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(P − 2w)/2

GRADING NOTE l = P/2 − w is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q1/5

Solve y = mx + b for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q2/5

963

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(y − b)/m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q2/5

Solve $A = \frac{1}{2}bh$ for h.

A. $h = 2Ab$

B. $h = \frac{A}{2b}$

C. $h = \frac{b}{2A}$

D. $h = \frac{2A}{b}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q3/5

964

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. h = 2A/b

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q3/5

True or false?

Solving $d = rt$ for t gives $t = \frac{d}{r}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q4/5

965

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $C = 5n + 20$ for n gives $n = 5C - 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q5/5

966

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $n = (C - 20)/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

Grade 9 (High School) • Algebra: Creating Equations • HSA-CED.A.4 • Set 31 • FORWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Solve word problems that lead to equations of the form p(x + q) = r

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Evaluate expressions with parentheses	**5.OA.A.1**
BACKWARD 2	Use the distributive property to expand an expression	**6.EE.A.3**
FORWARD 1	Solve equations with the distributive property on both sides	**8.EE.C.7.b**
FORWARD 2	Solve quadratic equations by square roots or factoring	**HSA-REI.B.4.b**

SET 32 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle is 7 cm wide and (x + 3) cm long. Its area is 63 cm².

Write and solve an equation to find x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q1/5

969

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

7(x + 3) = 63; x = 6

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q1/5

Solve for x.

$-3(x - 4) = 27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q2/5

970

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

−5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q2/5

5 friends each buy a $4 snack and a drink that costs d dollars. They spend $32.50 in all.

Which equation finds d?

A. d + 5(4) = 32.5

B. 5(d + 4) = 32.5

C. 5d + 4 = 32.5

D. 4(d + 5) = 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q3/5

971

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5(d + 4) = 32.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q3/5

True or false?

$x = 2.5$ is the solution of $2(x + 1.5) = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q4/5

QUESTION ID

S32-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q4/5

True or false?

The solution of $\frac{1}{3}(x - 6) = 4$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q5/5

973

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 32 • MAIN • Q5/5

SET 32 • 7.EE.B.4.a

BACKWARD 1

Evaluate expressions with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$3 \times (8 + 4)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q1/5

975

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q1/5

Evaluate.

$(15 - 6) \times 2$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q2/5

976

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate.

$4 \times (9 - 2) + 1$

A. 25

B. 29

C. 36

D. 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q3/5

977

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

$2 \times (5 + 5) = 20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q4/5

978

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q4/5

True or false?

$(12 + 3) \times 2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q5/5

979

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

BACKWARD 2

Use the distributive property to expand an expression • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite using the distributive property.

9(m + 2)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q1/5

981

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9m + 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite using the distributive property.

4(3k + 5)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q2/5

982

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12k + 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q2/5

Which expression is equivalent to 6(x + 1.5)?

A. 6x + 1.5

B. 6x + 9

C. 6x + 7.5

D. 7x + 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q3/5

983

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q3/5

True or false?

$5(2 + y) = 10 + 5y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q4/5

984

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q4/5

True or false?

$8(p + 3) = 8p + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q5/5

985

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 1

Solve equations with the distributive property on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$3(x + 2) = 2(x + 7)$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q1/5

987

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q1/5

Solve for x.

$-2(x - 5) = 4(x + 1)$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q2/5

988

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$\frac{1}{2}(6x - 4) = 2(x + 3)$

A. x = 2

B. x = 8

C. x = −8

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q3/5

989

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q3/5

True or false?

$x = 4$ is the solution of $5(x - 1) = 3(x + 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q4/5

990

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q4/5

True or false?

$x = 3$ is the solution of $2(x + 6) = 4(x - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q5/5

991

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 32 • FORWARD 1 • Q5/5

SET 32 • 7.EE.B.4.a

FORWARD 2

Solve quadratic equations by square roots or factoring • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve for x.

$(x + 3)^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q1/5

993

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 or −10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$x^2 = 81$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q2/5

994

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9 or −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q2/5

Solve for x.

$(x - 2)(x + 5) = 0$

A. $x = -5$ only

B. $x = 2$ only

C. $x = -2$ or $x = 5$

D. $x = 2$ or $x = -5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q3/5

995

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 or x = −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q3/5

True or false?

The solutions of $2(x - 1)^2 = 18$ are $x = 4$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q4/5

996

QUESTION ID

S32-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 = 16$ has only one solution, $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q5/5

997

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 4$ or $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • FORWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a • EXPRESSIONS & EQUATIONS

Compare an algebraic solution with an arithmetic solution

MAIN	Grade 7 main question	**7.EE.B.4.a**
BACKWARD 1	Solve two-step word problems	**3.OA.D.8**
BACKWARD 2	Decide whether a value makes an equation true	**6.EE.B.5**
FORWARD 1	Solve systems of two linear equations algebraically and by inspec…	**8.EE.C.8.b**
FORWARD 2	Explain each step in solving an equation	**HSA-REI.A.1**

SET 33 • 7.EE.B.4.a

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Ron has $50. He buys 3 books that cost the same amount and has $14 left.

Arithmetic: (50 − 14) ÷ 3. Algebra: 3b + 14 = 50.

What is the price of one book?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q1/5

1000

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q1/5

An arithmetic solution to 4(x + 2) = 36 is 36 ÷ 4 − 2.

Describe the operations in order, and give x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q2/5

1001

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

Divide 36 by 4, then subtract 2; x = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which arithmetic steps solve 5x − 8 = 27?

A. Multiply 27 by 5, then add 8.

B. Add 8 to 27, then divide by 5.

C. Divide 27 by 5, then add 8.

D. Subtract 8 from 27, then divide by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q3/5

1002

QUESTION ID

S33-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Add 8 to 27, then divide by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q3/5

True or false?

Solving 2x + 6 = 20 with algebra and computing (20 − 6) ÷ 2 both give 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q4/5

1003

QUESTION ID

S33-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q4/5

True or false?

The arithmetic steps "subtract 4 from 15, then divide by 3" solve $3(x - 4) = 15$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q5/5

1004

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The correct steps are to divide 15 by 3, then add 4, so x = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 33 • MAIN • Q5/5

SET 33 • 7.EE.B.4.a

BACKWARD 1

Solve two-step word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

Jo has 45 stickers. She gives away 9. Then she shares the rest equally among 4 friends.

How many stickers does each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q1/5

1006

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

There are 6 boxes with 8 pens in each box. 5 pens are lost.

How many pens are left?

Pens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q2/5

1007

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q2/5

A book has 90 pages. Kim reads 20 pages on Monday and 35 pages on Tuesday.

How many pages are left?

A. 15

B. 55

C. 145

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q3/5

1008

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q3/5

True or false?

3 packs of 7 cards and 4 more cards make 25 cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q4/5

1009

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

40 apples are shared equally in 5 bags. Then 2 apples are taken out of each bag. Each bag now has 8 apples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q5/5

1010

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each bag has 6 apples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 33 • BACKWARD 1 • Q5/5

1010

SET 33 • 7.EE.B.4.a

BACKWARD 2

Decide whether a value makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 4 a solution of 3x + 2 = 14? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q1/5

1012

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 3(4) + 2 = 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q1/5

Which value of n makes n + 9 = 15 true?

A. 9

B. 24

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q2/5

1013

QUESTION ID

S33-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which of the values 2, 5, and 8 makes 7y = 35 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q3/5

1014

QUESTION ID

S33-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q3/5

True or false?

m = 3 makes 12 ÷ m = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q4/5

1015

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q4/5

True or false?

a = 5 makes 2a − 1 = 11 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q5/5

1016

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 1

Solve systems of two linear equations algebraically and by inspection • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve the system.

$y = 2x$

$x + y = 18$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q1/5

1018

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 6, y = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q1/5

Solve the system.

$y = x + 3$

$y = -2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q2/5

1019

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2, y = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q2/5

Look at the system. How many solutions does it have?

$3x + 2y = 5$

$3x + 2y = 6$

A. Infinitely many solutions

B. Two solutions

C. No solution

D. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q3/5

1020

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q3/5

True or false?

The system y = 4x and y = 4x + 1 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q4/5

1021

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 4$ and $y = 6$ is the solution of the system $x + y = 10$ and $x - y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q5/5

1022

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 6$, $y = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 33 • FORWARD 1 • Q5/5

SET 33 • 7.EE.B.4.a

FORWARD 2

Explain each step in solving an equation • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

In solving an equation, $5(x - 2) = 30$ is rewritten as $5x - 10 = 30$.

Which property is used?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q1/5

1024

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The distributive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q1/5

In solving an equation, $5x - 10 = 30$ is rewritten as $5x = 40$.

Explain this step.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q2/5

1025

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 is added to both sides (addition property of equality).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q2/5

Which step changes 5x = 40 into x = 8?

A. Add 5 to both sides.

B. Subtract 5 from both sides.

C. Divide both sides by 5.

D. Multiply both sides by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q3/5

1026

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Divide both sides by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q3/5

True or false?

Each step in solving a linear equation should give an equation with the same solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q4/5

1027

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q4/5

True or false?

Subtracting 3 from only one side of an equation keeps the equation true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q5/5

1028

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 33 • FORWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Solve word problems that lead to inequalities of the form px + q > r or px + q < r

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Write an inequality for a situation	**6.EE.B.8**
BACKWARD 2	Decide whether a value makes an inequality true	**6.EE.B.5**
FORWARD 1	Solve multistep equations with the variable on both sides — near…	**8.EE.C.7.b**
FORWARD 2	Solve linear inequalities with the variable on both sides	**HSA-REI.B.3**

SET 34 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Sam has $40. He buys a $12 shirt and some pairs of socks that cost $3.50 each.

Write and solve an inequality for the number of pairs of socks p he can buy.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q1/5

1031

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

3.5p + 12 ≤ 40;
p ≤ 8

GRADING NOTE He can buy at most 8 pairs. Any correct inequality earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q1/5

Solve.

$-2x + 5 > 13$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q2/5

1032

QUESTION ID

S34-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$3x - 7 \geq 11$

A. $x \geq \frac{4}{3}$

B. $x \leq \frac{4}{3}$

C. $x \geq 6$

D. $x \leq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q3/5

1033

QUESTION ID

S34-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x ≥ 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q3/5

True or false?

Dividing both sides of $-4x < 20$ by -4 gives $x > -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q4/5

1034

QUESTION ID

S34-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q4/5

True or false?

The solution of $5x + 2 < 17$ is $x > 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q5/5

1035

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x < 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • MAIN • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 1

Write an inequality for a situation • Grade 6 • Expressions & Equations • 6.EE.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Riders must be at least 48 inches tall.

Write an inequality for a rider's height h.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q1/5

1037

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

h ≥ 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an inequality for "a temperature t below −2°F."

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q2/5

1038

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

t < −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which inequality means "x is more than 7"?

A. $x < 7$

B. $x > 7$

C. $x \leq 7$

D. $x = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q3/5

1039

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x > 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q3/5

True or false?

$x < 3$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q4/5

1040

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q4/5

True or false?

5 is a solution of $x < 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q5/5

1041

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 7.EE.B.4.b

BACKWARD 2

Decide whether a value makes an inequality true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is 9 a solution of 2x > 16? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q1/5

1043

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(9) = 18, and 18 > 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q1/5

Which value makes y + 4 < 10 true?

A. 8

B. 10

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q2/5

1044

QUESTION ID

S34-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q2/5

Which of the values 3, 6, and 9 make $n - 2 \geq 4$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q3/5

1045

QUESTION ID

S34-B2-Q3

TYPE

Short answer

CORRECT ANSWER

6 and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q3/5

True or false?

x = 0 makes 3x < 1 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q4/5

1046

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q4/5

True or false?

k = 2 makes 5k > 10 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q5/5

1047

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 7.EE.B.4.b

FORWARD 1

Solve multistep equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Solve for x.

$0.5x + 7 = 2x - 2$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q1/5 • NEAREST RELATED

1049

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$\frac{2}{3}(x + 3) = x - 1$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q2/5 • NEAREST RELATED

1050

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q2/5 • NEAREST RELATED

Solve for x.

$6 - 3x = x + 22$

A. $x = -4$

B. $x = 7$

C. $x = -7$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q3/5 • NEAREST RELATED

1051

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$x = -2$ is the solution of $4(x + 5) = x + 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q4/5 • NEAREST RELATED

1052

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 1$ is the solution of $9x - 3 = 2x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q5/5 • NEAREST RELATED

1053

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 5/7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 34 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 34 • 7.EE.B.4.b

FORWARD 2

Solve linear inequalities with the variable on both sides • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve.

$5x - 3 > 2x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q1/5

1055

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x > 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q1/5

Solve.

$-4(x + 1) \leq 2x + 8$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q2/5

1056

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \geq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q2/5

Solve.

$7 - 2x < x - 8$

A. $x < -5$

B. $x > -5$

C. $x > 5$

D. $x < 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q3/5

1057

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x > 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q3/5

True or false?

$x = 0$ is a solution of $3(x - 2) < x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q4/5

1058

QUESTION ID

S34-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

The solution of $6x + 1 \geq 4x + 11$ is $x \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q5/5

1059

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x \geq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b • EXPRESSIONS & EQUATIONS

Graph the solution set of an inequality and interpret it in context

MAIN	Grade 7 main question	**7.EE.B.4.b**
BACKWARD 1	Graph an inequality on a number line	**6.EE.B.8**
BACKWARD 2	Read an inequality as positions on a number line	**6.NS.C.7.a**
FORWARD 1	Find the solution of a system of equations from its graph	**8.EE.C.8.a**
FORWARD 2	Graph the solutions of a linear inequality in two variables	**HSA-REI.D.12**

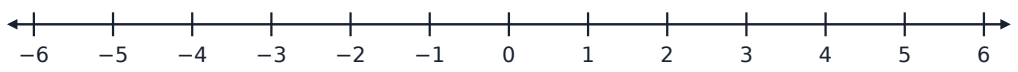
SET 35 • 7.EE.B.4.b

MAIN

Grade 7 main question • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Graph the solution set of $x > -3$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q1/5

1062

QUESTION ID

S35-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

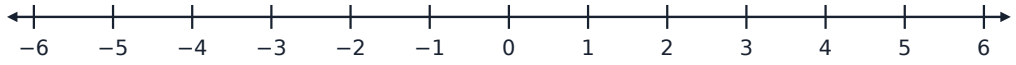
An open circle at −3 with an arrow or shading to the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 1 ≤ 9. Then graph the solution set on the number line.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q2/5

1063

QUESTION ID

S35-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A closed circle at 4 with an arrow or shading to the left

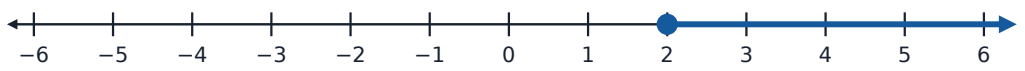
Written: x ≤ 4

GRADING NOTE Grade both: the solution x ≤ 4, and the graph with a closed circle at 4 shaded to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q2/5

Which inequality does the graph show?



A. $x > 2$

B. $x < 2$

C. $x \leq 2$

D. $x \geq 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q3/5

1064

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x \geq 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q3/5

A club has $30. It earns $5 for each ticket it sells. It needs more than $230.

Solve 5t + 30 > 230 and describe the solution in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q4/5

1065

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

t > 40. The club must sell more than 40 tickets (at least 41).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q4/5

True or false?

The graph shows the solution set of $-x < 2$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q5/5

1066

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

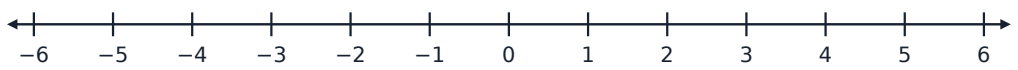
Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 35 • MAIN • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 1

Graph an inequality on a number line • Grade 6 • Expressions & Equations • 6.EE.B.8

Graph x < 4 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q1/5

1068

QUESTION ID

S35-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

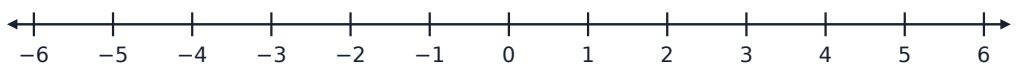
An open circle at 4 with an arrow or shading to the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q1/5

Graph x ≥ −1 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q2/5

1069

QUESTION ID

S35-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A closed circle at −1 with an arrow or shading to the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which inequality does the graph show?



A. $x < 0$

B. $x \geq 0$

C. $x \leq 0$

D. $x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q3/5

1070

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x < 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q3/5

True or false?

A closed circle on the graph of an inequality shows that the endpoint is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q4/5

1071

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The graph of $x > 5$ has a closed circle at 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q5/5

1072

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 7.EE.B.4.b

BACKWARD 2

Read an inequality as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

−4 is to the left of 1 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q1/5

1074

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−4 < 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q1/5

Which statement says that −2 is to the right of −6 on a number line?

A. −2 < −6

B. −2 = −6

C. −6 > −2

D. −2 > −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q2/5

1075

QUESTION ID

S35-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −2 > −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q2/5

What does $-8 < -3$ tell you about the positions of -8 and -3 on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q3/5

1076

QUESTION ID

S35-B2-Q3

TYPE

Short answer

CORRECT ANSWER

-8 is to the left of -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q3/5

True or false?

$-1 > -9$ because -1 is to the right of -9 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q4/5

1077

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q4/5

True or false?

$-5 > 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q5/5

1078

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 7.EE.B.4.b

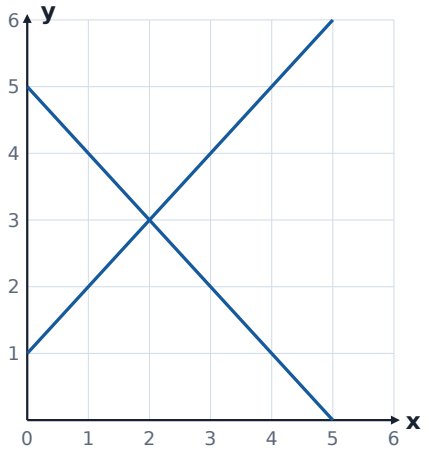
FORWARD 1

Find the solution of a system of equations from its graph • Grade 8 • Expressions & Equations • 8.EE.C.8.a

The graph shows y = x + 1 and y = −x + 5.

At what point do the lines intersect?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q1/5

1080

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q1/5

Two lines are graphed on the same grid.

What does their point of intersection represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q2/5

1081

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The pair (x, y) that makes both equations true; the solution of the system.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q2/5

The graphs of two linear equations are parallel lines.

How many solutions does the system have?

A. One solution

B. Two solutions

C. Infinitely many solutions

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q3/5

1082

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q3/5

True or false?

(1, 4) is the solution of the system y = 2x + 2 and y = 3x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q4/5

1083

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q4/5

True or false?

If two equations have the same graph, the system has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q5/5

1084

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 35 • FORWARD 1 • Q5/5

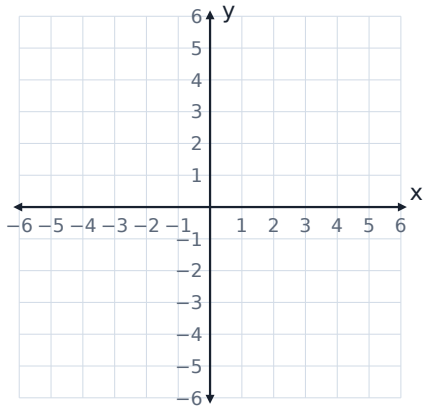
SET 35 • 7.EE.B.4.b

FORWARD 2

Graph the solutions of a linear inequality in two variables • Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Graph y > x + 1 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q1/5

1086

QUESTION ID

S35-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A dashed line through (0, 1) with slope 1, with the region above the line shaded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q1/5

Is (2, 5) a solution of y ≥ 2x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q2/5

1087

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 5 ≥ 2(2), because 5 ≥ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q2/5

Which point is in the solution set of $y < -x + 3$?

A. (0, 0)

B. (0, 5)

C. (3, 1)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q3/5

1088

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q3/5

True or false?

The boundary line of the graph of $y \leq 3x - 2$ is drawn solid.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q4/5

1089

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q4/5

True or false?

The solution set of $y > 2$ is the region below the line $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q5/5

1090

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

Grade 9 (High School) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 7.G.A.1 • GEOMETRY

Find actual lengths and areas from a scale drawing

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Convert measurement units	**6.RP.A.3.d**
BACKWARD 2	Find the area of a rectangle	**4.MD.A.3**
FORWARD 1	Describe similar figures with dilations	**8.G.A.4**
FORWARD 2	Decide whether two triangles are similar	**HSG-SRT.A.2**

SET 36 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A scale drawing of a garden uses the scale 1 cm : 2 m. The garden is 6 cm long on the drawing.

What is the actual length of the garden?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q1/5

1093

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q1/5

A floor plan uses the scale 1 inch : 4 feet. A room is 3 inches by 2.5 inches on the plan.

What is the actual area of the room?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q2/5

1094

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

120 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q2/5

A map uses the scale 1 cm : 25 km. Two towns are 4.4 cm apart on the map.

What is the actual distance between the towns?

A. 100 km

B. 110 km

C. 5.68 km

D. 29.4 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q3/5

1095

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 110 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q3/5

True or false?

A drawing uses the scale 1 cm : 3 m. A 5 cm by 2 cm rectangle on the drawing has an actual area of 90 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q4/5

1096

QUESTION ID

S36-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q4/5

True or false?

A drawing uses the scale 1 cm : 10 cm. The actual area is 10 times the area on the drawing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q5/5

1097

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The actual area is 100 times the area on the drawing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 36 • MAIN • Q5/5

SET 36 • 7.G.A.1

BACKWARD 1

Convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

How many inches are in 4.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q1/5

1099

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q1/5

How many meters are in 350 centimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q2/5

1100

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q2/5

How many yards are in 27 feet?

A. 24

B. 9

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q3/5

1101

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q3/5

True or false?

2.5 kilometers = 2,500 meters

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q4/5

1102

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

6 feet = 60 inches

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q5/5

1103

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6 feet = 72 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 7.G.A.1

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 9 m wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q1/5

1105

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

108 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q1/5

A rectangle has an area of 84 square feet. It is 12 feet long.

How wide is it?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q2/5

1106

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q2/5

A rectangle is 15 cm long and 6 cm wide.

What is its area?

A. 96 square centimeters

B. 90 square centimeters

C. 21 square centimeters

D. 42 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q3/5

1107

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q3/5

True or false?

A square with 7-inch sides has an area of 49 square inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q4/5

1108

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A rectangle 10 m long and 4 m wide has an area of 28 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q5/5

1109

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 40 square meters; 28 m is its perimeter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 7.G.A.1

FORWARD 1

Describe similar figures with dilations • Grade 8 • Geometry • 8.G.A.4

A triangle has sides 3, 4, and 5 units. It is dilated by a scale factor of 2.5.

What are the new side lengths?

Side lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q1/5

1111

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.5, 10, and 12.5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q1/5

A 4-by-6 rectangle is similar to a rectangle that is 10 by w. The 4 side matches the 10 side.

Find w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q2/5

1112

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q2/5

A figure is dilated by a scale factor of $\frac{1}{3}$. One side is 18 cm long.

How long is the matching side of the image?

A. $18\frac{1}{3}$ cm

B. 3 cm

C. 6 cm

D. 54 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q3/5

1113

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q3/5

True or false?

Dilations keep angle measures the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q4/5

1114

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q4/5

True or false?

A 2-by-3 rectangle and a 4-by-5 rectangle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q5/5

1115

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 7.G.A.1

FORWARD 2

Decide whether two triangles are similar • Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has angles of 40°, 60°, and 80°. Triangle DEF has angles of 40°, 60°, and 80°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q1/5

1117

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Their corresponding angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q1/5

One triangle has sides 6, 8, and 10. Another has sides 9, 12, and 15.

What is the scale factor from the first triangle to the second?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q2/5

1118

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 3/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q2/5

Which two triangles must be similar?

A. Two right triangles

B. Two triangles that each have a 5 cm side

C. Two isosceles triangles

D. Two triangles that each have a 50° angle and a 70° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q3/5

1119

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Two triangles that each have a 50° angle and a 70° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q3/5

True or false?

Two triangles are similar with a scale factor of 3. Each pair of corresponding sides has a ratio of 3 : 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q4/5

1120

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q4/5

True or false?

All rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q5/5

1121

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 36 • FORWARD 2 • Q5/5

SET 37 • 7.G.A.1 • GEOMETRY

Reproduce a scale drawing at a different scale

MAIN	Grade 7 main question	**7.G.A.1**
BACKWARD 1	Draw a polygon on the coordinate plane	**6.G.A.3**
BACKWARD 2	Compare the size of a product to the size of one factor	**5.NF.B.5.a**
FORWARD 1	Describe the effect of a dilation on coordinates	**8.G.A.3**
FORWARD 2	Verify the properties of dilations	**HSG-SRT.A.1**

SET 37 • 7.G.A.1

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.1

A drawing uses the scale 1 cm : 4 m. A wall is 5 cm long on the drawing.

How long should the wall be on a new drawing with the scale 1 cm : 2 m?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q1/5

1124

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

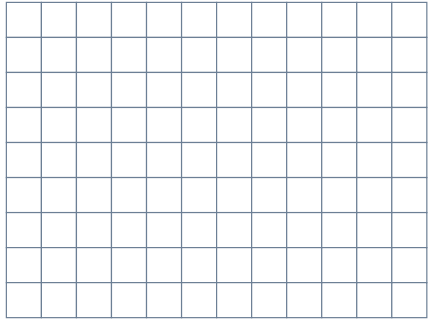
10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q1/5

A rectangle on a drawing is 4 units by 2 units.

Draw the rectangle at a scale twice as large on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q2/5

1125

QUESTION ID

S37-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A rectangle 8 units by 4 units, in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q2/5

A plan uses the scale 1 in : 6 ft. A table is 3 in long on the plan.

How long should the table be on a new plan with the scale 1 in : 3 ft?

A. 1.5 in

B. 18 in

C. 6 in

D. 9 in

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q3/5

1126

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6 in

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q3/5

True or false?

Changing a drawing's scale from 1 cm : 5 m to 1 cm : 10 m makes every length on the drawing half as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q4/5

1127

QUESTION ID

S37-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q4/5

True or false?

A 6 cm segment on a drawing with the scale 1 cm : 2 km becomes 9 cm on a drawing with the scale 1 cm : 3 km.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q5/5

1128

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It becomes 4 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 37 • MAIN • Q5/5

SET 37 • 7.G.A.1

BACKWARD 1

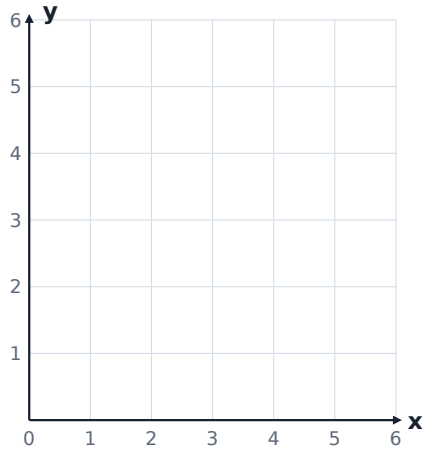
Draw a polygon on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Draw the polygon with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

1130

QUESTION ID

S37-B1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4)

Written: 4 units

GRADING NOTE Grade both: the rectangle drawn with the four given vertices, and the length 4 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

A rectangle has vertices (2, 1), (2, 6), (7, 6), and (7, 1).

How long is the side from (2, 1) to (7, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

1131

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

Which point makes a square with (1, 1), (4, 1), and (4, 4)?

A. (4, 7)

B. (3, 1)

C. (1, 3)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

1132

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

The points (0, 2) and (0, 7) are 5 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

1133

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

The side from (3, 2) to (3, 8) is 5 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

1134

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 6 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 7.G.A.1

BACKWARD 2

Compare the size of a product to the size of one factor • Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a

Without multiplying, is 3 × 12 greater than or less than 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q1/5

1136

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Greater than 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q1/5

Without multiplying, is $\frac{1}{2} \times 18$ greater than or less than 18?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q2/5

1137

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Less than 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q2/5

Which product is greater than 15?

A. 2×15

B. $\frac{3}{4} \times 15$

C. $\frac{1}{2} \times 15$

D. $\frac{1}{3} \times 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q3/5

1138

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 × 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} \times 9$ is less than 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q4/5

1139

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

4 × 8 is less than 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q5/5

1140

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.5.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 7.G.A.1

FORWARD 1

Describe the effect of a dilation on coordinates • Grade 8 • Geometry • 8.G.A.3

The point (4, −2) is dilated by a scale factor of 3 with the origin as the center.

What are the coordinates of the image?

Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q1/5

1142

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(12, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q1/5

A triangle has vertices (2, 2), (6, 2), and (2, 4). It is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the vertices of the image?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q2/5

1143

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 1), (3, 1), and (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q2/5

A dilation centered at the origin maps (5, 10) to (2, 4).

What is the scale factor?

A. $\frac{2}{5}$

B. $\frac{5}{2}$

C. -3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q3/5

1144

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A dilation by a scale factor of 2 with the origin as the center maps (−3, 1) to (−6, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q4/5

1145

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

A dilation by a scale factor of 4 with the origin as the center maps (2, 3) to (6, 7).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q5/5

1146

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It maps (2, 3) to (8, 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 7.G.A.1

FORWARD 2

Verify the properties of dilations • Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A segment 5 cm long is dilated by a scale factor of 1.8.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q1/5

1148

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q1/5

A line passes through the center of a dilation.

What is the image of the line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q2/5

1149

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The same line (it maps to itself).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q2/5

A line that does not pass through the center of a dilation is dilated.

What is its image?

A. The same line

B. A curve

C. A perpendicular line

D. A parallel line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q3/5

1150

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A parallel line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q3/5

True or false?

A dilation with a scale factor between 0 and 1 makes segments shorter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q4/5

1151

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q4/5

True or false?

A dilation always changes the slope of a line that does not pass through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q5/5

1152

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

Grade 9 (High School) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 7.G.A.2 • GEOMETRY

Draw triangles from given conditions

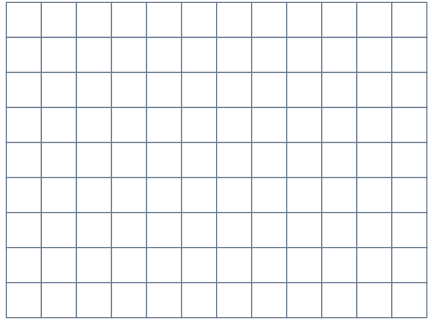
MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Use the Pythagorean theorem to find an unknown side length	**8.G.B.7**
FORWARD 2	Make formal constructions with a compass and straightedge	**HSG-CO.D.12**

SET 38 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

On the grid, draw a triangle with a right angle. The two sides that form the right angle should be 4 units and 6 units long.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q1/5

1155

QUESTION ID

S38-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Any right triangle whose two sides that form the right angle are 4 units and 6 units long, in any position or orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a ruler and a protractor. Draw a triangle with a 5 cm side that has a 50° angle at one end and a 60° angle at the other end.

Measure the third angle.

Third angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q2/5

1156

QUESTION ID

S38-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A triangle with a 5 cm side whose two end angles measure 50° and 60°, in any position or orientation

Written: 70°

GRADING NOTE Grade both: the triangle drawn with the given side and angles (within ±2 mm and ±2°), and the third angle 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q2/5

Which three side lengths can form a triangle?

A. 1 cm, 2 cm, 3 cm

B. 2 cm, 3 cm, 6 cm

C. 5 cm, 7 cm, 10 cm

D. 4 cm, 4 cm, 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q3/5

1157

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

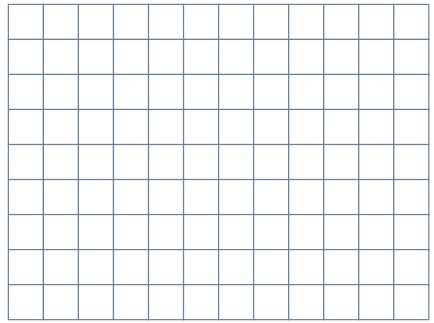
C. 5 cm, 7 cm, 10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q3/5

On the grid, draw an isosceles triangle that is not equilateral.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q4/5

1158

QUESTION ID

S38-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Any triangle with exactly two sides of equal length (the third side has a different length)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q4/5

True or false?

A triangle can have angles of 90°, 60°, and 40°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q5/5

1159

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The angles of a triangle add to 180°, and these add to 190°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 38 • MAIN • Q5/5

SET 38 • 7.G.A.2

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

Use a protractor to draw a 65° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q1/5

1161

QUESTION ID

S38-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

An angle that measures 65° (within ±2°), in any orientation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q1/5

One ray of an angle lines up with 0 on a protractor. The other ray passes through 120.

What is the measure of the angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q2/5

1162

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q2/5

Which angle is greater than a right angle?

A. 80°

B. 90°

C. 110°

D. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q3/5

1163

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q3/5

True or false?

A protractor measures angles in degrees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q4/5

1164

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A 30° angle is an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q5/5

1165

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 7.G.A.2

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has a 90° angle.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q1/5

1167

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q1/5

A triangle has angles of 100°, 50°, and 30°.

Classify the triangle by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q2/5

1168

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Obtuse triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q2/5

What kind of triangle has three angles that are each less than 90°?

A. Obtuse triangle

B. Acute triangle

C. Straight triangle

D. Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q3/5

1169

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Acute triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q3/5

True or false?

An obtuse triangle has exactly one obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q4/5

1170

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have two right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q5/5

1171

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 7.G.A.2

FORWARD 1

Use the Pythagorean theorem to find an unknown side length • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.

How long is the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q1/5

1173

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q1/5

A right triangle has a hypotenuse of 13 m and one leg of 5 m.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q2/5

1174

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q2/5

A 10-foot ladder leans against a wall. The top of the ladder is 8 feet above the ground.

How far is the bottom of the ladder from the wall?

A. 6 feet

B. 12.8 feet

C. 18 feet

D. 2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q3/5

1175

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q3/5

True or false?

A right triangle with legs of 9 and 12 has a hypotenuse of 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q4/5

1176

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs of 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q5/5

1177

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hypotenuse is √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 7.G.A.2

FORWARD 2

Make formal constructions with a compass and straightedge • Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12

To construct an equilateral triangle on segment AB, you draw two circles.

What are the centers and the radius of the circles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q1/5

1179

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The centers are A and B, and each radius is AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q1/5

To construct the perpendicular bisector of a segment, where do you place the compass point for the two arcs?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q2/5

1180

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At each endpoint of the segment, with the same radius (more than half the segment).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q2/5

1180

Which tool draws arcs and circles in a formal construction?

A. Protractor

B. Compass

C. Ruler with marks

D. Calculator

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q3/5

1181

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Compass

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q3/5

True or false?

The perpendicular bisector of a segment passes through its midpoint and makes right angles with it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q4/5

1182

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q4/5

True or false?

A formal construction may use a protractor to measure angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q5/5

1183

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.D.12 • Set 38 • FORWARD 2 • Q5/5

SET 39 • 7.G.A.2 • GEOMETRY

Decide whether given conditions make one triangle, more than one triangle, or no triangle

MAIN	Grade 7 main question	**7.G.A.2**
BACKWARD 1	Find an unknown angle when angles are joined	**4.MD.C.7**
BACKWARD 2	Find the perimeter of a polygon from its side lengths	**3.MD.D.8**
FORWARD 1	Use the converse of the Pythagorean theorem to test for a right tr…	**8.G.B.6**
FORWARD 2	Use triangle congruence criteria	**HSG-CO.B.8**

SET 39 • 7.G.A.2

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.2

How many triangles have sides of 3 cm, 4 cm, and 9 cm?

A. Exactly one triangle

B. More than one triangle

C. No triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q1/5

1186

QUESTION ID

S39-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. No triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q1/5

A triangle has angles of 30°, 60°, and 90°. No side lengths are given.

How many different triangles fit these conditions? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q2/5

1187

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

More than one. Triangles of different sizes can have the same three angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q2/5

How many different triangles have sides of 5 cm, 6 cm, and 7 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q3/5

1188

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

Exactly one triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q3/5

True or false?

Angles of 80° and 110° cannot both be angles of the same triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q4/5

1189

QUESTION ID

S39-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q4/5

True or false?

Three side lengths that form a triangle can make two triangles with different shapes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q5/5

1190

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 39 • MAIN • Q5/5

SET 39 • 7.G.A.2

BACKWARD 1

Find an unknown angle when angles are joined • Grade 4 • Measurement & Data • 4.MD.C.7

Angle ABC is split into two angles that measure 35° and 50°.

What is the measure of angle ABC?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q1/5

1192

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q1/5

A 90° angle is split into two angles. One measures 28°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q2/5

1193

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

62°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q2/5

A 120° angle is split into two angles. One measures 45°.

What does the other angle measure?

A. 45°

B. 165°

C. 75°

D. 85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q3/5

1194

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 75°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A 45° angle and a 45° angle joined together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q4/5

1195

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A 120° angle and a 70° angle joined together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q5/5

1196

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make a 190° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 7.G.A.2

BACKWARD 2

Find the perimeter of a polygon from its side lengths • Grade 3 • Measurement & Data • 3.MD.D.8

A triangle has sides of 7 cm, 9 cm, and 12 cm.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q1/5

1198

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

28 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q1/5

A triangle has a perimeter of 30 inches. Two of its sides are 8 inches and 10 inches.

How long is the third side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q2/5

1199

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q2/5

A triangle has sides of 5 m, 5 m, and 6 m.

What is its perimeter?

A. 11 m

B. 21 m

C. 150 m

D. 16 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q3/5

1200

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 16 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q3/5

True or false?

A triangle with sides of 4 feet, 6 feet, and 7 feet has a perimeter of 17 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q4/5

1201

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q4/5

True or false?

A triangle with three 9 cm sides has a perimeter of 18 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q5/5

1202

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 7.G.A.2

FORWARD 1

Use the converse of the Pythagorean theorem to test for a right triangle • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides of 7, 24, and 25 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q1/5

1204

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

$7^2 + 24^2 = 49 + 576 = 625 = 25^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q1/5

Is a triangle with sides of 6, 7, and 9 a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q2/5

1205

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. $6^2 + 7^2 = 85$, but $9^2 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q2/5

Which side lengths form a right triangle?

A. 8, 15, 17

B. 4, 5, 6

C. 5, 6, 8

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q3/5

1206

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8, 15, 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q3/5

True or false?

A triangle with sides of 9, 40, and 41 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q4/5

1207

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q4/5

True or false?

A triangle with sides of 3, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q5/5

1208

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 7.G.A.2

FORWARD 2

Use triangle congruence criteria • Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8

Two triangles have three pairs of equal sides.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q1/5

1210

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q1/5

Two triangles have two pairs of equal sides and equal angles between those sides.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q2/5

1211

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q2/5

Which information is NOT enough to show that two triangles are congruent?

A. Two angles and the side between them (ASA)

B. Two sides and the angle between them (SAS)

C. Three pairs of equal sides (SSS)

D. Three pairs of equal angles (AAA)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q3/5

1212

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Three pairs of equal angles (AAA)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q3/5

True or false?

SSS shows that three side lengths determine one triangle, up to congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q4/5

1213

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of equal sides are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q5/5

1214

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.B.8 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 7.G.A.3 • GEOMETRY

Describe the cross sections made by slicing right prisms and pyramids

MAIN	Grade 7 main question	**7.G.A.3**
BACKWARD 1	Identify the faces of solids from their nets	**6.G.A.4**
BACKWARD 2	Name shapes by their number of sides and faces	**2.G.A.1**
FORWARD 1	Find the volumes of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Identify cross sections of three-dimensional objects	**HSG-GMD.B.4**

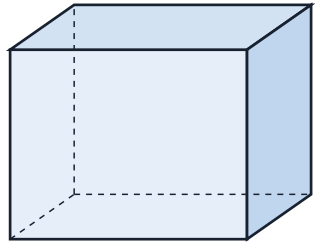
SET 40 • 7.G.A.3

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane parallel to its base.

What is the shape of the cross section?



Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q1/5

1217

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q1/5

A right square pyramid is sliced by a plane parallel to its base.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q2/5

1218

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

A square smaller than the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q2/5

A cube is sliced by a plane that is parallel to one of its faces.

What is the shape of the cross section?

A. A triangle

B. A pentagon

C. A square

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q3/5

1219

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q3/5

True or false?

A right square pyramid is sliced by a plane through its top vertex that is perpendicular to the base. The cross section is a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q4/5

1220

QUESTION ID

S40-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q4/5

True or false?

A right rectangular prism sliced by a plane parallel to its base can have a triangle as its cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q5/5

1221

QUESTION ID

S40-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 40 • MAIN • Q5/5

SET 40 • 7.G.A.3

BACKWARD 1

Identify the faces of solids from their nets • Grade 6 • Geometry • 6.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q1/5

A net has 1 square and 4 triangles.

What solid does it fold into?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A square pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q2/5

Which solid has a net of 2 triangles and 3 rectangles?

A. Square pyramid

B. Rectangular prism

C. Triangular prism

D. Cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Triangular prism

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

The net of a cube has 6 squares.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A triangular pyramid has 5 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 4 faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 7.G.A.3

BACKWARD 2

Name shapes by their number of sides and faces • Grade 2 • Geometry • 2.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

How many sides does a pentagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q1/5

A shape has 6 sides and 6 angles.

What is the name of the shape?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q2/5

Which shape has 4 sides?

A. Quadrilateral

B. Hexagon

C. Pentagon

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Quadrilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q3/5

True or false?

A cube has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q4/5

True or false?

A triangle has 4 angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 7.G.A.3

FORWARD 1

Find the volumes of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 2 cm and a height of 9 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q1/5

1235

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

113.04 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q1/5

A cone has a radius of 6 inches and a height of 5 inches.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q2/5

1236

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q2/5

A sphere has a radius of 3 m.

What is its volume in terms of π?

A. 12π cubic meters

B. 108π cubic meters

C. 27π cubic meters

D. 36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q3/5

1237

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q3/5

True or false?

A cone and a cylinder have the same base and height. The cone's volume is $\frac{1}{3}$ of the cylinder's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q4/5

1238

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q5/5

1239

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 20π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 7.G.A.3

FORWARD 2

Identify cross sections of three-dimensional objects • Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A cylinder is sliced by a plane parallel to its base.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q1/5

1241

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q1/5

A cone is sliced by a plane through its top point (apex) that is perpendicular to its base.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q2/5

1242

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle (an isosceles triangle)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q2/5

A sphere is sliced by a plane.

What is the shape of the cross section?

A. A triangle

B. A square

C. An oval that is not a circle

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q3/5

1243

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q3/5

True or false?

Spinning a rectangle around one of its sides makes a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q4/5

1244

QUESTION ID

S40-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q4/5

True or false?

A cube can never have a cross section shaped like a hexagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q5/5

1245

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A slice through the midpoints of six edges makes a hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 7.G.B.4 • GEOMETRY

Find the circumference of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the perimeter of a rectangle	**4.MD.A.3**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Find the volume of a cylinder	**8.G.C.9**
FORWARD 2	Find the length of an arc of a circle	**HSG-C.B.5**

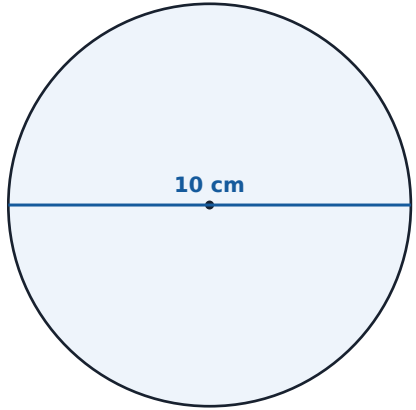
SET 41 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the circumference of the circle. Use 3.14 for π.

Circumference: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q1/5

1248

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

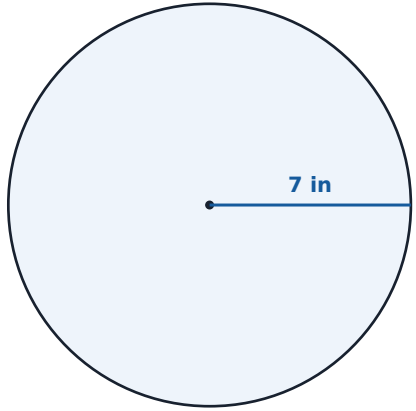
31.4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q1/5

Find the circumference of the circle. Use $\frac{22}{7}$ for π.



Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q2/5

1249

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

44 in

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q2/5

A circular track has a circumference of 62.8 m.

What is its diameter? Use 3.14 for π.

A. 31.4 m

B. 20 m

C. 10 m

D. 197.2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q3/5

1250

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q3/5

True or false?

A wheel has a radius of 15 inches. In one full turn it rolls about 94.2 inches. (Use 3.14 for π.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q4/5

1251

QUESTION ID

S41-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q4/5

True or false?

A circle with a radius of 4 m has a circumference of 4π m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q5/5

1252

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The circumference is 8π m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 41 • MAIN • Q5/5

SET 41 • 7.G.B.4

BACKWARD 1

Find the perimeter of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle is 9 m long and 4 m wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q1/5

1254

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

26 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q1/5

A square has 8 cm sides.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q2/5

1255

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q2/5

A rectangle is 12 feet long and 5 feet wide.

What is its perimeter?

A. 29 feet

B. 17 feet

C. 60 feet

D. 34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q3/5

1256

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q3/5

True or false?

A rectangle 6 inches long and 3 inches wide has a perimeter of 18 inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q4/5

1257

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q4/5

True or false?

A square with 10 m sides has a perimeter of 100 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q5/5

1258

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter is 40 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 7.G.B.4

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

3.14 × 6

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q1/5

1260

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q1/5

Multiply.

3.14 × 2.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q2/5

1261

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7.85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

3.14×12

A. 37.68

B. 36.48

C. 376.8

D. 15.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q3/5

1262

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 37.68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

$3.14 \times 10 = 31.4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q4/5

1263

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

$3.14 \times 4 = 12.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q5/5

1264

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 7.G.B.4

FORWARD 1

Find the volume of a cylinder • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 5 feet and a height of 4 feet.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q1/5

1266

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

100π cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q1/5

A cylinder has a diameter of 6 cm and a height of 10 cm.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q2/5

1267

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

282.6 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q2/5

A cylinder has a volume of 90π cubic meters and a radius of 3 m.

What is its height?

A. 30 m

B. 3 m

C. 10 m

D. 15 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q3/5

1268

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q3/5

True or false?

Doubling the height of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q4/5

1269

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 1 and a height of 1 has a volume of 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q5/5

1270

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 7.G.B.4

FORWARD 2

Find the length of an arc of a circle • Grade 9 (High School) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 6 cm.

Find the length of an arc with a central angle of 90°, in terms of π.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q1/5

1272

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3π cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q1/5

A circle has a circumference of 40 inches.

How long is an arc with a central angle of 45°?

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q2/5

1273

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q2/5

A circle has a radius of 10 m.

What is the length of an arc with a central angle of 60°?

A. $\frac{20}{3}\pi$ m

B. $\frac{5}{3}\pi$ m

C. $\frac{10}{3}\pi$ m

D. 60π m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q3/5

1274

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10/3π m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q3/5

True or false?

An arc with a central angle of 180° is half of the circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q4/5

1275

QUESTION ID

S41-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 3, an arc with a central angle of 120° has a length of 3π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q5/5

1276

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The arc length is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 41 • FORWARD 2 • Q5/5

SET 42 • 7.G.B.4 • GEOMETRY

Find the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Evaluate powers with whole-number exponents	**6.EE.A.1**
BACKWARD 2	Evaluate a formula	**6.EE.A.2.c**
FORWARD 1	Find the volume of a cone	**8.G.C.9**
FORWARD 2	Find the area of a sector of a circle	**HSG-C.B.5**

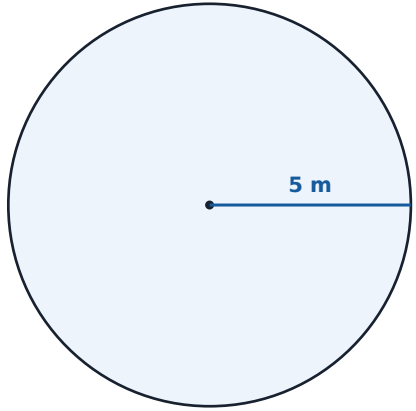
SET 42 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

Find the area of the circle.
Use 3.14 for π.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q1/5

1279

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

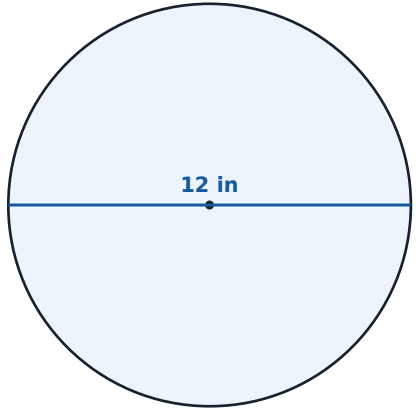
78.5 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q1/5

Find the area of the circle in terms of π.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q2/5

1280

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

36π square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q2/5

A circular rug has a radius of 3 feet.

What is its area? Use 3.14 for π.

A. 113.04 square feet

B. 28.26 square feet

C. 9.42 square feet

D. 18.84 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q3/5

1281

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 28.26 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q3/5

True or false?

A circle with a radius of 10 cm has an area of 100π square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q4/5

1282

QUESTION ID

S42-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q4/5

True or false?

A circle with a diameter of 8 m has an area of 64π square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q5/5

1283

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The radius is 4 m, so the area is 16π square meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 42 • MAIN • Q5/5

SET 42 • 7.G.B.4

BACKWARD 1

Evaluate powers with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q1/5

1285

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q2/5

1286

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q2/5

Find the value.

4^3

A. 7

B. 64

C. 12

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q3/5

1287

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q4/5

1288

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q5/5

1289

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 7.G.B.4

BACKWARD 2

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Use $A = s^2$.

Find A when s = 11.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q1/5

1291

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q1/5

Use $A = 3.14r^2$.

Find A when r = 2.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q2/5

1292

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12.56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $A = \frac{1}{2}bh$.

Find A when b = 8 and h = 5.

A. 13

B. 20

C. 40

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q3/5

1293

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q3/5

True or false?

Use $V = s^3$. When $s = 2$, $V = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q4/5

1294

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Use A = lw. When l = 3.5 and w = 2, A = 5.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q5/5

1295

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 7.G.B.4

FORWARD 1

Find the volume of a cone • Grade 8 • Geometry • 8.G.C.9

A cone has a radius of 3 cm and a height of 8 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q1/5

1297

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

24π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q1/5

A cone has a radius of 6 feet and a height of 7 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q2/5

1298

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

263.76 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q2/5

A cone has a base area of 30 square inches and a height of 9 inches.

What is its volume?

A. 270 cubic inches

B. 90 cubic inches

C. 30 cubic inches

D. 39 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q3/5

1299

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

A cone with a radius of 3 and a height of 3 has a volume of 9π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q4/5

1300

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q4/5

True or false?

The volume of a cone is $\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q5/5

1301

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The volume of a cone is (1/3)$\pi r^2 h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 7.G.B.4

FORWARD 2

Find the area of a sector of a circle • Grade 9 (High School) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 4 m.

Find the area of a sector with a central angle of 90°, in terms of π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q1/5

1303

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q1/5

A pizza with a radius of 8 inches is cut into 8 equal slices.

What is the area of one slice, in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q2/5

1304

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8π square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q2/5

A circle has a radius of 6 cm.

What is the area of a sector with a central angle of 120°?

A. 12π square centimeters **B.** 24π square centimeters

C. 4π square centimeters **D.** 36π square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q3/5

1305

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12π square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q3/5

True or false?

A sector with a central angle of 180° has half the area of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q4/5

1306

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

In a circle with a radius of 6, a sector with a central angle of 60° has an area of 12π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q5/5

1307

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area is 6π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

Grade 9 (High School) • Geometry: Circles • HSG-C.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 43 • 7.G.B.4 • GEOMETRY

Explain the relationship between the circumference and the area of a circle

MAIN	Grade 7 main question	**7.G.B.4**
BACKWARD 1	Find the area of a parallelogram	**6.G.A.1**
BACKWARD 2	Identify equivalent expressions	**6.EE.A.4**
FORWARD 1	Use square roots to solve equations of the form $x^2 = p$	**8.EE.A.2**
FORWARD 2	Give an informal argument for circle and volume formulas	**HSG-GMD.A.1**

SET 43 • 7.G.B.4

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle with radius r is cut into many equal wedges. The wedges are rearranged into a shape close to a parallelogram. Its base is half the circumference.

Write the base in terms of r.

Base: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q1/5

1310

QUESTION ID
S43-M-Q1

TYPE
Short answer

CORRECT ANSWER
πr

GRADING NOTE Half of 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q1/5

A circle with radius r is cut into many equal wedges and rearranged into a shape close to a parallelogram.

What is the height of the parallelogram in terms of r?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q2/5

1311

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q2/5

A circle rearranged into a near-parallelogram has a base of πr and a height of r.

What is the area?

A. πr^2

B. $2\pi r^2$

C. πr

D. $2\pi r$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q3/5

1312

QUESTION ID

S43-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. πr^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q3/5

True or false?

A circle has circumference C and radius r. Its area equals $\frac{1}{2} \times C \times r$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q4/5

1313

QUESTION ID

S43-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q4/5

True or false?

A circle with a circumference of 10π has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q5/5

1314

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its radius is 5, so its area is 25π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 43 • MAIN • Q5/5

SET 43 • 7.G.B.4

BACKWARD 1

Find the area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

A parallelogram has a base of 9 cm and a height of 4 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A parallelogram has a base of 12 m and a height of 6.5 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

78 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A triangle is cut off one end of a parallelogram and moved to the other end to make a new shape.

What shape does it make?

A. A pentagon

B. A trapezoid that is not a rectangle

C. A circle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

A parallelogram and a rectangle with the same base and the same height have the same area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

A parallelogram has a base of 10 and slanted sides of 6. Its area is 60.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area uses the height, not the slanted side, so it is less than 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 7.G.B.4

BACKWARD 2

Identify equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Is $\frac{1}{2} \times 2 \times r$ equivalent to r? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 1/2 × 2 = 1, and 1 × r = r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q1/5

Which expression is equivalent to 3 × r × r?

A. 6r

B. $3 + r^2$

C. $(3r)^2$

D. $3r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S43-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $3r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression equivalent to $x \cdot x \cdot x$ using an exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S43-B2-Q3

TYPE

Short answer

CORRECT ANSWER

x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q3/5

True or false?

2(4y) and 8y are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q4/5

True or false?

a^2 and $2a$ are equivalent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 7.G.B.4

FORWARD 1

Use square roots to solve equations of the form $x^2 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x.

$x^2 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q1/5

1328

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 or −8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q1/5

A circle has an area of 49π square centimeters.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q2/5

1329

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q2/5

Solve for x.

$x^3 = 27$

A. x = 3

B. x = 9

C. x = 6

D. x = 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q3/5

1330

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{144} = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q4/5

1331

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q4/5

True or false?

√50 is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q5/5

1332

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 7.G.B.4

FORWARD 2

Give an informal argument for circle and volume formulas • Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Use the wedge argument to explain why the area of a circle is πr^2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q1/5

1334

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The wedges form a shape close to a parallelogram with base πr (half the circumference) and height r, so the area is $\pi r \times r = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q1/5

Two stacks of 20 identical coins have the same height. One stack is straight and one is slanted.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q2/5

1335

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. At every height the cross sections have the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q2/5

The circumference of a circle is $C = \pi d$.

What is the ratio of C to d?

A. 2π

B. π

C. 2

D. r

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q3/5

1336

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q3/5

True or false?

As a circle is cut into more and more wedges, the rearranged shape gets closer to a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q4/5

1337

QUESTION ID

S43-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q4/5

True or false?

Cavalieri's principle says that any two solids with the same height have the same volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q5/5

1338

QUESTION ID
S43-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The cross sections must also have equal areas at every height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 43 • FORWARD 2 • Q5/5

SET 44 • 7.G.B.5 • GEOMETRY

Write and solve equations for supplementary and complementary angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Find an unknown part of a right angle or a straight angle	**4.MD.C.7**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Use the angle sum and exterior angles of triangles	**8.G.A.5**
FORWARD 2	Prove theorems about angles formed by parallel lines	**HSG-CO.C.9**

SET 44 • 7.G.B.5

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

Two angles are complementary. One measures 37°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q1/5

1341

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

53°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q1/5

Angles that measure (2x + 10)° and 70° are supplementary.

Write and solve an equation for x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q2/5

1342

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

2x + 10 + 70 = 180; x = 50

GRADING NOTE Any correct equation earns credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q2/5

Two complementary angles measure x° and (x + 20)°.

What is the value of x?

A. 45

B. 35

C. 80

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q3/5

1343

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q3/5

True or false?

An angle of 125° and an angle of 55° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q4/5

1344

QUESTION ID

S44-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Angles of 45° and 55° are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q5/5

1345

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 100°, not 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 44 • MAIN • Q5/5

SET 44 • 7.G.B.5

BACKWARD 1

Find an unknown part of a right angle or a straight angle • Grade 4 • Measurement & Data • 4.MD.C.7

A straight angle (180°) is split into two angles. One measures 115°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q1/5

1347

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q1/5

A right angle is split into two angles. One measures 34°.

What does the other angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q2/5

1348

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

56°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q2/5

A right angle is split into three angles. Two of them measure 20° and 30°.

What does the third angle measure?

A. 50°

B. 60°

C. 40°

D. 10°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q3/5

1349

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q3/5

True or false?

Angles of 72° and 18° together make a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q4/5

1350

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q4/5

True or false?

Angles of 95° and 95° together make a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q5/5

1351

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 7.G.B.5

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 48 = 90

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q1/5

1353

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q1/5

Solve for x.

3x = 180

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q2/5

1354

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for y.

125 + y = 180

A. y = 55

B. y = 45

C. y = 65

D. y = 305

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q3/5

1355

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q3/5

True or false?

a = 27 is the solution of a + 63 = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q4/5

1356

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q4/5

True or false?

b = 80 is the solution of 2b = 180.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q5/5

1357

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 7.G.B.5

FORWARD 1

Use the angle sum and exterior angles of triangles • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle measure 48° and 75°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q1/5

1359

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

57°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 130°. One of the two remote interior angles measures 60°.

What does the other remote interior angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q2/5

1360

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q2/5

The angles of a triangle measure x°, 2x°, and 3x°.

What is the value of x?

A. 36

B. 30

C. 45

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q3/5

1361

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The interior angles of a triangle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q4/5

1362

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q5/5

1363

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 7.G.B.5

FORWARD 2

Prove theorems about angles formed by parallel lines • Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9

Two parallel lines are cut by a transversal. One angle measures 65°.

What does its alternate interior angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q1/5

1365

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q1/5

Two parallel lines are cut by a transversal. Two same-side interior angles measure 3x° and 2x°.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q2/5

1366

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q2/5

Which statement is a theorem?

A. Vertical angles are complementary.

B. Vertical angles are supplementary.

C. Vertical angles are congruent.

D. Vertical angles are always right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q3/5

1367

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Vertical angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q3/5

True or false?

When parallel lines are cut by a transversal, corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q4/5

1368

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

Any two lines cut by a transversal form congruent alternate interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q5/5

1369

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE This is true only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 7.G.B.5 • GEOMETRY

Use vertical and adjacent angles to find unknown angles

MAIN	Grade 7 main question	**7.G.B.5**
BACKWARD 1	Identify lines, rays, segments, and angles	**4.G.A.1**
BACKWARD 2	Understand angle measure as a fraction of a full turn	**4.MD.C.5.a**
FORWARD 1	Find angles formed by parallel lines cut by a transversal	**8.G.A.5**
FORWARD 2	Prove that vertical angles are congruent	**HSG-CO.C.9**

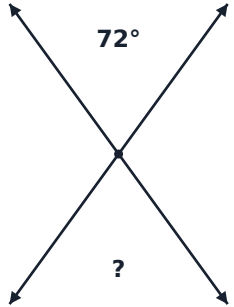
SET 45 • 7.G.B.5

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. One angle measures 72°.

What does the angle marked with a question mark measure?



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q1/5

1372

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

72°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q1/5

Two lines intersect. One angle measures (3x + 15)°. The angle vertical to it measures 75°.

What is the value of x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q2/5

1373

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q2/5

Two lines intersect. Angles a and b are adjacent and form a straight line. Angle a measures 118°.

What does angle b measure?

A. 72°

B. 118°

C. 62°

D. 242°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q3/5

1374

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 62°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q3/5

True or false?

Adjacent angles share a vertex and a side and do not overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q4/5

1375

QUESTION ID

S45-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q4/5

True or false?

Vertical angles are always adjacent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q5/5

1376

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 45 • MAIN • Q5/5

SET 45 • 7.G.B.5

BACKWARD 1

Identify lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

How many endpoints does a ray have?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q1/5

1378

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q1/5

What figure is formed by two rays that share an endpoint?

Figure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q2/5

1379

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

An angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q2/5

Which figure goes on forever in both directions?

A. Ray

B. Segment

C. Line

D. Angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q3/5

1380

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q3/5

True or false?

Two lines that cross meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q4/5

1381

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

A line segment has no endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q5/5

1382

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 7.G.B.5

BACKWARD 2

Understand angle measure as a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many degrees are in one full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q1/5

1384

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An angle turns through $\frac{1}{4}$ of a circle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q2/5

1385

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q2/5

An angle turns through $\frac{1}{2}$ of a circle.

What is its measure?

A. 90°

B. 45°

C. 360°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q3/5

1386

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q3/5

True or false?

An angle that turns through $\frac{1}{360}$ of a circle measures 1°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q4/5

1387

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q4/5

True or false?

An angle that turns through $\frac{1}{3}$ of a circle measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q5/5

1388

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It measures 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 7.G.B.5

FORWARD 1

Find angles formed by parallel lines cut by a transversal • Grade 8 • Geometry • 8.G.A.5

Two parallel lines are cut by a transversal. One angle measures 110°.

What does the corresponding angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q1/5

1390

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. Alternate interior angles measure (4x − 10)° and 70°.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q2/5

1391

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q2/5

Two parallel lines are cut by a transversal. One same-side interior angle measures 75°.

What does the other measure?

A. 15°

B. 105°

C. 75°

D. 285°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q3/5

1392

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 105°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q3/5

True or false?

When parallel lines are cut by a transversal, alternate exterior angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q4/5

1393

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q4/5

True or false?

When parallel lines are cut by a transversal, corresponding angles add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q5/5

1394

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 7.G.B.5

FORWARD 2

Prove that vertical angles are congruent • Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 form a linear pair. Angles 2 and 3 form a linear pair.

Explain why angle 1 and angle 3 have the same measure.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q1/5

1396

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each one equals 180° minus the measure of angle 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q1/5

Two lines intersect. Vertical angles measure $(5x - 20)°$ and $(3x + 30)°$.

Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q2/5

1397

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q2/5

Angles 1 and 2 form a linear pair.

Which reason supports angle 1 + angle 2 = 180°?

A. The angles of a triangle add to 180°.

B. Vertical angles are congruent.

C. Corresponding angles are congruent.

D. Linear pairs are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q3/5

1398

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Linear pairs are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q3/5

True or false?

Two angles that are each supplementary to the same angle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q4/5

1399

QUESTION ID

S45-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q4/5

True or false?

Two intersecting lines always form vertical angles that add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q5/5

1400

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

Grade 9 (High School) • Geometry: Congruence • HSG-CO.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 7.G.B.6 • GEOMETRY

Find the area of composite figures made of triangles and quadrilaterals

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the area of a triangle	**6.G.A.1**
BACKWARD 2	Find areas of rectilinear figures by breaking them into rectangles	**3.MD.C.7.d**
FORWARD 1	Find the volume of a solid made of cylinders, cones, and spheres	**8.G.C.9**
FORWARD 2	Use geometric shapes to model real objects	**HSG-MG.A.1**

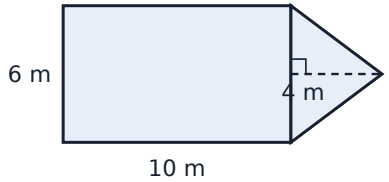
SET 46 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

The figure is a 10 m by 6 m rectangle and a triangle with a base of 6 m (the rectangle's side) and a height of 4 m. The parts do not overlap.

What is the total area?



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q1/5

1403

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

72 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q1/5

An L-shaped floor is made of two rectangles that do not overlap. One is 8 feet by 5 feet and the other is 4 feet by 3 feet.

What is the area of the floor?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q2/5

1404

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

52 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q2/5

A trapezoid has parallel sides of 8 cm and 12 cm. The distance between them (the height) is 5 cm.

What is its area?

A. 100 square centimeters

B. 480 square centimeters

C. 50 square centimeters

D. 60 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q3/5

1405

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 50 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q3/5

True or false?

A 4 cm by 4 cm square is cut from one corner of a 12 cm by 8 cm rectangle. The area that is left is 80 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q4/5

1406

QUESTION ID

S46-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q4/5

True or false?

A 9 m by 4 m rectangle and a right triangle with legs of 4 m and 3 m are joined without overlapping. Their total area is 48 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q5/5

1407

QUESTION ID

S46-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total area is 36 + 6 = 42 square meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • MAIN • Q5/5

SET 46 • 7.G.B.6

BACKWARD 1

Find the area of a triangle • Grade 6 • Geometry • 6.G.A.1

A triangle has a base of 14 cm and a height of 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q1/5

1409

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

35 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q1/5

A right triangle has legs of 9 m and 6 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q2/5

1410

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q2/5

A triangle has a base of 7 feet and a height of 10 feet.

What is its area?

A. 17 square feet

B. 35 square feet

C. 140 square feet

D. 70 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q3/5

1411

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

A triangle with a base of 8 and a height of 3 has an area of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q4/5

1412

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

A triangle with a base of 6 and a height of 6 has an area of 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q5/5

1413

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 7.G.B.6

BACKWARD 2

Find areas of rectilinear figures by breaking them into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

A shape is made of a 5-by-3 rectangle and a 2-by-2 square that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q1/5

1415

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

19 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q1/5

A shape is made of a 6-by-2 rectangle and a 3-by-4 rectangle that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q2/5

1416

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q2/5

A shape is made of a 4-by-4 square and a 4-by-1 rectangle that do not overlap.

What is its area?

A. 16 square units

B. 20 square units

C. 13 square units

D. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q3/5

1417

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q3/5

True or false?

A 3-by-5 rectangle and a 2-by-5 rectangle that do not overlap have a total area of 25 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q4/5

1418

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q4/5

True or false?

A 7-by-2 rectangle and a 1-by-3 rectangle that do not overlap have a total area of 13 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q5/5

1419

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total area is 17 square units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 7.G.B.6

FORWARD 1

Find the volume of a solid made of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

ANSWER KEY Reduced copy of the question page, with its answer

A cone sits on top of a cylinder. Both have a radius of 3. The cylinder is 10 tall and the cone is 4 tall.

Find the total volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q1/5

1421

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

102π

GRADING NOTE 90π + 12π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q1/5

A hemisphere (half of a sphere) has a radius of 6 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q2/5

1422

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

144π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q2/5

A capsule is a cylinder with radius 2 and height 5 with half of a sphere of radius 2 on each end.

What is its volume?

A. $\frac{52}{3}\pi$

B. 20π

C. $\frac{92}{3}\pi$

D. $\frac{32}{3}\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q3/5

1423

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 92/3π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q3/5

True or false?

A sphere with a radius of 3 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q4/5

1424

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q4/5

True or false?

A cone with a radius of 3 and a height of 4 has a volume of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q5/5

1425

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 12π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 7.G.B.6

FORWARD 2

Use geometric shapes to model real objects • Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1

A tree trunk is modeled as a cylinder with a diameter of 2 feet and a height of 30 feet.

Estimate its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q1/5

1427

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

30π cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q1/5

A round tabletop is modeled as a circle with a circumference of 9.42 feet.

What is its radius? Use 3.14 for π.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q2/5

1428

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q2/5

Which shape is the best model for a basketball?

A. Cylinder

B. Sphere

C. Cube

D. Cone

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q3/5

1429

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q3/5

True or false?

A soup can is well modeled by a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q4/5

1430

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q4/5

True or false?

An ice cream cone is best modeled by a rectangular prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q5/5

1431

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 7.G.B.6 • GEOMETRY

Find the volume of right prisms

MAIN	Grade 7 main question	**7.G.B.6**
BACKWARD 1	Find the volume of a rectangular prism with fractional edges	**6.G.A.2**
BACKWARD 2	Find volume with $V = l \times w \times h$	**5.MD.C.5.b**
FORWARD 1	Compare the volumes of cylinders and prisms	**8.G.C.9**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

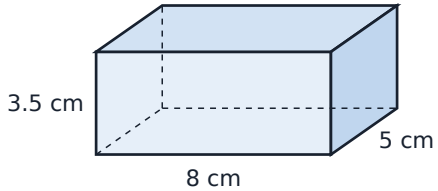
SET 47 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the volume of the right rectangular prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q1/5

1434

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

140 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q1/5

A right triangular prism has triangular bases with a base of 6 m and a height of 4 m. The prism is 10 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q2/5

1435

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

120 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q2/5

The base of a right prism has an area of 24 square inches. The prism is 7.5 inches tall.

What is its volume?

A. 90 cubic inches

B. 180 cubic inches

C. 360 cubic inches

D. 31.5 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q3/5

1436

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 180 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q3/5

True or false?

A cube with $\frac{1}{2}$-foot edges has a volume of $\frac{1}{8}$ cubic foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q4/5

1437

QUESTION ID

S47-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q4/5

True or false?

Doubling the height of a right prism doubles the area of its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q5/5

1438

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It doubles the volume; the base does not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • MAIN • Q5/5

SET 47 • 7.G.B.6

BACKWARD 1

Find the volume of a rectangular prism with fractional edges • Grade 6 • Geometry • 6.G.A.2

A right rectangular prism is $2\frac{1}{2}$ feet by 2 feet by 4 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q1/5

1440

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q1/5

A cube has $\frac{3}{4}$-inch edges.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q2/5

1441

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27/64 cubic inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q2/5

A right rectangular prism is $\frac{1}{2}$ m by 3 m by 6 m.

What is its volume?

A. 9.5 cubic meters

B. 4.5 cubic meters

C. 9 cubic meters

D. 18 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q3/5

1442

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A right rectangular prism $1\frac{1}{2}$ cm by 2 cm by 2 cm has a volume of 6 cubic centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q4/5

1443

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A right rectangular prism $\frac{1}{2}$ foot by $\frac{1}{2}$ foot by 4 feet has a volume of 2 cubic feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q5/5

1444

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 1 cubic foot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 7.G.B.6

BACKWARD 2

Find volume with V = l × w × h • Grade 5 • Measurement & Data • 5.MD.C.5.b

A rectangular prism is 7 units long, 4 units wide, and 3 units tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q1/5

1446

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

84 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q1/5

A box is 10 cm long, 6 cm wide, and 5 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q2/5

1447

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

300 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q2/5

A rectangular prism is 9 feet by 2 feet by 3 feet.

What is its volume?

A. 27 cubic feet

B. 14 cubic feet

C. 108 cubic feet

D. 54 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q3/5

1448

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 54 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q4/5

1449

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q4/5

True or false?

A box 5 m by 5 m by 2 m has a volume of 12 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q5/5

1450

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 50 cubic meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 7.G.B.6

FORWARD 1

Compare the volumes of cylinders and prisms • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 4 inches and a height of 5 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q1/5

1452

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

251.2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q1/5

A cylinder's base has an area of 12.56 square centimeters. The cylinder is 10 cm tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q2/5

1453

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

125.6 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q2/5

A cylinder has a radius of 1 m and a height of 3 m.

What is its volume in terms of π?

A. π cubic meters

B. 9π cubic meters

C. 6π cubic meters

D. 3π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q3/5

1454

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q3/5

True or false?

The volume of a cylinder is its base area times its height, just like a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q4/5

1455

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 2 has a volume of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q5/5

1456

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 8π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 7.G.B.6

FORWARD 2

Use volume formulas to solve problems • Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A pyramid has a square base with 6 m sides. It is 10 m tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q1/5

1458

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q1/5

A cone-shaped cup has a radius of 3 cm and holds 30π cubic centimeters.

How tall is the cup?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q2/5

1459

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q2/5

A sphere has a volume of 288π cubic centimeters.

What is its radius?

A. 6 cm

B. 12 cm

C. 4 cm

D. 8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q3/5

1460

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q3/5

True or false?

A pyramid has $\frac{1}{3}$ the volume of a prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q4/5

1461

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q4/5

True or false?

A sphere with a radius of 3 has a volume of 27π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q5/5

1462

QUESTION ID

S47-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its volume is 36π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

Grade 9 (High School) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 47 • FORWARD 2 • Q5/5

SET 48 • 7.G.B.6 • GEOMETRY

Find the surface area of right prisms

MAIN	Grade 7 main question	7.G.B.6
BACKWARD 1	Use a net to find surface area	6.G.A.4
BACKWARD 2	Find the area of a rectangle	4.MD.A.3
FORWARD 1	Use the Pythagorean theorem in three-dimensional figures — near…	8.G.B.7
FORWARD 2	Apply geometric methods to solve design problems	HSG-MG.A.3

SET 48 • 7.G.B.6

MAIN

Grade 7 main question • Grade 7 • Geometry • 7.G.B.6

Find the surface area of a right rectangular prism that is 6 cm by 4 cm by 3 cm.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q1/5

1465

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

108 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q1/5

Find the surface area of a cube with 5-inch edges.

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q2/5

1466

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

150 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q2/5

A right triangular prism has triangular bases with sides of 3 cm, 4 cm, and 5 cm and a right angle between the 3 cm and 4 cm sides. The prism is 10 cm long.

What is its surface area?

A. 126 square centimeters **B.** 60 square centimeters

C. 132 square centimeters **D.** 120 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q3/5

1467

QUESTION ID

S48-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 132 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q3/5

True or false?

A box 10 in by 2 in by 2 in needs exactly 88 square inches of paper to cover it with no overlap.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q4/5

1468

QUESTION ID

S48-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q4/5

True or false?

A right rectangular prism 2 m by 3 m by 4 m has a surface area of 24 square meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q5/5

1469

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its surface area is 52 square meters; 24 cubic meters is its volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • MAIN • Q5/5

SET 48 • 7.G.B.6

BACKWARD 1

Use a net to find surface area • Grade 6 • Geometry • 6.G.A.4

The net of a cube has 6 squares. Each square is 4 cm by 4 cm.

What is the surface area of the cube?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q1/5

1471

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

96 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q1/5

The net of a rectangular prism has two 5-by-3 rectangles, two 5-by-2 rectangles, and two 3-by-2 rectangles.

What is the surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q2/5

1472

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

62 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q2/5

The net of a square pyramid has a 6-by-6 square and 4 triangles. Each triangle has a base of 6 and a height of 5.

What is the surface area?

A. 186 square units

B. 66 square units

C. 156 square units

D. 96 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q3/5

1473

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 96 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q3/5

True or false?

A net shows all the faces of a solid laid flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q4/5

1474

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q4/5

True or false?

The net of a rectangular prism has 4 rectangles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q5/5

1475

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 6 rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 7.G.B.6

BACKWARD 2

Find the area of a rectangle • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 8 cm by 5 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q1/5

1477

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

40 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q1/5

A rectangle is 14 inches by 3 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q2/5

1478

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

42 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q2/5

A rectangle is 11 m by 6 m.

What is its area?

A. 34 square meters

B. 66 square meters

C. 17 square meters

D. 60 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q3/5

1479

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 66 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q3/5

True or false?

A rectangle 9 feet by 2 feet has an area of 18 square feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q4/5

1480

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q4/5

True or false?

A rectangle 5 cm by 5 cm has an area of 20 square centimeters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q5/5

1481

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 25 square centimeters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 7.G.B.6

FORWARD 1

Use the Pythagorean theorem in three-dimensional figures • Grade 8 • Geometry • 8.G.B.7

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A box is 4 units long, 3 units wide, and 12 units tall.

How long is a diagonal of its 4-by-3 base?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q1/5 • NEAREST RELATED

1483

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q1/5 • NEAREST RELATED

A box is 4 units long, 3 units wide, and 12 units tall.

How long is the diagonal from a bottom corner to the opposite top corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q2/5 • NEAREST RELATED

1484

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q2/5 • NEAREST RELATED

A square pyramid has base edges of 6 m and is 4 m tall.

What is the slant height (from the top to the midpoint of a base edge)?

A. 10 m

B. 7 m

C. 5 m

D. 2 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q3/5 • NEAREST RELATED

1485

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A rectangle that is 6 by 8 has a diagonal of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q4/5 • NEAREST RELATED

1486

QUESTION ID

S48-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A cube with edges of 1 has a diagonal from one corner to the opposite corner of length 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q5/5 • NEAREST RELATED

1487

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its length is √3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 48 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 48 • 7.G.B.6

FORWARD 2

Apply geometric methods to solve design problems • Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a box shaped like a cube with a volume of 64 cubic inches.

What is the edge length, and how much cardboard covers the box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q1/5

1489

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 inches; 96 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q1/5

One gallon of paint covers 350 square feet. The walls of a room total 1,225 square feet.

How many whole gallons should be bought?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q2/5

1490

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 3.5 gallons are needed, so 4 whole gallons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q2/5

Each box has a volume of 24 cubic centimeters.

Which box uses the least cardboard?

A. 1 cm × 4 cm × 6 cm

B. 1 cm × 1 cm × 24 cm

C. 2 cm × 3 cm × 4 cm

D. 1 cm × 2 cm × 12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q3/5

1491

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 cm × 3 cm × 4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q3/5

True or false?

Among rectangular boxes with the same volume, a cube has the least surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q4/5

1492

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q4/5

True or false?

Two boxes with the same volume always have the same surface area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q5/5

1493

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

Grade 9 (High School) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 48 • FORWARD 2 • Q5/5

SET 49 • 7.SP.A.1 • STATISTICS & PROBABILITY

Understand random sampling and representative samples

MAIN	Grade 7 main question	7.SP.A.1
BACKWARD 1	Recognize a statistical question	6.SP.A.1
BACKWARD 2	Report the number of observations	6.SP.B.5.a
FORWARD 1	Summarize survey data in a two-way table — nearest related (no…	8.SP.A.4
FORWARD 2	Recognize surveys, experiments, and observational studies	HSS-IC.B.3

SET 49 • 7.SP.A.1

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know students' favorite lunch.

Which sample is most likely to represent all the students in the school?

A. 20 students in one 7th grade class

B. 20 students on the basketball team

C. The first 20 students in the lunch line

D. 20 students chosen at random from the school list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q1/5

1496

QUESTION ID

S49-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. 20 students chosen at random from the school list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q1/5

A town surveys people at a gym about how often they exercise.

Is the sample likely to represent the whole town? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q2/5

1497

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. People at a gym probably exercise more than people in the whole town, so the sample is biased.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe one way to choose a random sample of 30 students from a school of 600 students.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q3/5

1498

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

Number the students 1 to 600 and use a random number generator to choose 30 different numbers.

GRADING NOTE Any method that gives every student the same chance of being chosen is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q3/5

True or false?

In a random sample, every member of the population has the same chance of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q4/5

1499

QUESTION ID

S49-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q4/5

True or false?

A large sample always represents the population, even if it is not chosen at random.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q5/5

1500

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 49 • MAIN • Q5/5

SET 49 • 7.SP.A.1

BACKWARD 1

Recognize a statistical question • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How tall are the students in my class?" a statistical question? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q1/5

1502

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The answers will vary from student to student.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q1/5

Which is a statistical question?

A. How many hours do 7th graders sleep on school nights?

B. What is 5 × 8?

C. How tall is the school flagpole?

D. What day is it today?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q2/5

1503

QUESTION ID

S49-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. How many hours do 7th graders sleep on school nights?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite "How old am I?" as a statistical question.

Question: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q3/5

1504

QUESTION ID

S49-B1-Q3

TYPE

Short answer

CORRECT ANSWER

How old are the students in my school?

GRADING NOTE Any question whose answers vary, such as one about a group's ages, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

"What are the shoe sizes of the players on the team?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q4/5

1505

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

"How many days are in March?" is a statistical question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q5/5

1506

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 7.SP.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

How many observations does the dot plot show?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q1/5

1508

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q1/5

A survey asks 25 students how many pets they have. Each student gives one answer.

How many observations are there?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q2/5

1509

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q2/5

A table lists the heights of 18 plants, one height for each plant.

How many observations are there?

A. 1

B. 36

C. 18

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q3/5

1510

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q3/5

True or false?

Each dot on a dot plot shows one observation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q4/5

1511

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q4/5

True or false?

A data set of 12 test scores has 24 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q5/5

1512

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 7.SP.A.1

FORWARD 1

Summarize survey data in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What fraction of the surveyed students play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q1/5 • NEAREST RELATED

1514

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 40/80 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What fraction of all surveyed students have a job and play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q2/5 • NEAREST RELATED

1515

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/20

GRADING NOTE 12/80, or 0.15, is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Among students who have a job, what fraction play a sport?

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. $\frac{3}{10}$ **B.** $\frac{3}{5}$ **C.** $\frac{2}{5}$ **D.** $\frac{3}{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q3/5 • NEAREST RELATED

1516

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, 56% of the students with no job play a sport.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q4/5 • NEAREST RELATED

1517

QUESTION ID

S49-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, more than half of all the students have a job.

	Plays a sport	No sport	Total
Has a job	12	18	30
No job	28	22	50
Total	40	40	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q5/5 • NEAREST RELATED

1518

QUESTION ID

S49-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 49 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 49 • 7.SP.A.1

FORWARD 2

Recognize surveys, experiments, and observational studies • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 50 plants to get a new plant food and 50 plants to get none.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q1/5

1520

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

An experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q1/5

A study records the screen time of 200 teens without changing anything they do.

Is this a survey, an experiment, or an observational study?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q2/5

1521

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

An observational study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q2/5

Why is random assignment used in an experiment?

A. To avoid collecting data

B. To make the sample larger

C. To make the groups alike, so differences can be linked to the treatment

D. To let people choose their own group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q3/5

1522

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. To make the groups alike, so differences can be linked to the treatment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q3/5

True or false?

A survey asks a sample of people questions in order to estimate facts about a population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q4/5

1523

QUESTION ID

S49-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q4/5

True or false?

An observational study can prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q5/5

1524

QUESTION ID

S49-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use data from a random sample to draw inferences about a population

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find what percent a part is of a whole	**6.RP.A.3.c**
BACKWARD 2	Find a missing value in equivalent ratios	**6.RP.A.3.a**
FORWARD 1	Use a linear model to make a prediction — nearest related (no di…	**8.SP.A.3**
FORWARD 2	Use a margin of error to describe a population estimate	**HSS-IC.B.4**

SET 50 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 50 students, 18 prefer the art club.

Predict how many of the 800 students in the school prefer the art club.

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q1/5

1527

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q1/5

In a random sample of 40 light bulbs, 2 do not work.

Estimate how many bulbs in a shipment of 2,000 do not work.

Bulbs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q2/5

1528

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q2/5

In a random sample of 60 voters, 33 support a plan.

What is the best estimate of the percent of all voters who support it?

A. 45%

B. 33%

C. 60%

D. 55%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q3/5

1529

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 55%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q3/5

True or false?

In a random sample of 25 fish from a lake, 5 are trout. A good estimate is that 20% of the fish in the lake are trout.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q4/5

1530

QUESTION ID

S50-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q4/5

True or false?

In a random sample of 30 homes, 9 have a dog. A good estimate is that 900 of 1,500 homes have a dog.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q5/5

1531

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A good estimate is 450 homes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 50 • MAIN • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 1

Find what percent a part is of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

18 out of 50 is what percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q1/5

1533

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q1/5

10 is what percent of 40?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q2/5

1534

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

7 out of 20 is what percent?

A. 35%

B. 7%

C. 20%

D. 70%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q3/5

1535

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 35%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q3/5

True or false?

12 out of 48 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q4/5

1536

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

9 out of 30 is 9%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q5/5

1537

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 out of 30 is 30%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 7.SP.A.2

BACKWARD 2

Find a missing value in equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Find the missing value.

2 : 5 = ? : 100

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q1/5

1539

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q1/5

Find the missing value.

$4 : 9 = 28 : ?$

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q2/5

1540

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 6 : 15?

A. 3 : 5

B. 6 : 5

C. 2 : 5

D. 1 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q3/5

1541

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 : 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

5 : 8 = 25 : 40

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q4/5

1542

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$3 : 7 = 9 : 14$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q5/5

1543

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 7.SP.A.2

FORWARD 1

Use a linear model to make a prediction • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A line of fit is y = 2.5x + 40, where x is hours studied and y is a test score.

Predict the score for 8 hours of study.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q1/5 • NEAREST RELATED

1545

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q1/5 • NEAREST RELATED

A line of fit is y = 2.5x + 40, where x is hours studied and y is a test score.

What does 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q2/5 • NEAREST RELATED

1546

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Each extra hour of study goes with a score about 2.5 points higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q2/5 • NEAREST RELATED

A line of fit is y = −3x + 90.

What does it predict for x = 10?

A. 120

B. 93

C. 87

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q3/5 • NEAREST RELATED

1547

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

For the line of fit y = 0.5x + 12, the predicted y-value at x = 0 is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q4/5 • NEAREST RELATED

1548

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

The line of fit $y = 4x + 1$ predicts $y = 41$ when $x = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q5/5 • NEAREST RELATED

1549

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It predicts $y = 37$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 50 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 50 • 7.SP.A.2

FORWARD 2

Use a margin of error to describe a population estimate • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample shows 52% support for a plan, with a margin of error of 4%.

What interval of values is likely for the whole population?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q1/5

1551

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

48% to 56%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q1/5

A random sample has a mean of 6.8 hours with a margin of error of 0.5 hour.

What interval of values is likely for the population mean?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q2/5

1552

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6.3 to 7.3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q2/5

As a random sample gets larger, what usually happens to the margin of error?

A. It gets larger.

B. It becomes 0 once the sample is over 10.

C. It gets smaller.

D. It stays the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q3/5

1553

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. It gets smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

A margin of error describes how far a sample estimate is likely to be from the population value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q4/5

1554

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

A poll result of 52% with a 4% margin of error proves that more than half of the population supports the plan.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q5/5

1555

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The interval 48% to 56% includes values below 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 7.SP.A.2 • STATISTICS & PROBABILITY

Use several random samples to judge how much estimates vary

MAIN	Grade 7 main question	**7.SP.A.2**
BACKWARD 1	Find the mean of a data set	**6.SP.B.5.c**
BACKWARD 2	Find the range as a measure of variation	**6.SP.A.3**
FORWARD 1	Describe the association shown in a scatter plot — nearest relate…	**8.SP.A.1**
FORWARD 2	Decide whether a result is consistent with a model by simulation	**HSS-IC.A.2**

SET 51 • 7.SP.A.2

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.A.2

Four random samples of 20 students give these mean hours of sleep: 7.8, 8.2, 8.0, and 8.4.

What is a good estimate of the mean for all the students? Use the mean of the samples.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q1/5

1558

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

8.1 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q1/5

Four random samples give mean estimates of 7.8, 8.2, 8.0, and 8.4 hours.

What is the range of the estimates?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q2/5

1559

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.6 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q2/5

Three random samples of 50 people show that 22%, 26%, and 24% prefer blue.

Which conclusion is best?

A. Exactly 22% of the population prefers blue.

B. Exactly 26% of the population prefers blue.

C. The samples are useless because they differ.

D. About 24% of the population prefers blue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q3/5

1560

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. About 24% of the population prefers blue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q3/5

True or false?

Different random samples from the same population usually give somewhat different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q4/5

1561

QUESTION ID

S51-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q4/5

True or false?

Small random samples usually give estimates that vary less than large random samples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q5/5

1562

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 51 • MAIN • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 7, 9, 10

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q1/5

1564

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q1/5

Find the mean.

12, 15, 18

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q2/5

1565

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q2/5

What is the mean of 3, 3, 6, and 8?

A. 5

B. 6

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q3/5

1566

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q3/5

True or false?

The mean of 2, 4, 6, and 8 is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q4/5

1567

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q4/5

True or false?

The mean of 10, 20, and 60 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q5/5

1568

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The mean is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 7.SP.A.2

BACKWARD 2

Find the range as a measure of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the range.

12, 18, 9, 15

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q1/5

1570

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q1/5

Find the range.

3.5, 7, 4.25, 6

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q2/5

1571

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which data set has the greatest range?

A. 1, 3, 2

B. 7, 9, 8

C. 4, 6, 5

D. 2, 10, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q3/5

1572

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2, 10, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q3/5

True or false?

The range of 20, 25, and 30 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q4/5

1573

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q4/5

True or false?

A data set whose values are all the same has a range of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q5/5

1574

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its range is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 7.SP.A.2

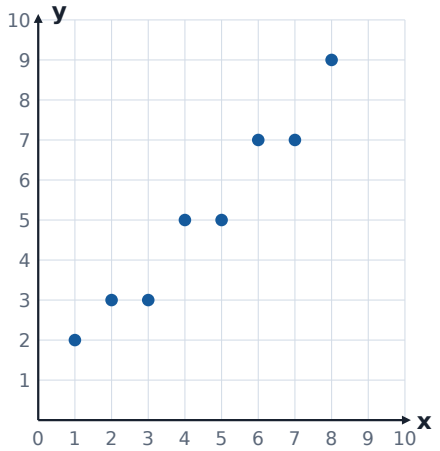
FORWARD 1

Describe the association shown in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

What kind of association does the scatter plot show?

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q1/5 • NEAREST RELATED

1576

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot shows hours of practice (x) and free throws made (y). The points rise from left to right in a roughly straight pattern.

Describe the association.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q2/5 • NEAREST RELATED

1577

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q2/5 • NEAREST RELATED

A scatter plot shows the age of a car (x) and its value (y).

What association is most likely?

A. Negative association

B. Every point on one line

C. No association

D. Positive association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q3/5 • NEAREST RELATED

1578

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Negative association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a scatter plot with no association, the points show no clear trend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q4/5 • NEAREST RELATED

1579

QUESTION ID

S51-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Points that fall from left to right in a scatter plot show a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q5/5 • NEAREST RELATED

1580

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 51 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 51 • 7.SP.A.2

FORWARD 2

Decide whether a result is consistent with a model by simulation • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A coin is said to be fair. In 100 simulated sets of 20 fair flips, 15 or more heads happened only 2 times. A real coin gives 15 heads in 20 flips.

Is the fair-coin model believable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q1/5

1582

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Probably not. A result that extreme happens only about 2% of the time with a fair coin, so the result casts doubt on the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q1/5

Simulations of a spinner model usually give 3 to 7 reds in 20 spins. A real spinner gives 5 reds in 20 spins.

What does this show?

A. The spinner is proven unfair.

B. The result is impossible.

C. The result is consistent with the model.

D. The model is proven wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q2/5

1583

QUESTION ID

S51-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The result is consistent with the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q2/5

In 100 simulations of a model, a certain result happened 40 times.

Is that result unusual for the model? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q3/5

1584

QUESTION ID

S51-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No. It happened about 40% of the time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q3/5

True or false?

A result that rarely happens in simulations of a model suggests that the model may be wrong.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q4/5

1585

QUESTION ID

S51-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q4/5

True or false?

A result that happens even once in 1,000 simulations is a common result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q5/5

1586

QUESTION ID

S51-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 51 • FORWARD 2 • Q5/5

SET 52 • 7.SP.B.3 • STATISTICS & PROBABILITY

Describe how much two distributions overlap using the difference in means as a multiple of the MAD

MAIN	Grade 7 main question	**7.SP.B.3**
BACKWARD 1	Find the mean absolute deviation (MAD)	**6.SP.B.5.c**
BACKWARD 2	Describe the shape of a distribution	**6.SP.A.2**
FORWARD 1	Compare groups using relative frequencies in a two-way table —…	**8.SP.A.4**
FORWARD 2	Compare the center and spread of two data sets	**HSS-ID.A.2**

SET 52 • 7.SP.B.3

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.3

Team A's heights have a mean of 160 cm. Team B's heights have a mean of 172 cm. Both teams have a MAD of 4 cm.

The difference in the means is how many MADs?

MADs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q1/5

1589

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q1/5

Class 1 has a mean quiz score of 82. Class 2 has a mean of 76. Each class has a MAD of 3.

Write the difference in means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q2/5

1590

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

2 MADs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q2/5

Two data sets have means of 50 and 54. Each has a MAD of 8.

How much do the distributions overlap?

A. A lot, because the difference is only half of a MAD

B. Not at all, because the difference is 4 MADs

C. Very little, because the difference is 8 MADs

D. Not at all, because the means are different

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q3/5

1591

QUESTION ID

S52-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A lot, because the difference is only half of a MAD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q3/5

True or false?

When the difference in means is several MADs, the two distributions overlap very little.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q4/5

1592

QUESTION ID

S52-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q4/5

True or false?

Two data sets have means of 30 and 40 and a MAD of 10 each. The difference in means is 10 MADs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q5/5

1593

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 1 MAD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 52 • MAIN • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 1

Find the mean absolute deviation (MAD) • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the MAD.

2, 4, 6, 8

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q1/5

1595

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q1/5

Find the MAD.

5, 5, 9, 13

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q2/5

1596

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the MAD of 10, 10, 10, 10?

A. 0

B. 40

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q3/5

1597

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

The MAD of 1, 3, 5, 7 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q4/5

1598

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

The MAD of 0 and 10 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q5/5

1599

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The MAD is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 7.SP.B.3

BACKWARD 2

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

A dot plot has most of its dots on the left and a long tail to the right.

How is its shape described?

A. Uniform

B. Skewed right

C. Skewed left

D. Symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q1/5

1601

QUESTION ID

S52-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is the distribution in the dot plot symmetric?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q2/5

1602

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q2/5

A dot plot has a tall stack of dots in the middle and short, equal stacks on each side.

What is the shape of the distribution?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q3/5

1603

QUESTION ID

S52-B2-Q3

TYPE

Short answer

CORRECT ANSWER

Symmetric (mound-shaped)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A distribution that looks the same on both sides of its center is symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q4/5

1604

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A dot plot with all of its dots stacked above one value is skewed left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q5/5

1605

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 7.SP.B.3

FORWARD 1

Compare groups using relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows how students get to school.

What percent of sixth graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q1/5 • NEAREST RELATED

1607

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows how students get to school.

What percent of seventh graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q2/5 • NEAREST RELATED

1608

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows how students get to school.

Is there evidence that grade and walking are related?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

A. No, because each grade has 60 students.

B. No, because more students ride than walk in both grades.

C. Yes, because 40% of sixth graders walk but only 25% of seventh graders do.

D. Yes, because 24 is greater than 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q3/5 • NEAREST RELATED

1609

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes, because 40% of sixth graders walk but only 25% of seventh graders do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

Relative frequencies make it possible to compare groups of different sizes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q4/5 • NEAREST RELATED

1610

QUESTION ID

S52-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If 40% of sixth graders and 40% of seventh graders walk, there is a strong relationship between grade and walking.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q5/5 • NEAREST RELATED

1611

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 52 • 7.SP.B.3

FORWARD 2

Compare the center and spread of two data sets • Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Data set A has a mean of 20 and a standard deviation of 2. Data set B has a mean of 20 and a standard deviation of 6.

Which data set is more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q1/5

1613

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Data set B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q1/5

Data set A has a median of 45 and an IQR of 10. Data set B has a median of 52 and an IQR of 10.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q2/5

1614

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

B's median is 7 greater; the spreads (IQRs) are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q2/5

Which measure of spread is best for a skewed data set with outliers?

A. Interquartile range

B. Mean

C. Range

D. Standard deviation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q3/5

1615

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q3/5

True or false?

Standard deviation measures how spread out the data are around the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q4/5

1616

QUESTION ID

S52-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q5/5

1617

QUESTION ID

S52-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 7.SP.B.4 • STATISTICS & PROBABILITY

Compare two populations using measures of center and variability from random samples

MAIN	Grade 7 main question	**7.SP.B.4**
BACKWARD 1	Find the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Read a box plot	**6.SP.B.4**
FORWARD 1	Identify clusters and outliers in a scatter plot — nearest related (n…	**8.SP.A.1**
FORWARD 2	Interpret differences in center and spread, including the effect of…	**HSS-ID.A.3**

SET 53 • 7.SP.B.4

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.4

Random samples of quiz scores: Class A has a median of 78 and an IQR of 10. Class B has a median of 70 and an IQR of 10.

Which class tends to score higher, and by how much (median)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q1/5

1620

QUESTION ID

S53-M-Q1

TYPE

Short answer

CORRECT ANSWER

Class A, by 8 points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q1/5

A random sample of 7th graders has a mean of 6.5 hours of screen time and a MAD of 1.2. A random sample of 8th graders has a mean of 7.7 hours and a MAD of 1.2.

Write the difference in means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q2/5

1621

QUESTION ID

S53-M-Q2

TYPE

Short answer

CORRECT ANSWER

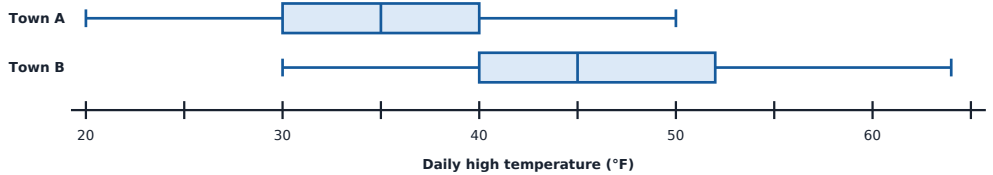
1 MAD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q2/5

The box plots show random samples of daily high temperatures in two towns.

Which statement is true?



A. The two towns have the same range.

B. The IQRs differ by 10 degrees.

C. The median for Town B is 10 degrees greater than the median for Town A.

D. Town A has the greater median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q3/5

1622

QUESTION ID

S53-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The median for Town B is 10 degrees greater than the median for Town A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q3/5

True or false?

Random samples are a good way to compare two populations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q4/5

1623

QUESTION ID

S53-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q4/5

True or false?

Sample A has a greater mean than Sample B. So every value in Sample A is greater than every value in Sample B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q5/5

1624

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q5/5

SET 53 • 7.SP.B.4

BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

A data set has Q1 = 12 and Q3 = 20.

What is the IQR?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q1/5

1626

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q1/5

Find the IQR.

2, 4, 6, 8, 10, 12, 14, 16

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q2/5

1627

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q2/5

A box plot has Q1 = 30 and Q3 = 55.

What is the IQR?

A. 30

B. 85

C. 25

D. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q3/5

1628

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q3/5

True or false?

The IQR is the difference between Q3 and Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q4/5

1629

QUESTION ID

S53-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q4/5

True or false?

A data set with Q1 = 7 and Q3 = 15 has an IQR of 22.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q5/5

1630

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The IQR is 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q5/5

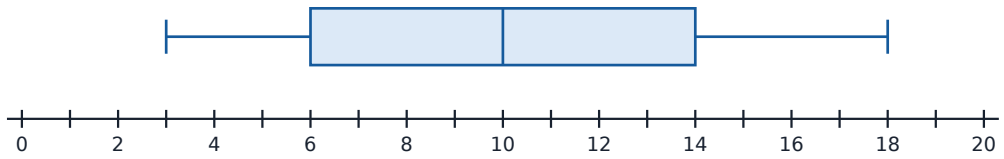
SET 53 • 7.SP.B.4

BACKWARD 2

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

What is the median of the data in the box plot?



Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

1632

QUESTION ID

S53-B2-Q1

TYPE

Short answer

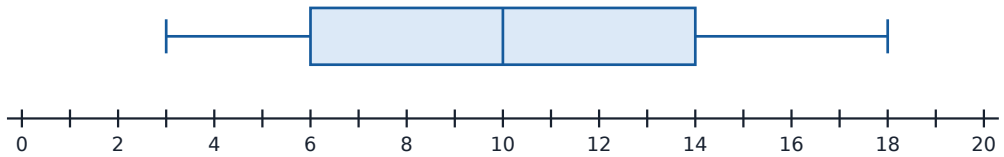
CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

What is the first quartile (Q1) of the data in the box plot?



Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

1633

QUESTION ID

S53-B2-Q2

TYPE

Short answer

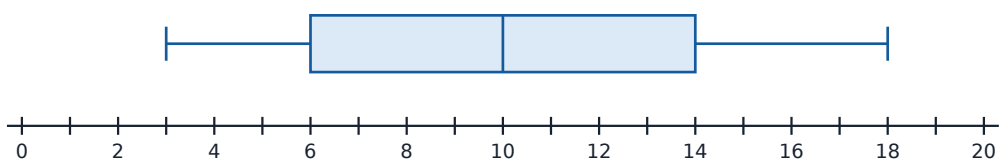
CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

What is the maximum of the data in the box plot?



A. 14

B. 18

C. 3

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

1634

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

The box of a box plot shows the middle half of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

1635

QUESTION ID

S53-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

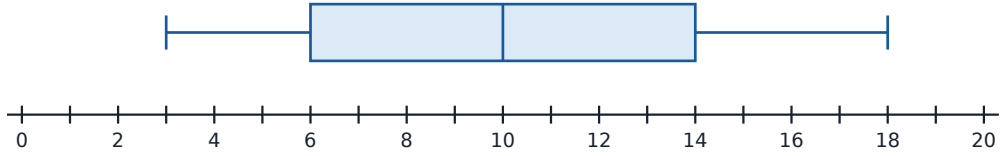
ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The minimum of the data in the box plot is 6.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

1636

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minimum is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 7.SP.B.4

FORWARD 1

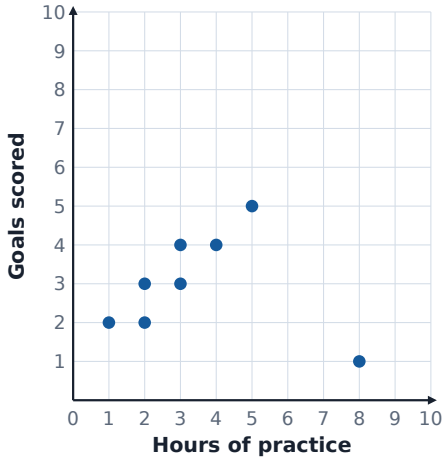
Identify clusters and outliers in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

Which point in the scatter plot is an outlier?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q1/5 • NEAREST RELATED

1638

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(8, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot has a group of points close together.

What is the group called?

A. A cluster

B. An outlier

C. An intercept

D. A slope

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q2/5 • NEAREST RELATED

1639

QUESTION ID

S53-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

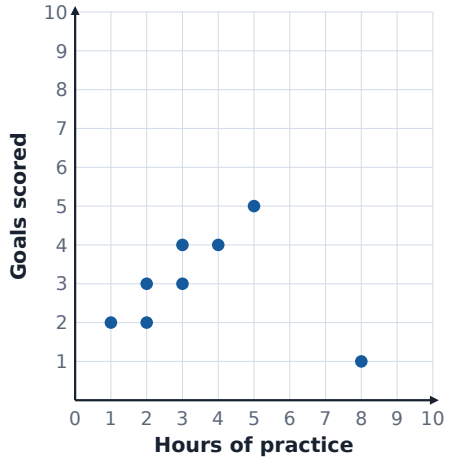
A. A cluster

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q2/5 • NEAREST RELATED

Leaving out the outlier, describe the association in the scatter plot.

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q3/5 • NEAREST RELATED

1640

QUESTION ID

S53-F1-Q3

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

An outlier is a point that is far from the overall pattern of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q4/5 • NEAREST RELATED

1641

QUESTION ID

S53-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Every scatter plot has exactly one outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q5/5 • NEAREST RELATED

1642

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 53 • 7.SP.B.4

FORWARD 2

Interpret differences in center and spread, including the effect of outliers • Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set is 4, 5, 5, 6, 7, 40.

Which better describes its center, the mean or the median? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q1/5

1644

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The median, because the outlier 40 pulls the mean up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q1/5

A data set is 4, 5, 5, 6, 7, 40.

What is the mean with the outlier, and what is the mean without it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q2/5

1645

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 11.2 with the outlier; 5.4 without it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q2/5

Which measure is affected most by an outlier?

A. Mean

B. IQR

C. Mode

D. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q3/5

1646

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q3/5

True or false?

The median is resistant to outliers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q4/5

1647

QUESTION ID

S53-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q4/5

True or false?

Removing a very large outlier makes the mean larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q5/5

1648

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q5/5

SET 54 • 7.SP.C.5 • STATISTICS & PROBABILITY

Understand probability as a number from 0 to 1 that shows how likely an event is

MAIN	Grade 7 main question	**7.SP.C.5**
BACKWARD 1	Compare two fractions	**4.NF.A.2**
BACKWARD 2	Place fractions on a number line from 0 to 1	**3.NF.A.2**
FORWARD 1	Find relative frequencies in a two-way table — nearest related (no…	**8.SP.A.4**
FORWARD 2	Describe events as subsets of a sample space	**HSS-CP.A.1**

SET 54 • 7.SP.C.5

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.5

Which probability describes an event that is likely to happen?

A. 0.1

B. 0.8

C. 0.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q1/5

1651

QUESTION ID

S54-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 5 red marbles and no other marbles.

What is the probability of drawing a red marble?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q2/5

1652

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

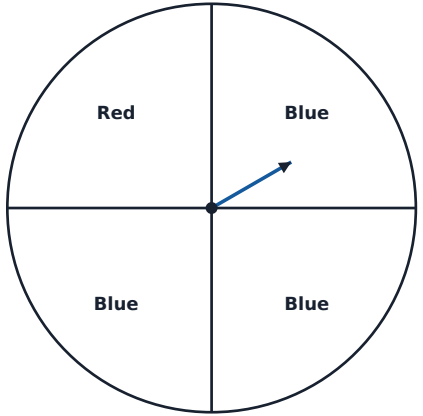
Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The spinner has 4 equal sections.

Is landing on Blue likely, unlikely, or neither likely nor unlikely?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q3/5

1653

QUESTION ID

S54-M-Q3

TYPE

Short answer

CORRECT ANSWER

Likely

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q3/5

True or false?

An event with a probability of $\frac{1}{2}$ is just as likely to happen as not to happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q4/5

1654

QUESTION ID

S54-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A probability of 1.3 means that an event is very likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q5/5

1655

QUESTION ID

S54-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Probabilities are from 0 to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 1

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is greater, $\frac{3}{8}$ or $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q1/5

1657

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q1/5

Write >, <, or =.

$\frac{5}{10}$ ___ $\frac{1}{2}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q2/5

1658

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

=

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which fraction is less than $\frac{1}{2}$?

A. $\frac{3}{4}$

B. $\frac{5}{8}$

C. $\frac{3}{8}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q3/5

1659

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

$\frac{7}{12} > \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q4/5

1660

QUESTION ID

S54-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

$\frac{2}{5} > \frac{3}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q5/5

1661

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 2

Place fractions on a number line from 0 to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

Plot $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q1/5

1663

QUESTION ID

S54-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A point at the third tick mark after 0 (3/4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q1/5

A number line from 0 to 1 is split into 4 equal parts.

What fraction is at the first tick mark after 0?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q2/5

1664

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q2/5

Which fraction is halfway between 0 and 1 on a number line?

A. $\frac{1}{2}$

B. $\frac{1}{4}$

C. $\frac{2}{1}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q3/5

1665

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q3/5

True or false?

$\frac{4}{4}$ is at the same point as 1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q4/5

1666

QUESTION ID

S54-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3}$ is farther from 0 than $\frac{2}{3}$ is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q5/5

1667

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 7.SP.C.5

FORWARD 1

Find relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

Among Grade 9 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

1669

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 18/60 or 30% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

Among Grade 10 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

1670

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 12/40 or 30% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What is the relative frequency of all students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.3 **B.** 0.18 **C.** 0.12 **D.** 0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

1671

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, students in both grades play music at the same rate.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

1672

QUESTION ID

S54-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the table, Grade 9 students are more likely than Grade 10 students to play music, because 18 > 12.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

1673

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both rates are 30%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • 7.SP.C.5

FORWARD 2

Describe events as subsets of a sample space • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A number cube is rolled.

List the outcomes in the event "an even number."

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q1/5

1675

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q1/5

A number cube is rolled.

List the outcomes in the complement of the event "a number greater than 4."

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q2/5

1676

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q2/5

A coin is flipped twice.

What is the sample space?

A. HH, HT, TH, TT

B. HH, TT

C. HH, HT, TT

D. H, T

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q3/5

1677

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. HH, HT, TH, TT

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q3/5

True or false?

The probability of an event plus the probability of its complement is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q4/5

1678

QUESTION ID

S54-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q4/5

True or false?

When a number cube is rolled, the events "even number" and "odd number" together include no outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q5/5

1679

QUESTION ID
S54-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Together they include all six outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 7.SP.C.6 • STATISTICS & PROBABILITY

Estimate a probability from relative frequency and make predictions

MAIN	Grade 7 main question	**7.SP.C.6**
BACKWARD 1	Use a rate to make a prediction	**6.RP.A.3.b**
BACKWARD 2	Compare decimals to thousandths	**5.NBT.A.3.b**
FORWARD 1	Use relative frequencies in a two-way table to make predictions…	**8.SP.A.4**
FORWARD 2	Find conditional probabilities from a two-way table	**HSS-CP.A.4**

SET 55 • 7.SP.C.6

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A spinner landed on green 36 times in 120 spins.

Estimate the probability of landing on green.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q1/5

1682

QUESTION ID

S55-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 36/120 or 3/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q1/5

A basketball player made 42 of her last 60 free throws.

Predict how many she will make in her next 150 free throws.

Free throws: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q2/5

1683

QUESTION ID

S55-M-Q2

TYPE

Short answer

CORRECT ANSWER

105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q2/5

A coin was flipped 200 times and landed heads 94 times.

What is the relative frequency of heads?

A. 0.94

B. 0.47

C. 0.53

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q3/5

1684

QUESTION ID

S55-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q3/5

True or false?

A number cube is rolled 300 times. About 50 of the rolls should be a 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q4/5

1685

QUESTION ID

S55-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q4/5

True or false?

A spinner landed on red 9 out of 10 times. The probability of red must be exactly 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q5/5

1686

QUESTION ID

S55-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Relative frequency only estimates the probability, and 10 spins is a small number of trials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 1

Use a rate to make a prediction • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

A team scores 3 goals every 2 games.

At this rate, how many goals does it score in 10 games?

Goals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q1/5

1688

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q1/5

4 out of every 10 apples in a crate are green.

At this rate, how many of 250 apples are green?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q2/5

1689

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q2/5

A machine fills 6 bottles every 4 seconds.

At this rate, how many bottles does it fill in 30 seconds?

A. 45

B. 20

C. 24

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q3/5

1690

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q3/5

True or false?

At a rate of 2 out of every 5, there are 40 out of every 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q4/5

1691

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q4/5

True or false?

At a rate of 3 out of every 8, there are 30 out of every 64.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q5/5

1692

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are 24 out of every 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 2

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Which is greater, 0.47 or 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q1/5

1694

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write >, <, or =.

0.305 ___ 0.35

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q2/5

1695

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

<

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q2/5

Which decimal is the greatest?

A. 0.059

B. 0.62

C. 0.506

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q3/5

1696

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q3/5

True or false?

0.250 = 0.25

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q4/5

1697

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q4/5

True or false?

0.09 > 0.1

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q5/5

1698

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 7.SP.C.6

FORWARD 1

Use relative frequencies in a two-way table to make predictions • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

Among teens, what is the relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q1/5 • NEAREST RELATED

1700

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.225

GRADING NOTE 9/40 or 22.5% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q1/5 • NEAREST RELATED

Use the adults row of the table.

Predict how many of 200 adults like rain.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Adults: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q2/5 • NEAREST RELATED

1701

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Which group has the greater relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. Adults

B. Teens

C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q3/5 • NEAREST RELATED

1702

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of liking rain among all 100 people is 0.27.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q4/5 • NEAREST RELATED

1703

QUESTION ID

S55-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of disliking rain among adults is 0.42.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q5/5 • NEAREST RELATED

1704

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 42/60 = 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 55 • 7.SP.C.6

FORWARD 2

Find conditional probabilities from a two-way table • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The table shows students' favorite pets. A student is chosen at random.

Find P(dog | Grade 9).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q1/5

1706

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

GRADING NOTE 25/40 or 0.625 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows students' favorite pets. A student is chosen at random.

Find P(Grade 10 | cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q2/5

1707

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4/7

GRADING NOTE 20/35 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q2/5

The table shows students' favorite pets. A student is chosen at random.

Find P(cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. $\frac{1}{2}$ **B.** $\frac{9}{16}$ **C.** $\frac{7}{16}$ **D.** $\frac{3}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q3/5

1708

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 7/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q3/5

True or false?

In the table, P(dog | Grade 10) = $\frac{1}{2}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q4/5

1709

QUESTION ID

S55-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the table, P(Grade 9 | dog) = $\frac{25}{40}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q5/5

1710

QUESTION ID

S55-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE P(Grade 9 | dog) = 25/45 = 5/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q5/5

1710

SET 56 • 7.SP.C.7.a • STATISTICS & PROBABILITY

Develop a uniform probability model and use it to find probabilities

MAIN	Grade 7 main question	**7.SP.C.7.a**
BACKWARD 1	Write equivalent fractions	**4.NF.A.1**
BACKWARD 2	Write a fraction for part of a whole	**3.NF.A.1**
FORWARD 1	Understand a function as a rule that gives exactly one output for e…	**8.F.A.1**
FORWARD 2	Use the addition rule for probability	**HSS-CP.B.7**

SET 56 • 7.SP.C.7.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A number cube has faces numbered 1 to 6.

What is the probability of rolling a number greater than 4?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q1/5

1713

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 2/6 is also correct.

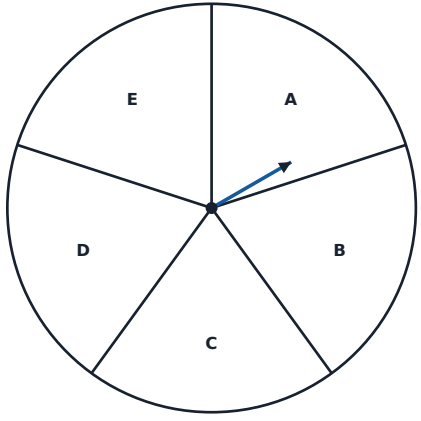
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q1/5

The spinner has 5 equal sections.

What is the probability of landing on a vowel?

Probability: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q2/5

1714

QUESTION ID

S56-M-Q2

TYPE

Short answer

CORRECT ANSWER

2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q2/5

A class has 12 girls and 18 boys. One student is chosen at random.

What is the probability that a girl is chosen?

A. $\frac{2}{5}$

B. $\frac{2}{3}$

C. $\frac{1}{12}$

D. $\frac{3}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q3/5

1715

QUESTION ID

S56-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q3/5

True or false?

In a uniform probability model, every outcome has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q4/5

1716

QUESTION ID

S56-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q4/5

True or false?

A letter is chosen at random from the word MATH.

The probability of choosing a vowel is $\frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q5/5

1717

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The probability is 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q5/5

SET 56 • 7.SP.C.7.a

BACKWARD 1

Write equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write a fraction equivalent to $\frac{3}{12}$ with a denominator of 4.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q1/5

1719

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q1/5

Write a fraction equivalent to $\frac{10}{15}$ with a denominator of 3.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q2/5

1720

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{2}{3}$

B. $\frac{6}{4}$

C. $\frac{4}{12}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q3/5

1721

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q3/5

True or false?

$\frac{5}{10} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q4/5

1722

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{2}{8} = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q5/5

1723

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 7.SP.C.7.a

BACKWARD 2

Write a fraction for part of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A spinner has 8 equal parts. 3 of the parts are red.

What fraction of the spinner is red?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q1/5

1725

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 6 equal-size tiles. 1 tile is blue.

What fraction of the tiles is blue?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q2/5

1726

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q2/5

A pizza is cut into 4 equal slices. 3 slices are eaten.

What fraction of the pizza is eaten?

A. $\frac{3}{7}$

B. $\frac{4}{3}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q3/5

1727

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q3/5

True or false?

One of 5 equal parts is $\frac{1}{5}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q4/5

1728

QUESTION ID

S56-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q4/5

True or false?

Two of 6 equal parts is $\frac{2}{4}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q5/5

1729

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 7.SP.C.7.a

FORWARD 1

Understand a function as a rule that gives exactly one output for each input • Grade 8 • Functions • 8.F.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Is the set of pairs (1, 3), (2, 5), (3, 7) a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q1/5 • NEAREST RELATED

1731

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Each input has exactly one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q1/5 • NEAREST RELATED

Is the set of pairs (2, 4), (2, 6), (3, 8) a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q2/5 • NEAREST RELATED

1732

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 2 has two different outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which table shows a function?

A. x: 1, 1, 2 and y: 3, 5, 7

B. x: 1, 2, 3 and y: 4, 4, 4

C. x: 0, 0, 1 and y: 0, 1, 1

D. x: 5, 5, 5 and y: 1, 2, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q3/5 • NEAREST RELATED

1733

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x: 1, 2, 3 and y: 4, 4, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A probability model gives exactly one probability to each outcome, just as a function gives exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q4/5 • NEAREST RELATED

1734

QUESTION ID

S56-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In a function, one input can have two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q5/5 • NEAREST RELATED

1735

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 56 • 7.SP.C.7.a

FORWARD 2

Use the addition rule for probability • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A card is chosen at random from cards numbered 1 to 10.

Find P(even or greater than 7).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q1/5

1737

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE 6/10 or 0.6 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q1/5

P(A) = 0.4, P(B) = 0.5, and P(A and B) = 0.2.

Find P(A or B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q2/5

1738

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q2/5

A number cube is rolled.

What is P(less than 3 or odd)?

A. $\frac{5}{6}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q3/5

1739

QUESTION ID

S56-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q3/5

True or false?

If A and B cannot both happen, then P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q4/5

1740

QUESTION ID

S56-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(A or B) is always equal to P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q5/5

1741

QUESTION ID

S56-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You subtract P(A and B) when A and B can happen together.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b • STATISTICS & PROBABILITY

Develop a probability model from observed frequencies

MAIN	Grade 7 main question	**7.SP.C.7.b**
BACKWARD 1	Read a scaled bar graph	**3.MD.B.3**
BACKWARD 2	Write a part of a whole as a percent	**6.RP.A.3.c**
FORWARD 1	Construct a two-way table from collected data — nearest related…	**8.SP.A.4**
FORWARD 2	Summarize categorical data with relative frequencies	**HSS-ID.B.5**

SET 57 • 7.SP.C.7.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

A bottle cap is tossed 80 times. It lands top-up 52 times.

Estimate the probability that it lands top-up.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q1/5

1744

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.65

GRADING NOTE 52/80 or 13/20 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner with three unequal sections is spun 200 times: red 90 times, blue 70 times, and green 40 times.

Write an estimated probability for each color.

Probabilities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q2/5

1745

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

Red 0.45, blue 0.35, green 0.2

GRADING NOTE Equivalent fractions or percents are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q2/5

A spinner landed on red 90 times, blue 70 times, and green 40 times in 200 spins.

Which is the best prediction for 1,000 spins?

A. About 900 reds

B. About 333 reds

C. About 90 reds

D. About 450 reds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q3/5

1746

QUESTION ID

S57-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. About 450 reds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q3/5

True or false?

A probability estimated from observed frequencies can change when more trials are done.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q4/5

1747

QUESTION ID

S57-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q4/5

True or false?

A model built from 10 tosses is just as reliable as a model built from 1,000 tosses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q5/5

1748

QUESTION ID

S57-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q5/5

SET 57 • 7.SP.C.7.b

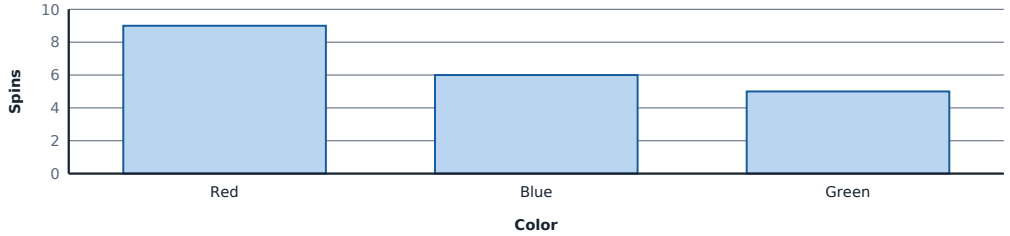
BACKWARD 1

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

ANSWER KEY Reduced copy of the question page, with its answer

The bar graph shows the results of some spins.

How many spins landed on red?



Spins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q1/5

1750

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

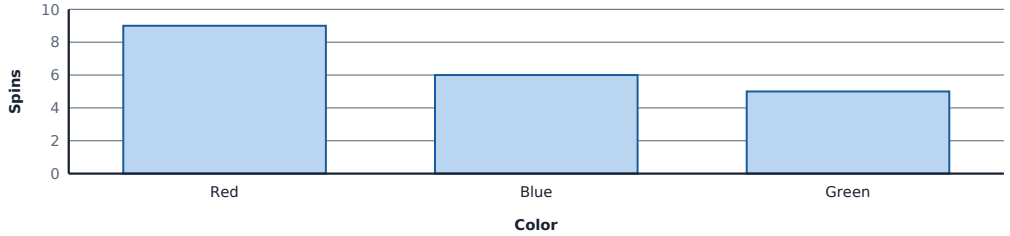
Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q1/5

1750

ANSWER KEY Reduced copy of the question page, with its answer

The bar graph shows the results of some spins.

How many more spins landed on red than on green?



Spins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q2/5

1751

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

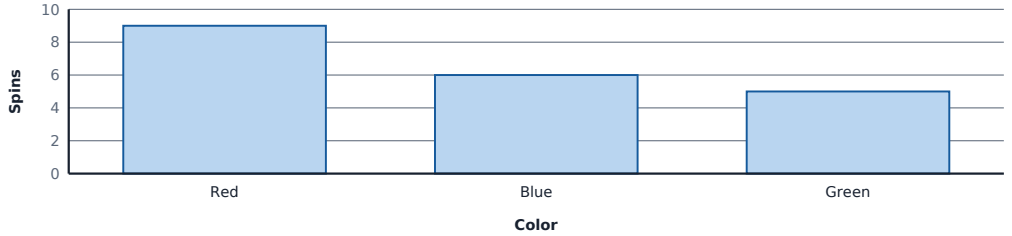
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The bar graph shows the results of some spins.

How many spins are shown in all?



A. 9 **B.** 11 **C.** 15 **D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q3/5

1752

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

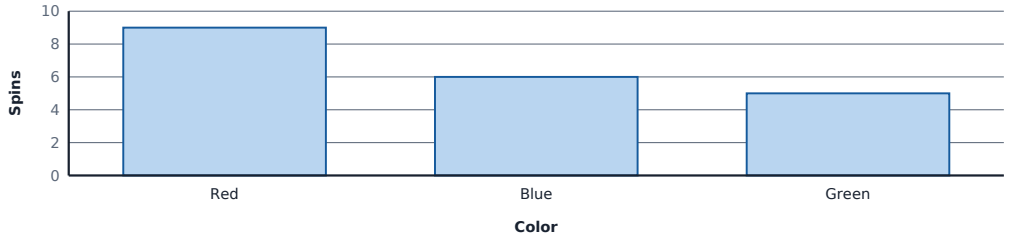
D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q3/5

True or false?

The bar graph shows that blue was spun 6 times.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q4/5

1753

QUESTION ID

S57-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

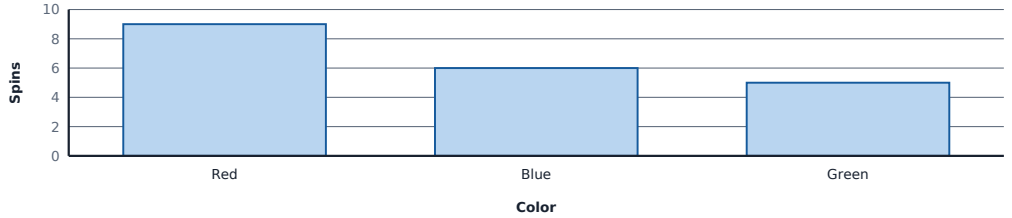
ID S57-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The bar graph shows that green was spun more times than blue.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q5/5

1754

QUESTION ID

S57-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 7.SP.C.7.b

BACKWARD 2

Write a part of a whole as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Write 45 out of 60 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q1/5

1756

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q1/5

Write 13 out of 50 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q2/5

1757

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

26%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q2/5

What percent is 72 out of 120?

A. 72%

B. 48%

C. 60%

D. 40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q3/5

1758

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q3/5

True or false?

9 out of 36 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q4/5

1759

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

14 out of 40 is 14%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q5/5

1760

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 14 out of 40 is 35%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b

FORWARD 1

Construct a two-way table from collected data • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

50 students were surveyed. 30 have a phone. 18 of the students with a phone also have a tablet. 8 of the students without a phone have a tablet.

How many students have neither a phone nor a tablet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q1/5 • NEAREST RELATED

1762

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

50 students were surveyed. 30 have a phone. 18 of the students with a phone also have a tablet. 8 of the students without a phone have a tablet.

How many students have a tablet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q2/5 • NEAREST RELATED

1763

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q2/5 • NEAREST RELATED

A two-way table has the rows "Phone" and "No phone" and the columns "Tablet" and "No tablet."

How many cells show counts for both variables (not totals)?

A. 2

B. 9

C. 6

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q3/5 • NEAREST RELATED

1764

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A two-way table shows how often each combination of two categories happens.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q4/5 • NEAREST RELATED

1765

QUESTION ID

S57-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In a two-way table, the row totals and the column totals add to different grand totals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q5/5 • NEAREST RELATED

1766

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 57 • 7.SP.C.7.b

FORWARD 2

Summarize categorical data with relative frequencies • Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What is the joint relative frequency of Grade 9 students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q1/5

1768

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.18

GRADING NOTE 18/100 or 18% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What is the marginal relative frequency of students who do not play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q2/5

1769

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 70/100 or 70% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q2/5

The table shows the results of a survey.

What is the conditional relative frequency of playing music among Grade 10 students?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.7 **B.** 0.4 **C.** 0.3 **D.** 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q3/5

1770

QUESTION ID

S57-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

A marginal relative frequency uses a row total or a column total divided by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q4/5

1771

QUESTION ID

S57-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

A conditional relative frequency always divides by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q5/5

1772

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 7.SP.C.7 • STATISTICS & PROBABILITY

Compare probabilities from a model with observed frequencies and explain any differences

MAIN	Grade 7 main question	**7.SP.C.7**
BACKWARD 1	Find a fraction of a whole number	**5.NF.B.4.a**
BACKWARD 2	Divide whole numbers to find equal shares	**4.NBT.B.6**
FORWARD 1	Compare a linear model's predictions with observed data — near…	**8.SP.A.3**
FORWARD 2	Decide whether observed results are consistent with a model	**HSS-IC.A.2**

SET 58 • 7.SP.C.7

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7

ANSWER KEY Reduced copy of the question page, with its answer

A fair number cube is rolled 60 times.

How many times would you expect to roll a 5?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q1/5

1775

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q1/5

A fair coin is flipped 100 times and lands heads 61 times.

Compare this with the model. Give a possible reason for the difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q2/5

1776

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

The model predicts about 50 heads. Random variation can explain some of the difference, but 61 heads might also mean the coin is not fair.

GRADING NOTE Any answer that compares with the expected 50 and gives a reasonable cause (chance variation or an unfair coin).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q2/5

A spinner has 4 equal sections, one of them yellow. In 80 spins, it lands on yellow 35 times.

Which conclusion is best?

A. The model predicts 40 yellows.

B. Yellow came up much more than the expected 20 times, so the spinner may not be fair.

C. Yellow should come up 35 times in every 80 spins.

D. The results match the model exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q3/5

1777

QUESTION ID

S58-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Yellow came up much more than the expected 20 times, so the spinner may not be fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q3/5

True or false?

A good probability model does not have to match observed frequencies exactly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q4/5

1778

QUESTION ID

S58-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q4/5

True or false?

A coin that lands heads 7 times in 10 flips must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q5/5

1779

QUESTION ID

S58-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 1

Find a fraction of a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Find the value.

$\frac{1}{6} \times 60$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

1781

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\frac{1}{4}$ of 80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

1782

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

What is $\frac{2}{5}$ of 50?

A. 25

B. 125

C. 20

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

1783

QUESTION ID

S58-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3}$ of 90 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

1784

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{3}{4}$ of 40 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

1785

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3/4 of 40 is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 2

Divide whole numbers to find equal shares • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

240 ÷ 4

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q1/5

1787

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q1/5

Divide.

300 ÷ 6

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q2/5

1788

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q2/5

Divide.

1,000 ÷ 8

A. 105

B. 150

C. 125

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q3/5

1789

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q3/5

True or false?

500 ÷ 5 = 100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q4/5

1790

QUESTION ID

S58-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q4/5

True or false?

720 ÷ 3 = 24

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q5/5

1791

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 7.SP.C.7

FORWARD 1

Compare a linear model's predictions with observed data • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A model is y = 2x + 1. At x = 4, the observed value is y = 11.

How far is the observed value from the prediction?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q1/5 • NEAREST RELATED

1793

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE The model predicts 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q1/5 • NEAREST RELATED

A model is y = 5x. At x = 3, the observed value is y = 13.

Is the model's prediction too high or too low, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q2/5 • NEAREST RELATED

1794

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Too high, by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which observed point is closest to the model y = x + 3?

A. (2, 7)

B. (2, 9)

C. (2, 3)

D. (2, 5.2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q3/5 • NEAREST RELATED

1795

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5.2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A linear model can be useful even when the data points do not all lie on the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q4/5 • NEAREST RELATED

1796

QUESTION ID

S58-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If one data point is far from the line of fit, the model must be thrown out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q5/5 • NEAREST RELATED

1797

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 58 • 7.SP.C.7

FORWARD 2

Decide whether observed results are consistent with a model • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on red 25% of the time. In 200 spins it lands on red 52 times.

Is this consistent with the claim? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q1/5

1799

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 52 is close to the expected 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q1/5

A number cube is claimed to be fair. In 300 rolls, 6 comes up 120 times.

Is this consistent with the claim? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q2/5

1800

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The expected number is about 50, and 120 is far more than that.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q2/5

A model says P(win) = 0.1. Which number of wins in 50 games would cast the most doubt on the model?

A. 5

B. 4

C. 6

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q3/5

1801

QUESTION ID

S58-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q3/5

True or false?

A simulation can show the range of results that a model usually produces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q4/5

1802

QUESTION ID

S58-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q4/5

True or false?

Any difference between observed and expected results proves that a model is wrong.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q5/5

1803

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a • STATISTICS & PROBABILITY

Find the probability of a compound event as the fraction of outcomes in the sample space

MAIN	Grade 7 main question	**7.SP.C.8.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Use multiplication to count combinations in equal groups	**3.OA.A.3**
FORWARD 1	Find joint relative frequencies in a two-way table — nearest relate…	**8.SP.A.4**
FORWARD 2	Use the multiplication rule for independent events	**HSS-CP.A.2**

SET 59 • 7.SP.C.8.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

A coin is flipped and a number cube is rolled.

What is the probability of getting heads and a 6?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q1/5

1806

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two coins are flipped.

What is the probability that both land tails?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q2/5

1807

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q2/5

The spinner has 3 equal sections. It is spun twice.

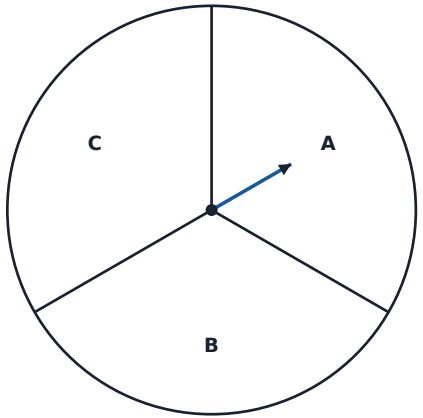
What is the probability of landing on A both times?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{9}$

D. $\frac{1}{3}$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q3/5

1808

QUESTION ID

S59-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q3/5

True or false?

Two number cubes are rolled. The probability that the sum is 7 is $\frac{6}{36} = \frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q4/5

1809

QUESTION ID

S59-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Two coins are flipped. The probability of getting one head and one tail is $\frac{1}{4}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q5/5

1810

QUESTION ID

S59-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The probability is 2/4 = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{2} \times \frac{1}{6}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q1/5

1812

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{2}{3} \times \frac{3}{4}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q2/5

1813

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 6/12 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{4} \times \frac{1}{4}$

A. $\frac{2}{4}$

B. $\frac{1}{8}$

C. $\frac{1}{16}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q3/5

1814

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q4/5

1815

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{5} \times \frac{1}{5} = \frac{2}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q5/5

1816

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 2

Use multiplication to count combinations in equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

A café has 3 kinds of sandwiches and 4 kinds of drinks.

How many different sandwich-and-drink pairs can be made?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q1/5

1818

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q1/5

Ty has 4 shirts and 5 pairs of pants.

How many different shirt-and-pants outfits can he make?

Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q2/5

1819

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q2/5

A shop has 6 ice cream flavors and 2 kinds of cones.

How many different flavor-and-cone choices are there?

A. 12

B. 8

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q3/5

1820

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q3/5

1820

True or false?

3 rows of 7 chairs is 21 chairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q4/5

1821

QUESTION ID

S59-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q4/5

True or false?

5 hats and 3 scarves make 8 different hat-and-scarf pairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q5/5

1822

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make 15 pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a

FORWARD 1

Find joint relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of an even number and heads?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q1/5 • NEAREST RELATED

1824

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of tails?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q2/5 • NEAREST RELATED

1825

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows 100 trials of flipping a coin and rolling a number cube.

Which result happened most often?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. Even and tails

B. Odd and tails

C. Even and heads

D. Odd and heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q3/5 • NEAREST RELATED

1826

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Odd and heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of an odd number and tails is 0.23.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q4/5 • NEAREST RELATED

1827

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The relative frequencies of the four results in the table add to 2.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q5/5 • NEAREST RELATED

1828

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 59 • 7.SP.C.8.a

FORWARD 2

Use the multiplication rule for independent events • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

Events A and B are independent. P(A) = 0.3 and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q1/5

1830

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q1/5

A fair coin is flipped 3 times.

What is the probability that all 3 flips are heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q2/5

1831

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q2/5

A fair number cube is rolled twice.

What is the probability of rolling two 6s?

A. $\frac{1}{6}$

B. $\frac{1}{36}$

C. $\frac{1}{12}$

D. $\frac{2}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q3/5

1832

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q3/5

True or false?

Events A and B are independent when P(A and B) = P(A) × P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q4/5

1833

QUESTION ID

S59-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q4/5

True or false?

Drawing two cards from a deck without putting the first card back are independent events.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q5/5

1834

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b • STATISTICS & PROBABILITY

Represent the sample space of a compound event with a list, a table, or a tree diagram

MAIN	Grade 7 main question	**7.SP.C.8.b**
BACKWARD 1	Use repeated addition to find the total in an array	**2.OA.C.4**
BACKWARD 2	Form ordered pairs from two numerical patterns	**5.OA.B.3**
FORWARD 1	Organize the outcomes of two categorical variables in a two-way…	**8.SP.A.4**
FORWARD 2	Find a conditional probability as P(A and B) ÷ P(B)	**HSS-CP.A.3**

SET 60 • 7.SP.C.8.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A coin is flipped and a spinner with sections 1, 2, and 3 is spun.

List the sample space.

Sample space: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q1/5

1837

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

H1, H2, H3, T1, T2, T3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A café has 3 sandwiches and 4 drinks.

How many outcomes are in the sample space for choosing one sandwich and one drink?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q2/5

1838

QUESTION ID

S60-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q2/5

A tree diagram shows choosing a shirt (red or blue) and then pants (black, tan, or gray).

How many branches are at the end of the tree?

A. 3

B. 2

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q3/5

1839

QUESTION ID

S60-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q3/5

True or false?

"Roll a 3 and get tails" is one outcome in the sample space for rolling a number cube and flipping a coin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q4/5

1840

QUESTION ID

S60-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q4/5

True or false?

The sample space for flipping two coins is HH, HT, TT.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q5/5

1841

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE TH is missing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 1

Use repeated addition to find the total in an array • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

An array has 3 rows with 4 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q1/5

1843

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 4 + 4 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q1/5

An array has 2 rows with 5 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q2/5

1844

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 + 5 = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q2/5

Which equation gives the total of an array with 4 rows of 3?

A. 4 + 3 = 7

B. 3 + 3 + 3 + 3 = 12

C. 3 + 3 + 3 = 9

D. 3 + 4 + 3 = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q3/5

1845

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 + 3 + 3 + 3 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q3/5

True or false?

An array with 3 rows of 5 has 5 + 5 + 5 = 15 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q4/5

1846

QUESTION ID

S60-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q4/5

True or false?

An array with 2 rows of 4 has 4 + 2 = 6 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q5/5

1847

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 2

Form ordered pairs from two numerical patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 4.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q1/5

1849

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 0), (2, 4), (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Pattern A starts at 1 and adds 1. Pattern B starts at 3 and adds 3.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q2/5

1850

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 3), (2, 6), (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q2/5

Pattern A is 0, 3, 6, 9. Pattern B is 0, 6, 12, 18.

Which ordered pair (A, B) comes from these patterns?

A. (3, 3)

B. (12, 6)

C. (6, 9)

D. (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q3/5

1851

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

In the patterns A: 0, 5, 10 and B: 0, 10, 20, each term in B is twice the matching term in A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q4/5

1852

QUESTION ID

S60-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The patterns A: 0, 2, 4 and B: 0, 3, 6 make the ordered pair (4, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q5/5

1853

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make the pair (4, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b

FORWARD 1

Organize the outcomes of two categorical variables in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

In a class, 14 students have a sibling and 10 do not. 8 of the students with a sibling have a pet. 3 of the students without a sibling have a pet.

How many students have a sibling but no pet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q1/5 • NEAREST RELATED

1855

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q1/5 • NEAREST RELATED

In a class, 14 students have a sibling and 10 do not. 8 of the students with a sibling have a pet. 3 of the students without a sibling have a pet.

How many students have a pet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q2/5 • NEAREST RELATED

1856

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q2/5 • NEAREST RELATED

A two-way table has the rows "Sibling" and "No sibling" and the columns "Pet" and "No pet."

How many outcome cells (not totals) does it have?

A. 4

B. 2

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q3/5 • NEAREST RELATED

1857

QUESTION ID

S60-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A two-way table can show the counts for two categorical variables at the same time.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q4/5 • NEAREST RELATED

1858

QUESTION ID

S60-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A two-way table with 3 rows and 2 columns of outcomes has 5 outcome cells.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q5/5 • NEAREST RELATED

1859

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 3 × 2 = 6 outcome cells.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 60 • 7.SP.C.8.b

FORWARD 2

Find a conditional probability as P(A and B) ÷ P(B) • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.12 and P(B) = 0.4.

Find P(A | B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q1/5

1861

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q1/5

Two coins are flipped. At least one landed heads.

What is the probability that both landed heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q2/5

1862

QUESTION ID

S60-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q2/5

A number cube is rolled, and the result is even.

What is the probability that it is a 6?

A. $\frac{1}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q3/5

1863

QUESTION ID

S60-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q3/5

True or false?

If A and B are independent, then P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q4/5

1864

QUESTION ID

S60-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q5/5

1865

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q5/5

SET 61 • 7.SP.C.8.c • STATISTICS & PROBABILITY

Design and use a simulation to estimate the probability of a compound event

MAIN	Grade 7 main question	**7.SP.C.8.c**
BACKWARD 1	Display data in a dot plot	**6.SP.B.4**
BACKWARD 2	Compare fractions with the same numerator or the same denomin…	**3.NF.A.3.d**
FORWARD 1	Construct and interpret a scatter plot — nearest related (no direct…	**8.SP.A.1**
FORWARD 2	Use a simulation to decide whether a difference between treatme…	**HSS-IC.B.5**

SET 61 • 7.SP.C.8.c

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.c

ANSWER KEY Reduced copy of the question page, with its answer

A player makes 60% of her free throws.

Describe how to use random digits 0 to 9 to simulate one trial of 2 free throws.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q1/5

1868

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

Let the digits 0 to 5 stand for a make and 6 to 9 stand for a miss. Choose 2 random digits; each pair of digits is one trial.

GRADING NOTE Any plan where 6 of the 10 digits stand for a make, and each trial uses 2 digits, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a simulation with 50 trials, 18 trials showed both free throws made.

Estimate the probability that both are made.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q2/5

1869

QUESTION ID
S61-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.36

GRADING NOTE 18/50 or 9/25 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q2/5

Each of 3 children is equally likely to be a boy or a girl.

Which tool can simulate the 3 children?

A. Spin a spinner with 3 unequal parts once

B. Flip a coin 3 times

C. Draw one card from a deck

D. Roll one number cube once

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q3/5

1870

QUESTION ID

S61-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Flip a coin 3 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q3/5

True or false?

Running more trials of a simulation usually gives a better estimate of a probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q4/5

1871

QUESTION ID

S61-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q4/5

True or false?

A single spin of a spinner with 4 equal sections can simulate an event with a probability of $\frac{1}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q5/5

1872

QUESTION ID

S61-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Four equal sections give probabilities in fourths; use 3 equal sections instead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 1

Display data in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

Make a dot plot of the numbers of heads in 8 trials:

1, 2, 2, 3, 1, 2, 0, 2



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q1/5

1874

QUESTION ID

S61-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Dots above 0: 1, above 1: 2, above 2: 4, above 3: 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q1/5

The dot plot shows the number of heads in 10 trials.

In how many trials were there 2 or more heads?



Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q2/5

1875

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q2/5

Which display shows each data value as a dot above a number line?

A. Dot plot

B. Histogram

C. Bar graph

D. Box plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q3/5

1876

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A dot plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q4/5

1877

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A box plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q5/5

1878

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 2

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

Which is greater, $\frac{3}{8}$ or $\frac{5}{8}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q1/5

1880

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write >, <, or =.

$\frac{2}{3}$ ___ $\frac{2}{6}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q2/5

1881

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

>

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q2/5

Which fraction is less than $\frac{4}{6}$?

A. $\frac{5}{6}$

B. $\frac{6}{6}$

C. $\frac{3}{6}$

D. $\frac{4}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q3/5

1882

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{4} < \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q4/5

1883

QUESTION ID

S61-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{3}{5} > \frac{4}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q5/5

1884

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 7.SP.C.8.c

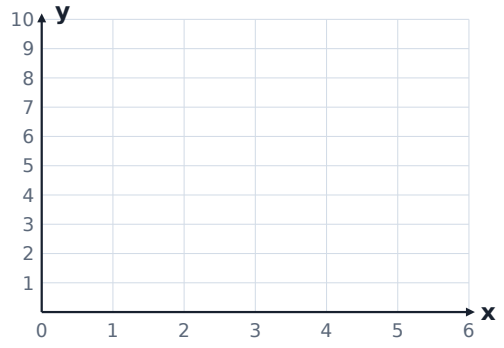
FORWARD 1

Construct and interpret a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Make a scatter plot of the data in the table.

x	1	2	3	4	5
y	2	4	5	7	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q1/5 • NEAREST RELATED

1886

QUESTION ID

S61-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), (3, 5), (4, 7), and (5, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows x-values 1, 2, 3, 4, 5 and y-values 2, 4, 5, 7, 9.

Describe the association between x and y.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q2/5 • NEAREST RELATED

1887

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q2/5 • NEAREST RELATED

What does each point in a scatter plot stand for?

A. An outlier

B. One pair of values from the data

C. The mean of the data

D. The line of fit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q3/5 • NEAREST RELATED

1888

QUESTION ID

S61-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. One pair of values from the data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A scatter plot shows the relationship between two numerical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q4/5 • NEAREST RELATED

1889

QUESTION ID

S61-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

All the points in a scatter plot must lie on a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q5/5 • NEAREST RELATED

1890

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 61 • 7.SP.C.8.c

FORWARD 2

Use a simulation to decide whether a difference between treatments is significant • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plants in Group A grew a mean of 12 cm and plants in Group B grew a mean of 9 cm. In 100 re-randomizations of the data, a difference of 3 cm or more happened 2 times.

Is the difference likely due to the treatment? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q1/5

1892

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. A difference that large happened by chance only about 2% of the time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q1/5

In 100 re-randomizations of the data, a difference of 3 cm or more happened 40 times.

What does this suggest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q2/5

1893

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The difference could easily be due to chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q2/5

Why are the data re-randomized many times in the simulation?

A. To remove outliers

B. To see what differences happen by chance alone

C. To make the mean larger

D. To make the treatment work better

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q3/5

1894

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. To see what differences happen by chance alone

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q3/5

True or false?

Random assignment helps make two treatment groups alike before an experiment begins.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q4/5

1895

QUESTION ID

S61-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q4/5

True or false?

A difference between two groups in one experiment is always caused by the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q5/5

1896

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q5/5